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// 文件：backend/app/api/auth.py
import base64
import html
import secrets
import time
import uuid
from fastapi import APIRouter, Depends, HTTPException, Request, Response
from sqlalchemy import delete, select
from sqlalchemy.orm import Session
from app.core.db import get_db
from app.core.config import get_settings
from app.core.security import current_user, hash_password, issue_token, sign_captcha_answer, ver
    ↳ ify_password
from app.core.rate_limit import enforce_auth_rate_limit
from app.models import CaptchaChallenge, User, UserProfile
from app.schemas import AuthInput, RegisterInput
router = APIRouter(prefix="/auth", tags=["auth"])
@router.get("/captcha")
def registration_captcha(request: Request, response: Response, db: Session = Depends(get_db)):
    enforce_auth_rate_limit(request, db, "captcha")
    now = int(time.time())
    db.execute(delete(CaptchaChallenge).where(CaptchaChallenge.expires_at < now))
    challenge_id = str(uuid.uuid4())
    alphabet = "ABCDEFGHJKLMNPQRSTUVWXYZ23456789"
    answer = "".join(secrets.choice(alphabet) for _ in range(5))
    db.add(CaptchaChallenge(
        id=challenge_id,
        answer_hash=sign_captcha_answer(challenge_id, answer),
        expires_at=now + 300,
        attempts=0,
        created_at=now,
    ))
    db.commit()
    colors = ["#5436a8", "#704bd0", "#42366c", "#8a69db"]
    glyphs = []
    for index, char in enumerate(answer):
        x = 25 + index * 29
        y = secrets.randbelow(9) + 31
        angle = secrets.randbelow(31) - 15
        color = secrets.choice(colors)
        glyphs.append(f'<text x="{x}" y="{y}" fill="{color}" transform="rotate({angle} {x} {y})"
    ↳ >{html.escape(char)}</text>')
    noise = []
    for _ in range(7):
        x1, x2 = secrets.randbelow(160), secrets.randbelow(160)
        y1, y2 = secrets.randbelow(52), secrets.randbelow(52)
        noise.append(f'<path d="M{x1} {y1} Q{(x1+x2)//2} {secrets.randbelow(52)} {x2} {y2}"/>')
    svg = (
        '<svg xmlns="http://www.w3.org/2000/svg" width="160" height="52" viewBox="0 0 160 52">'
        '<rect width="160" height="52" rx="11" fill="#f5f0ff"/>'
        '<g fill="none" stroke="#baa6ec" stroke-width="1" opacity=".55">' + "".join(noise) + "</
    ↳ g>"
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        '<g font-family="Arial,sans-serif" font-size="25" font-weight="700" letter-spacing="2">'
    ↳  + "".join(glyphs) + "</g></svg>"
    )
    response.headers["Cache-Control"] = "no-store, max-age=0"
    response.headers["Pragma"] = "no-cache"
    return {"captcha_id": challenge_id, "image_data": "" + base64.b64e
    ↳ ncode(svg.encode()).decode(), "expires_in": 300}
def _consume_registration_captcha(db: Session, challenge_id: str, answer: str) -> None:
    now = int(time.time())
    challenge = db.scalar(select(CaptchaChallenge).where(CaptchaChallenge.id == challenge_id).wi
    ↳ th_for_update())
    invalid = HTTPException(400, "验证码错误或已失效，请刷新后重试")
    if challenge is None:
        raise invalid
    if challenge.expires_at <= now:
        db.delete(challenge)
        db.commit()
        raise invalid
    expected = sign_captcha_answer(challenge_id, answer)
    if not secrets.compare_digest(challenge.answer_hash, expected):
        challenge.attempts += 1
        if challenge.attempts >= 5:
            db.delete(challenge)
        db.commit()
        raise invalid
    db.delete(challenge)
    db.commit()
@router.post("/register", status_code=201)
def register(payload: RegisterInput, request: Request, db: Session = Depends(get_db)):
    enforce_auth_rate_limit(request, db, "register", payload.username)
    if payload.password != payload.password_confirm:
        raise HTTPException(400, "两次输入的密码不一致")
    _consume_registration_captcha(db, payload.captcha_id, payload.captcha_code)
    if db.scalar(select(User).where(User.username == payload.username)):
        raise HTTPException(409, "用户名已被使用")
    user = User(id=str(uuid.uuid4()), username=payload.username, password_hash=hash_password(pay
    ↳ load.password))
    db.add_all([user, UserProfile(user_id=user.id)])
    db.commit()
    return {"access_token": issue_token(user.id), "token_type": "bearer", "username": user.usern
    ↳ ame, "user_id": user.id}
@router.post("/login")
def login(payload: AuthInput, request: Request, db: Session = Depends(get_db)):
    enforce_auth_rate_limit(request, db, "login", payload.username)
    user = db.scalar(select(User).where(User.username == payload.username))
    if not user or not verify_password(payload.password, user.password_hash):
        raise HTTPException(401, "用户名或密码不正确")
    return {"access_token": issue_token(user.id), "token_type": "bearer", "username": user.usern
    ↳ ame, "user_id": user.id}
@router.get("/me")
def me(user: User = Depends(current_user)):
```

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// 文件：backend/app/api/auth.py
    admin_usernames = {
        username.strip()
        for username in get_settings().content_admin_usernames.split(",")
        if username.strip()
    }
    return {"user_id": user.id, "username": user.username, "role": "admin" if user.username in a
    ↳ dmin_usernames else "learner"}
// 文件：backend/app/api/content_admin.py
import hmac
from fastapi import APIRouter, Depends, Header, HTTPException
from pydantic import BaseModel, Field
from sqlalchemy import select
from sqlalchemy.orm import Session
from app.core.config import get_settings
from app.core.db import get_db
from app.core.security import current_user
from app.models import KnowledgeEdge, KnowledgeNode, LearningCourse, QuizQuestion, User
from app.api.learning import ensure_default_question_for_node
router = APIRouter(prefix="/api/admin", tags=["content-admin"])
def require_content_admin(x_content_admin_token: str | None = Header(default=None), user: User =
    ↳  Depends(current_user)):
    settings = get_settings()
    configured = settings.content_admin_token
    if not configured:
        raise HTTPException(503, "课程内容管理未启用，请在服务端配置 CONTENT_ADMIN_TOKEN")
    admin_usernames = {
        username.strip()
        for username in getattr(settings, "content_admin_usernames", "").split(",")
        if username.strip()
    }
    if user.username not in admin_usernames:
        raise HTTPException(403, "当前账号不是课程内容管理员")
    if not x_content_admin_token or not hmac.compare_digest(x_content_admin_token, configured):
        raise HTTPException(403, "无权管理课程内容")
    return user
class CourseBody(BaseModel):
    id: str = Field(min_length=1, max_length=80)
    name: str = Field(min_length=1, max_length=160)
    category: str = Field(default="", max_length=80)
    points: int = Field(default=0, ge=0, le=100000)
    chapter: str = Field(default="", max_length=240)
    color: str = Field(default="#7357dc", max_length=20)
    icon: str = Field(default="", max_length=20)
class NodeBody(BaseModel):
    id: str = Field(min_length=1, max_length=100)
    name: str = Field(min_length=1, max_length=180)
    type: str = Field(default="concept", max_length=24)
    description: str = Field(default="", max_length=5000)
class EdgeBody(BaseModel):
    source: str = Field(min_length=1, max_length=100)
    target: str = Field(min_length=1, max_length=100)
class QuestionBody(BaseModel):
```

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// 文件：backend/app/api/content_admin.py
    course_id: str = Field(min_length=1, max_length=80)
    type: str = Field(default="单选题", max_length=30)
    difficulty: str = Field(default="基础", max_length=30)
    topic: str = Field(default="", max_length=160)
    concept_id: str = Field(default="", max_length=100)
    title: str = Field(min_length=1, max_length=10000)
    options: list[str] = Field(default_factory=list, max_length=20)
    answer: str | list[str]
    explanation: str = Field(default="", max_length=10000)
    misconceptions: dict = Field(default_factory=dict)
    transfer_question_ids: list[int] = Field(default_factory=list, max_length=20)
    evidence: dict | None = None
def _course_out(row: LearningCourse):
    return {"id": row.id, "name": row.name, "category": row.category, "points": row.points,
            "chapter": row.chapter, "color": row.color, "icon": row.icon}
@router.post("/courses", status_code=201)
def create_course(payload: CourseBody, db: Session = Depends(get_db), _admin: User = Depends(req
    ↳ uire_content_admin)):
    if db.get(LearningCourse, payload.id):
        raise HTTPException(409, "课程 ID 已存在")
    row = LearningCourse(**payload.model_dump())
    db.add(row)
    db.commit()
    return _course_out(row)
@router.patch("/courses/{course_id}")
def update_course(course_id: str, payload: CourseBody, db: Session = Depends(get_db), _admin: Us
    ↳ er = Depends(require_content_admin)):
    row = db.get(LearningCourse, course_id)
    if row is None:
        raise HTTPException(404, "课程不存在")
    if payload.id != course_id:
        raise HTTPException(422, "课程 ID 不允许修改")
    for key, value in payload.model_dump().items():
        setattr(row, key, value)
    db.commit()
    return _course_out(row)
@router.delete("/courses/{course_id}", status_code=204)
def delete_course(course_id: str, db: Session = Depends(get_db), _admin: User = Depends(require_
    ↳ content_admin)):
    row = db.get(LearningCourse, course_id)
    if row is None:
        raise HTTPException(404, "课程不存在")
    db.delete(row)
    db.commit()
@router.get("/courses/{course_id}/graph")
def get_admin_graph(course_id: str, db: Session = Depends(get_db), _admin: User = Depends(requir
    ↳ e_content_admin)):
    if not db.get(LearningCourse, course_id):
        raise HTTPException(404, "课程不存在")
    nodes = db.scalars(select(KnowledgeNode).where(KnowledgeNode.course_id == course_id, Knowled
    ↳ geNode.owner_id.is_(None)).order_by(KnowledgeNode.id)).all()
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// 文件：backend/app/api/content_admin.py
    edges = db.scalars(select(KnowledgeEdge).where(KnowledgeEdge.course_id == course_id, Knowled
    ↳ geEdge.owner_id.is_(None))).all()
    return {"nodes": [{"id": item.id, "name": item.name, "type": item.type, "description": item.
    ↳ description} for item in nodes],
            "edges": [{"source": item.source, "target": item.target} for item in edges]}
@router.post("/courses/{course_id}/nodes", status_code=201)
def create_node(course_id: str, payload: NodeBody, db: Session = Depends(get_db), _admin: User =
    ↳  Depends(require_content_admin)):
    if not db.get(LearningCourse, course_id):
        raise HTTPException(404, "课程不存在")
    if db.get(KnowledgeNode, payload.id):
        raise HTTPException(409, "知识点 ID 已存在")
    row = KnowledgeNode(course_id=course_id, **payload.model_dump())
    db.add(row)
    db.flush()
    ensure_default_question_for_node(db, row)
    db.commit()
    return {"id": row.id, "courseId": row.course_id, "name": row.name, "type": row.type, "descri
    ↳ ption": row.description}
@router.patch("/courses/{course_id}/nodes/{node_id}")
def update_node(course_id: str, node_id: str, payload: NodeBody, db: Session = Depends(get_db),
    ↳ _admin: User = Depends(require_content_admin)):
    row = db.scalar(select(KnowledgeNode).where(KnowledgeNode.id == node_id, KnowledgeNode.cours
    ↳ e_id == course_id, KnowledgeNode.owner_id.is_(None)))
    if row is None:
        raise HTTPException(404, "知识点不存在")
    if payload.id != node_id:
        raise HTTPException(422, "知识点 ID 不允许修改")
    row.name, row.type, row.description = payload.name, payload.type, payload.description
    ensure_default_question_for_node(db, row)
    db.commit()
    return {"id": row.id, "courseId": row.course_id, "name": row.name, "type": row.type, "descri
    ↳ ption": row.description}
@router.delete("/courses/{course_id}/nodes/{node_id}", status_code=204)
def delete_node(course_id: str, node_id: str, db: Session = Depends(get_db), _admin: User = Depe
    ↳ nds(require_content_admin)):
    row = db.scalar(select(KnowledgeNode).where(KnowledgeNode.id == node_id, KnowledgeNode.cours
    ↳ e_id == course_id, KnowledgeNode.owner_id.is_(None)))
    if row is None:
        raise HTTPException(404, "知识点不存在")
    db.delete(row)
    db.commit()
@router.put("/courses/{course_id}/edges")
def replace_course_edges(course_id: str, edges: list[EdgeBody], db: Session = Depends(get_db), _
    ↳ admin: User = Depends(require_content_admin)):
    if not db.get(LearningCourse, course_id):
        raise HTTPException(404, "课程不存在")
    if len(edges) > 2000:
        raise HTTPException(422, "关系数量超出限制")
    node_ids = set(db.scalars(select(KnowledgeNode.id).where(KnowledgeNode.course_id == course_i
    ↳ d, KnowledgeNode.owner_id.is_(None))).all())
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// 文件：backend/app/api/content_admin.py
    pairs = [(edge.source, edge.target) for edge in edges]
    if len(pairs) != len(set(pairs)) or any(source not in node_ids or target not in node_ids or
    ↳ source == target for source, target in pairs):
        raise HTTPException(422, "图谱关系重复、指向无效节点或形成自环")
    db.query(KnowledgeEdge).filter(KnowledgeEdge.course_id == course_id, KnowledgeEdge.owner_id.
    ↳ is_(None)).delete(synchronize_session=False)
    db.add_all([KnowledgeEdge(course_id=course_id, source=source, target=target) for source, tar
    ↳ get in pairs])
    db.commit()
    return {"courseId": course_id, "edgeCount": len(pairs)}
@router.post("/questions", status_code=201)
def create_question(payload: QuestionBody, db: Session = Depends(get_db), _admin: User = Depends
    ↳ (require_content_admin)):
    if not db.get(LearningCourse, payload.course_id):
        raise HTTPException(404, "课程不存在")
    row = QuizQuestion(**payload.model_dump())
    db.add(row)
    db.commit()
    return {"id": row.id, "courseId": row.course_id}
@router.get("/questions")
def list_admin_questions(course_id: str | None = None, db: Session = Depends(get_db), _admin: Us
    ↳ er = Depends(require_content_admin)):
    query = select(QuizQuestion).where(QuizQuestion.owner_id.is_(None)).order_by(QuizQuestion.id
    ↳ )
    if course_id:
        query = query.where(QuizQuestion.course_id == course_id)
    return [{"id": row.id, "courseId": row.course_id, "type": row.type, "difficulty": row.diffic
    ↳ ulty,
             "topic": row.topic, "conceptId": row.concept_id, "title": row.title, "options": row
    ↳ .options,
             "answer": row.answer, "explanation": row.explanation, "misconceptions": row.misconc
    ↳ eptions,
             "transferQuestionIds": row.transfer_question_ids, "evidence": row.evidence}
            for row in db.scalars(query).all()]
@router.patch("/questions/{question_id}")
def update_question(question_id: int, payload: QuestionBody, db: Session = Depends(get_db), _adm
    ↳ in: User = Depends(require_content_admin)):
    row = db.get(QuizQuestion, question_id)
    if row is None:
        raise HTTPException(404, "题目不存在")
    if not db.get(LearningCourse, payload.course_id):
        raise HTTPException(404, "课程不存在")
    for key, value in payload.model_dump().items():
        setattr(row, key, value)
    db.commit()
    return {"id": row.id, "courseId": row.course_id}
@router.delete("/questions/{question_id}", status_code=204)
def delete_question(question_id: int, db: Session = Depends(get_db), _admin: User = Depends(requ
    ↳ ire_content_admin)):
    row = db.get(QuizQuestion, question_id)
    if row is None:
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// 文件: backend/app/api/content_admin.py
        raise HTTPException(404, "题目不存在")
    db.delete(row)
    db.commit()
// 文件: backend/app/api/documents.py
import hashlib
import re
import threading
import time
from datetime import datetime, timedelta, timezone
import uuid
from pathlib import Path
from fastapi import APIRouter, Depends, File, HTTPException, UploadFile
from fastapi.responses import FileResponse
from sqlalchemy import select, update, delete
from sqlalchemy.orm import Session
from app.core.config import DATA_DIR, get_settings
from app.core.db import SessionLocal, get_db
from app.core.security import current_user
from app.models import Document, DocumentChunk, IndexJob, KnowledgeBase, LearningCourse, Knowled
    ↳ geEdge, KnowledgeNode, QuizQuestion, User, UserMistake
from app.rag.chunker import chunk_segments
from app.rag.parsers import parse_document
from app.rag.vector_store import delete_chunks, delete_document, upsert_chunks
router = APIRouter(tags=["documents"])
ALLOWED = {"pdf": ("PDF", "application/pdf"), "docx": ("DOCX", "application/vnd.openxmlformats-o
    ↳ fficedocument.wordprocessingml.document"), "txt": ("TXT", "text/plain"), "md": ("MD", "text/mark
    ↳ down")}
_index_guard = threading.Lock()
_document_locks: dict[str, threading.Lock] = {}
def serialize(doc: Document):
    return {"id": doc.id, "knowledge_base_id": doc.knowledge_base_id, "file_name": doc.file_name
    ↳ , "file_type": doc.file_type, "file_size": doc.file_size, "status": doc.status, "error_message":
    ↳  doc.error_message, "chunk_count": doc.chunk_count, "created_at": doc.created_at}
def _generate_document_learning_assets(db: Session, doc: Document, chunks: list) -> int:
    """Create private graph concepts and one source-grounded question per section."""
    kb = db.get(KnowledgeBase, doc.knowledge_base_id)
    if not kb:
        return 0
    labels = [kb.course_name, kb.name, doc.file_name]
    normalized_labels = [re.sub(r"[^0-9a-z\u4e00-\u9fff]+", "", value.lower()) for value in labe
    ↳ ls if value]
    courses = db.scalars(select(LearningCourse)).all()
    course = next((item for item in courses if re.sub(r"[^0-9a-z\u4e00-\u9fff]+", "", item.name.
    ↳ lower()) in normalized_labels), None)
    if course is None:
        course = next((item for item in sorted(courses, key=lambda row: len(row.name), reverse=T
    ↳ rue)
                       if len(item.name) >= 3 and any(item.name in value or value in item.name f
    ↳ or value in labels if value)), None)
    if course is None:
        return 0
    grouped: dict[str, list[str]] = {}
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// 文件：backend/app/api/documents.py
    for chunk in chunks:
        text = (chunk.content or "").strip()
        if not text:
            continue
        title = (chunk.section_title or "").strip()
        if not title or title.lower() in {"正文", "内容", "未分类", "document"}:
            title = re.split(r"[。！？.!?\n]", text, maxsplit=1)[0].strip()[:40] or f"资料片段 {
    ↳ len(grouped) + 1}"
        if title not in grouped and len(grouped) >= 40:
            continue
        grouped.setdefault(title[:180], []).append(text)
    sections = [(title, "\n".join(parts)) for title, parts in grouped.items()]
    if not sections:
        return 0
    # Re-indexing replaces this document's generated assets without affecting other uploads.
    old_question_ids = db.scalars(select(QuizQuestion.id).where(QuizQuestion.source_document_id
    ↳ == doc.id, QuizQuestion.owner_id == doc.owner_id)).all()
    if old_question_ids:
        db.execute(delete(UserMistake).where(UserMistake.user_id == doc.owner_id, UserMistake.qu
    ↳ estion_id.in_(old_question_ids)))
    db.execute(delete(QuizQuestion).where(QuizQuestion.source_document_id == doc.id, QuizQuestio
    ↳ n.owner_id == doc.owner_id))
    db.execute(delete(KnowledgeEdge).where(KnowledgeEdge.source_document_id == doc.id, Knowledge
    ↳ Edge.owner_id == doc.owner_id))
    db.execute(delete(KnowledgeNode).where(KnowledgeNode.source_document_id == doc.id, Knowledge
    ↳ Node.owner_id == doc.owner_id))
    db.flush()
    descriptions: list[str] = []
    for _title, text in sections:
        sentence = re.split(r"(?<=[。！？.!?])\s*|\n+", text.strip(), maxsplit=1)[0].strip()
        descriptions.append((sentence or text).replace("\x00", "")[:700])
    created = 0
    for index, ((title, _text), description) in enumerate(zip(sections, descriptions)):
        node_id = f"doc-{doc.id.replace('-', '')[:24]}-{index + 1}"
        node = KnowledgeNode(id=node_id, course_id=course.id, name=title, type="concept", descri
    ↳ ption=description,
                             owner_id=doc.owner_id, source_document_id=doc.id)
        db.add(node)
        db.flush()
        db.add(KnowledgeEdge(course_id=course.id, source=course.id, target=node.id,
                             owner_id=doc.owner_id, source_document_id=doc.id))
        options = [description]
        for distractor in descriptions:
            if distractor != description and distractor not in options:
                options.append(distractor)
            if len(options) >= 4:
                break
        while len(options) < 4:
            options.append(f"资料未将第 {len(options) + 1} 项内容作为该主题的核心表述。")
        db.add(QuizQuestion(course_id=course.id, owner_id=doc.owner_id, source_document_id=doc.i
    ↳ d,
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// 文件：backend/app/api/documents.py
                            type="单选题", difficulty="基础", topic=title, concept_id=node.id,
                            title=f"根据《{doc.file_name}》，关于“{title}”的核心表述是哪一项？
    ↳ ",
                            options=options, answer=description, explanation=description,
                            evidence={"title": doc.file_name, "section": title, "excerpt": descr
    ↳ iption}))
            created += 1
    return created
def backfill_ready_document_learning_assets(db: Session) -> int:
    """Generate assets for already indexed uploads that predate this feature."""
    documents = db.scalars(select(Document).where(Document.status == "READY").order_by(Document.
    ↳ created_at)).all()
    generated = 0
    for doc in documents:
        existing = db.scalar(select(KnowledgeNode.id).where(
            KnowledgeNode.source_document_id == doc.id,
            KnowledgeNode.owner_id == doc.owner_id,
        ).limit(1))
        if existing:
            continue
        chunks = db.scalars(select(DocumentChunk).where(
            DocumentChunk.document_id == doc.id,
            DocumentChunk.owner_id == doc.owner_id,
        ).order_by(DocumentChunk.chunk_index)).all()
        generated += _generate_document_learning_assets(db, doc, chunks)
    db.commit()
    return generated
def index_document(document_id: str):
    with _index_guard:
        lock = _document_locks.setdefault(document_id, threading.Lock())
    with lock:
        _index_document_unlocked(document_id)
def _index_document_unlocked(document_id: str):
    db = SessionLocal()
    try:
        doc = db.get(Document, document_id)
        if not doc:
            return
        if doc.status != "UPLOADED":
            return
        source = DATA_DIR / "uploads" / doc.owner_id / doc.storage_name
        doc.status, doc.error_message = "PARSING", ""
        db.commit()
        segments = parse_document(source, doc.file_type)
        doc.extracted_text = "\n\n".join(segment.text for segment in segments)
        if not segments:
            if doc.file_type == "PDF":
                raise ValueError("PDF 未提取到可搜索文字；可能是扫描件，需要 OCR 处理")
            raise ValueError("文档没有可索引的文字内容")
        doc.status = "CHUNKING"
        db.commit()
```

```
// 文件：backend/app/api/documents.py
        chunks = chunk_segments(segments, document_id=doc.id, file_hash=doc.file_hash)
        if not chunks:
            raise ValueError("没有生成有效文本分块")
        doc.status = "EMBEDDING"
        db.commit()
        from app.core.config import get_settings
        payloads = [{"chunk_id": chunk.id, "document_id": doc.id, "knowledge_base_id": doc.knowl
    ↳ edge_base_id, "owner_id": doc.owner_id, "content": chunk.content, "page_start": chunk.page_start
    ↳ , "page_end": chunk.page_end, "section_title": chunk.section_title, "chunk_index": chunk.chunk_i
    ↳ ndex, "token_count": chunk.token_count, "content_hash": chunk.content_hash, "document_name": doc
    ↳ .file_name} for chunk in chunks]
        upsert_chunks(payloads)
        doc.status = "INDEXING"
        db.commit()
        old = list(db.scalars(select(DocumentChunk).where(DocumentChunk.document_id == doc.id)).
    ↳ all())
        wanted = {chunk.id for chunk in chunks}
        stale = [chunk.id for chunk in old if chunk.id not in wanted]
        if stale:
            delete_chunks(stale)
        for chunk in old:
            db.delete(chunk)
        from app.rag.embeddings import model
        for chunk in chunks:
            db.add(DocumentChunk(id=chunk.id, document_id=doc.id, knowledge_base_id=doc.knowledg
    ↳ e_base_id, owner_id=doc.owner_id, content=chunk.content, page_start=chunk.page_start, page_end=c
    ↳ hunk.page_end, section_title=chunk.section_title, chunk_index=chunk.chunk_index, token_count=chu
    ↳ nk.token_count, content_hash=chunk.content_hash, model_version=get_settings().embedding_model))
        doc.chunk_count, doc.status, doc.error_message = len(chunks), "READY", ""
        _generate_document_learning_assets(db, doc, chunks)
        db.commit()
    except Exception as error:
        db.rollback()
        doc = db.get(Document, document_id)
        if doc:
            try:
                delete_document(doc.id, doc.owner_id)
            except Exception:
                pass
            doc.status, doc.error_message = "FAILED", f"{type(error).__name__}: {str(error)[:400
    ↳ ]}"
            db.commit()
    finally:
        db.close()
_worker_stop = threading.Event()
_worker_thread: threading.Thread | None = None
def _claim_job() -> str | None:
    db = SessionLocal()
    try:
        job = db.scalar(select(IndexJob).where(IndexJob.status == "PENDING").order_by(IndexJob.c
    ↳ reated_at).limit(1))
```

```
// 文件：backend/app/api/documents.py
        if not job:
            return None
        claimed = db.execute(update(IndexJob).where(IndexJob.id == job.id, IndexJob.status == "P
    ↳ ENDING").values(status="RUNNING", attempts=IndexJob.attempts + 1, updated_at=datetime.now(timezo
    ↳ ne.utc)))
        db.commit()
        return job.id if claimed.rowcount else None
    except Exception:
        db.rollback()
        return None
    finally:
        db.close()
def _process_job(job_id: str) -> None:
    db = SessionLocal()
    try:
        job = db.get(IndexJob, job_id)
        if not job or job.status != "RUNNING":
            return
        document_id = job.document_id
    finally:
        db.close()
    index_document(document_id)
    db = SessionLocal()
    try:
        job = db.get(IndexJob, job_id)
        doc = db.get(Document, document_id)
        if job:
            job.status = "DONE" if doc and doc.status == "READY" else "FAILED"
            job.updated_at = datetime.now(timezone.utc)
            db.commit()
    finally:
        db.close()
def start_index_worker() -> None:
    global _worker_thread
    _worker_stop.clear()
    db = SessionLocal()
    try:
        cutoff = datetime.now(timezone.utc) - timedelta(hours=1)
        expired = list(db.scalars(select(IndexJob).where(IndexJob.status == "RUNNING", IndexJob.
    ↳ updated_at < cutoff)).all())
        for job in expired:
            doc = db.get(Document, job.document_id)
            if doc and doc.status != "READY":
                doc.status, doc.error_message = "UPLOADED", ""
            job.status, job.updated_at = "PENDING", datetime.now(timezone.utc)
        db.commit()
    finally:
        db.close()
    def run():
        while not _worker_stop.is_set():
            job_id = _claim_job()
```

```
// 文件：backend/app/api/documents.py
            if job_id:
                _process_job(job_id)
            else:
                _worker_stop.wait(1.0)
    _worker_thread = threading.Thread(target=run, name="studypath-index-worker", daemon=True)
    _worker_thread.start()
def stop_index_worker() -> None:
    _worker_stop.set()
    if _worker_thread and _worker_thread.is_alive():
        _worker_thread.join(timeout=10)
@router.post("/knowledge-bases/{knowledge_base_id}/documents", status_code=202)
async def upload_document(knowledge_base_id: str, file: UploadFile = File(...), db: Session = De
    ↳ pends(get_db), user: User = Depends(current_user)):
    kb = db.scalar(select(KnowledgeBase).where(KnowledgeBase.id == knowledge_base_id, KnowledgeB
    ↳ ase.owner_id == user.id))
    if not kb:
        raise HTTPException(404, "知识库不存在或无权访问")
    original_name = re.split(r"[\\/]", file.filename or "")[-1].strip().replace("\x00", "").repl
    ↳ ace("\r", "").replace("\n", "")
    ext = Path(original_name).suffix.lower().lstrip(".")
    if ext not in ALLOWED:
        raise HTTPException(415, "仅支持 PDF、DOCX、TXT、Markdown")
    mime = (file.content_type or "").split(";")[0].lower()
    accepted_mime = {ALLOWED[ext][1], "application/octet-stream"}
    if ext == "pdf": accepted_mime.add("application/x-pdf")
    if ext == "md": accepted_mime.add("text/plain")
    if mime and mime not in accepted_mime:
        raise HTTPException(415, "文件 MIME 类型与扩展名不匹配")
    limit = get_settings().max_upload_mb * 1024 * 1024
    raw = await file.read(limit + 1)
    if len(raw) > limit:
        raise HTTPException(413, f"文件不能超过 {get_settings().max_upload_mb} MB")
    if not raw:
        raise HTTPException(400, "文件为空")
    if ext == "pdf" and not raw.startswith(b"%PDF-"):
        raise HTTPException(415, "文件内容与 PDF 扩展名不匹配")
    if ext == "docx" and not raw.startswith(b"PK\x03\x04"):
        raise HTTPException(415, "文件内容与 DOCX 扩展名不匹配")
    if ext in ("txt", "md") and b"\x00" in raw:
        raise HTTPException(415, "文本文件包含二进制内容")
    digest = hashlib.sha256(raw).hexdigest()
    if db.scalar(select(Document).where(Document.owner_id == user.id, Document.knowledge_base_id
    ↳  == kb.id, Document.file_hash == digest)):
        raise HTTPException(409, "该知识库已上传相同文件")
    doc_id = str(uuid.uuid4())
    storage_name = f"{doc_id}.{ext}"
    directory = DATA_DIR / "uploads" / user.id
    directory.mkdir(parents=True, exist_ok=True)
    destination = directory / storage_name
    destination.write_bytes(raw)
    doc = Document(id=doc_id, owner_id=user.id, knowledge_base_id=kb.id, file_name=original_name
```

```
// 文件：backend/app/api/documents.py
    ↳ [:255], storage_name=storage_name, file_type=ALLOWED[ext][0], file_size=len(raw), file_hash=dige
    ↳ st, status="UPLOADED")
    db.add(doc)
    db.flush()
    db.add(IndexJob(id=str(uuid.uuid4()), document_id=doc.id, status="PENDING"))
    db.commit()
    return serialize(doc)
@router.get("/knowledge-bases/{knowledge_base_id}/documents")
def list_documents(knowledge_base_id: str, db: Session = Depends(get_db), user: User = Depends(c
    ↳ urrent_user)):
    kb = db.scalar(select(KnowledgeBase).where(KnowledgeBase.id == knowledge_base_id, KnowledgeB
    ↳ ase.owner_id == user.id))
    if not kb:
        raise HTTPException(404, "知识库不存在")
    return [serialize(item) for item in db.scalars(select(Document).where(Document.knowledge_bas
    ↳ e_id == kb.id, Document.owner_id == user.id).order_by(Document.created_at.desc())).all()]
@router.get("/documents/{document_id}/status")
def document_status(document_id: str, db: Session = Depends(get_db), user: User = Depends(curren
    ↳ t_user)):
    doc = db.scalar(select(Document).where(Document.id == document_id, Document.owner_id == user
    ↳ .id))
    if not doc:
        raise HTTPException(404, "文档不存在")
    return serialize(doc)
@router.post("/documents/{document_id}/reindex", status_code=202)
def reindex_document(document_id: str, db: Session = Depends(get_db), user: User = Depends(curre
    ↳ nt_user)):
    doc = db.scalar(select(Document).where(Document.id == document_id, Document.owner_id == user
    ↳ .id))
    if not doc:
        raise HTTPException(404, "文档不存在")
    if doc.status in {"PARSING", "CHUNKING", "EMBEDDING", "INDEXING"}:
        raise HTTPException(409, "文档正在处理中")
    doc.status, doc.error_message = "UPLOADED", ""
    db.add(IndexJob(id=str(uuid.uuid4()), document_id=doc.id, status="PENDING"))
    db.commit()
    return serialize(doc)
@router.delete("/documents/{document_id}", status_code=204)
def delete_doc(document_id: str, db: Session = Depends(get_db), user: User = Depends(current_use
    ↳ r)):
    doc = db.scalar(select(Document).where(Document.id == document_id, Document.owner_id == user
    ↳ .id))
    if not doc:
        raise HTTPException(404, "文档不存在")
    try:
        delete_document(doc.id, user.id)
    except Exception:
        raise HTTPException(503, "向量索引暂不可用，文档未删除；请恢复服务后重试") from None
    (DATA_DIR / "uploads" / user.id / doc.storage_name).unlink(missing_ok=True)
    question_ids = db.scalars(select(QuizQuestion.id).where(QuizQuestion.source_document_id == d
    ↳ oc.id, QuizQuestion.owner_id == user.id)).all()
```

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// 文件: backend/app/api/documents.py
    if question_ids:
        db.execute(delete(UserMistake).where(UserMistake.user_id == user.id, UserMistake.questio
    ↳ n_id.in_(question_ids)))
        db.execute(delete(QuizQuestion).where(QuizQuestion.id.in_(question_ids), QuizQuestion.ow
    ↳ ner_id == user.id))
    db.execute(delete(KnowledgeEdge).where(KnowledgeEdge.source_document_id == doc.id, Knowledge
    ↳ Edge.owner_id == user.id))
    db.execute(delete(KnowledgeNode).where(KnowledgeNode.source_document_id == doc.id, Knowledge
    ↳ Node.owner_id == user.id))
    db.delete(doc)
    db.commit()
@router.get("/documents/{document_id}/preview")
def preview_document(document_id: str, chunk_id: str | None = None, page: int | None = None, db:
    ↳  Session = Depends(get_db), user: User = Depends(current_user)):
    doc = db.scalar(select(Document).where(Document.id == document_id, Document.owner_id == user
    ↳ .id))
    if not doc:
        raise HTTPException(404, "文档不存在")
    if chunk_id:
        chunk = db.scalar(select(DocumentChunk).where(DocumentChunk.id == chunk_id, DocumentChun
    ↳ k.document_id == doc.id, DocumentChunk.owner_id == user.id))
        if not chunk:
            raise HTTPException(404, "原文片段不存在")
        original_text = chunk.content
        if doc.file_type == "PDF" and chunk.page_start:
            import pymupdf
            with pymupdf.open(DATA_DIR / "uploads" / user.id / doc.storage_name) as pdf:
                if 1 <= chunk.page_start <= pdf.page_count:
                    original_text = pdf[chunk.page_start - 1].get_text("text")
        return {"document_id": doc.id, "document_name": doc.file_name, "page": chunk.page_start,
    ↳  "page_end": chunk.page_end, "section_title": chunk.section_title, "text": original_text, "highl
    ↳ ight": chunk.content}
    if doc.file_type == "PDF":
        import fitz
        with fitz.open(DATA_DIR / "uploads" / user.id / doc.storage_name) as pdf:
            page_num = max(1, min(page or 1, pdf.page_count))
            text = pdf[page_num - 1].get_text("text")
        return {"document_id": doc.id, "document_name": doc.file_name, "page": page_num, "text":
    ↳  text, "highlight": ""}
    return {"document_id": doc.id, "document_name": doc.file_name, "page": None, "text": doc.ext
    ↳ racted_text[:16000], "highlight": ""}
@router.get("/documents/{document_id}/file")
def download_document(document_id: str, db: Session = Depends(get_db), user: User = Depends(curr
    ↳ ent_user)):
    doc = db.scalar(select(Document).where(Document.id == document_id, Document.owner_id == user
    ↳ .id))
    if not doc:
        raise HTTPException(404, "文档不存在")
    return FileResponse(DATA_DIR / "uploads" / user.id / doc.storage_name, filename=doc.file_nam
    ↳ e)
// 文件: backend/app/api/knowledge_bases.py
import shutil
```

```
// 文件：backend/app/api/knowledge_bases.py
import uuid
from pathlib import Path
from fastapi import APIRouter, Depends, HTTPException
from sqlalchemy import func, select
from sqlalchemy.orm import Session
from app.core.config import DATA_DIR
from app.core.db import get_db
from app.core.security import current_user
from app.models import Document, KnowledgeBase, User
from app.rag.vector_store import delete_knowledge_base as delete_knowledge_base_vectors
from app.schemas import KnowledgeBaseInput
router = APIRouter(prefix="/knowledge-bases", tags=["knowledge-bases"])
@router.get("")
def list_knowledge_bases(db: Session = Depends(get_db), user: User = Depends(current_user)):
    rows = db.execute(select(KnowledgeBase, func.count(Document.id)).outerjoin(Document).where(K
    ↳ nowledgeBase.owner_id == user.id).group_by(KnowledgeBase.id).order_by(KnowledgeBase.created_at.d
    ↳ esc())).all()
    return [{"id": kb.id, "name": kb.name, "course_name": kb.course_name, "document_count": coun
    ↳ t, "created_at": kb.created_at} for kb, count in rows]
@router.get("/{knowledge_base_id}")
def get_knowledge_base(knowledge_base_id: str, db: Session = Depends(get_db), user: User = Depen
    ↳ ds(current_user)):
    result = db.execute(select(KnowledgeBase, func.count(Document.id)).outerjoin(Document).where
    ↳ (KnowledgeBase.id == knowledge_base_id, KnowledgeBase.owner_id == user.id).group_by(KnowledgeBas
    ↳ e.id)).first()
    if not result:
        raise HTTPException(404, "知识库不存在或无权访问")
    kb, count = result
    return {"id": kb.id, "name": kb.name, "course_name": kb.course_name, "document_count": count
    ↳ , "created_at": kb.created_at}
@router.post("", status_code=201)
def create_knowledge_base(payload: KnowledgeBaseInput, db: Session = Depends(get_db), user: User
    ↳  = Depends(current_user)):
    kb = KnowledgeBase(id=str(uuid.uuid4()), owner_id=user.id, name=payload.name.strip(), course
    ↳ _name=payload.course_name.strip())
    db.add(kb)
    db.commit()
    return {"id": kb.id, "name": kb.name, "course_name": kb.course_name, "document_count": 0, "c
    ↳ reated_at": kb.created_at}
@router.patch("/{knowledge_base_id}")
def update_knowledge_base(knowledge_base_id: str, payload: KnowledgeBaseInput, db: Session = Dep
    ↳ ends(get_db), user: User = Depends(current_user)):
    kb = db.scalar(select(KnowledgeBase).where(KnowledgeBase.id == knowledge_base_id, KnowledgeB
    ↳ ase.owner_id == user.id))
    if not kb:
        raise HTTPException(404, "知识库不存在")
    kb.name, kb.course_name = payload.name.strip(), payload.course_name.strip()
    db.commit()
    return {"id": kb.id, "name": kb.name, "course_name": kb.course_name}
@router.delete("/{knowledge_base_id}", status_code=204)
def delete_knowledge_base(knowledge_base_id: str, db: Session = Depends(get_db), user: User = De
```

```
// 文件: backend/app/api/knowledge_bases.py
    ↳ pends(current_user)):
    kb = db.scalar(select(KnowledgeBase).where(KnowledgeBase.id == knowledge_base_id, KnowledgeB
    ↳ ase.owner_id == user.id))
    if not kb:
        raise HTTPException(404, "知识库不存在")
    docs = list(db.scalars(select(Document).where(Document.knowledge_base_id == kb.id, Document.
    ↳ owner_id == user.id)).all())
    try:
        delete_knowledge_base_vectors(kb.id, user.id)
    except Exception:
        raise HTTPException(503, "向量索引暂不可用，知识库未删除；请恢复服务后重试") from None
    for doc in docs:
        (DATA_DIR / "uploads" / user.id / doc.storage_name).unlink(missing_ok=True)
    db.delete(kb)
    db.commit()
// 文件: backend/app/api/learning.py
from datetime import datetime, timedelta, timezone
import json
from pathlib import Path
from fastapi import APIRouter, Depends, HTTPException
from sqlalchemy import select, func, or_
from sqlalchemy.exc import IntegrityError
from sqlalchemy.orm import Session
from app.core.db import get_db
from app.core.security import current_user
from app.models import (
    KnowledgeEdge, KnowledgeNode, LearningCourse, QuizAttempt, QuizQuestion,
    CourseProgress, KnowledgeMastery, LearningPathPreference, StudySession, StudyTaskRecord, Use
    ↳ r, UserMistake, UserNotification, UserProfile, utcnow,
)
from app.schemas import MistakeListInput, ProfileInput, QuizAttemptInput, StudyPlanInput
from app.curriculum_defaults import DEFAULT_CURRICULA
router = APIRouter(tags=["learning"])
def _sync_course_progress(db: Session, user_id: str, course_id: str, last_node_id: str = "") ->
    ↳ int:
    """Keep the course summary aligned with mastered graph concepts."""
    node_ids = db.scalars(
        select(KnowledgeNode.id).where(
            KnowledgeNode.course_id == course_id,
            KnowledgeNode.type.in_(["concept", "application"]),
            or_(KnowledgeNode.owner_id.is_(None), KnowledgeNode.owner_id == user_id),
        )
    ).all()
    mastery_by_node = {
        row.node_id: row.mastery
        for row in db.scalars(
            select(KnowledgeMastery).where(
                KnowledgeMastery.user_id == user_id,
                KnowledgeMastery.node_id.in_(node_ids),
            )
        ).all()
    } if node_ids else {}
```

```
// 文件：backend/app/api/learning.py
    progress = round(sum(mastery_by_node.get(node_id, 0) >= 80 for node_id in node_ids) / len(no
    ↳ de_ids) * 100) if node_ids else 0
    row = db.scalar(select(CourseProgress).where(CourseProgress.user_id == user_id, CourseProgre
    ↳ ss.course_id == course_id))
    if row is None:
        row = CourseProgress(user_id=user_id, course_id=course_id)
        db.add(row)
    row.progress = progress
    if last_node_id:
        row.last_node_id = last_node_id
    return progress
def ensure_default_question_for_node(db: Session, node: KnowledgeNode) -> None:
    """Guarantee at least one selectable, answerable practice question per concept."""
    if node.owner_id is not None or node.type not in {"concept", "application"} or db.scalar(
        select(QuizQuestion.id).where(QuizQuestion.concept_id == node.id)
    ):
        return
    description = (node.description or f"{node.name}的基本定义与作用。").strip()
    distractors = [
        value.strip() for value in db.scalars(select(KnowledgeNode.description).where(
            KnowledgeNode.course_id == node.course_id,
            KnowledgeNode.owner_id.is_(None),
            KnowledgeNode.id != node.id,
            KnowledgeNode.type.in_(["concept", "application"]),
        )).all() if value and value.strip() and value.strip() != description
    ]
    options = [description]
    for item in distractors:
        if item not in options:
            options.append(item)
        if len(options) == 4:
            break
    while len(options) < 4:
        options.append(f"该知识点与{node.name}无关，不能作为其定义。")
    course = db.get(LearningCourse, node.course_id)
    db.add(QuizQuestion(
        course_id=node.course_id, type="单选题", difficulty="基础", topic=node.name,
        concept_id=node.id, title=f"关于“{node.name}”，下列哪项表述正确？",
        options=options, answer=description, explanation=description,
        evidence={"title": f"{course.name if course else '课程'}知识图谱", "section": node.name,
    ↳  "excerpt": description},
    ))
def _profile(db: Session, user: User) -> UserProfile:
    profile = db.get(UserProfile, user.id)
    if profile is None:
        profile = UserProfile(user_id=user.id)
        db.add(profile)
        try:
            db.commit()
        except IntegrityError:
```

```
// 文件：backend/app/api/learning.py
            db.rollback()
            profile = db.get(UserProfile, user.id)
        db.refresh(profile)
    return profile
def _profile_dict(profile: UserProfile) -> dict:
    return {
        "name": profile.name, "school": profile.school, "major": profile.major,
        "grade": profile.grade, "bio": profile.bio,
        "weeklyGoalHours": profile.weekly_goal_hours,
        "dailyFocusHours": profile.daily_focus_hours,
        "learningReminders": profile.learning_reminders,
        "favoriteCourseIds": profile.favorite_course_ids or [],
    }
@router.get("/profile")
def get_profile(db: Session = Depends(get_db), user: User = Depends(current_user)):
    return _profile_dict(_profile(db, user))
@router.patch("/profile")
def update_profile(payload: ProfileInput, db: Session = Depends(get_db), user: User = Depends(cu
    ↳ rrent_user)):
    profile = _profile(db, user)
    mapping = {
        "name": "name", "school": "school", "major": "major", "grade": "grade", "bio": "bio",
        "weeklyGoalHours": "weekly_goal_hours", "dailyFocusHours": "daily_focus_hours",
        "learningReminders": "learning_reminders",
        "favoriteCourseIds": "favorite_course_ids",
    }
    for key, value in payload.model_dump(exclude_unset=True, by_alias=False).items():
        if key == "favorite_course_ids":
            valid_ids = set(db.scalars(select(LearningCourse.id).where(LearningCourse.id.in_(val
    ↳ ue))).all()) if value else set()
            if valid_ids != set(value):
                raise HTTPException(422, "收藏列表包含不存在的课程")
        setattr(profile, mapping.get(key, key), value.strip() if isinstance(value, str) else val
    ↳ ue)
    db.commit()
    db.refresh(profile)
    return _profile_dict(profile)
@router.get("/courses")
def list_courses(db: Session = Depends(get_db), user: User = Depends(current_user)):
    courses = db.scalars(select(LearningCourse).order_by(LearningCourse.name)).all()
    progress_rows = db.scalars(select(CourseProgress).where(CourseProgress.user_id == user.id)).
    ↳ all()
    progress = {row.course_id: row.progress for row in progress_rows}
    return [{
        "id": course.id, "name": course.name, "category": course.category,
        "progress": progress.get(course.id, 0),
        "points": course.points, "chapter": course.chapter, "color": course.color, "icon": cours
    ↳ e.icon,
    } for course in courses]
@router.get("/courses/{course_id}/graph")
```

```
// 文件：backend/app/api/learning.py
def get_course_graph(course_id: str, db: Session = Depends(get_db), user: User = Depends(current
    ↳ _user)):
    if db.get(LearningCourse, course_id) is None:
        raise HTTPException(404, "课程不存在")
    nodes = db.scalars(select(KnowledgeNode).where(KnowledgeNode.course_id == course_id,
        or_(KnowledgeNode.owner_id.is_(None), KnowledgeNode.owner_id == user.id)).order_by(Knowl
    ↳ edgeNode.id)).all()
    edges = db.scalars(select(KnowledgeEdge).where(KnowledgeEdge.course_id == course_id,
        or_(KnowledgeEdge.owner_id.is_(None), KnowledgeEdge.owner_id == user.id))).all()
    mastery = {row.node_id: row.mastery for row in db.scalars(select(KnowledgeMastery).where(Kno
    ↳ wledgeMastery.user_id == user.id, KnowledgeMastery.node_id.in_([node.id for node in nodes]))).al
    ↳ l()} if nodes else {}
    return {
        "nodes": [{"id": node.id, "name": node.name, "type": node.type, "mastery": mastery.get(n
    ↳ ode.id, 0), "description": node.description} for node in nodes],
        "links": [{"source": edge.source, "target": edge.target} for edge in edges],
    }
@router.get("/quizzes")
def list_quizzes(course_id: str | None = None, db: Session = Depends(get_db), user: User = Depen
    ↳ ds(current_user)):
    query = select(QuizQuestion).where((QuizQuestion.owner_id.is_(None)) | (QuizQuestion.owner_i
    ↳ d == user.id)).order_by(QuizQuestion.id)
    if course_id:
        query = query.where(QuizQuestion.course_id == course_id)
    questions = db.scalars(query).all()
    return [{
        "id": question.id, "type": question.type, "difficulty": question.difficulty,
        "topic": question.topic, "course": db.get(LearningCourse, question.course_id).name,
        "courseId": question.course_id, "conceptId": question.concept_id,
        "title": question.title, "options": question.options,
        "transferQuestionIds": question.transfer_question_ids, "evidence": question.evidence,
    } for question in questions]
@router.post("/quiz-attempts", status_code=201)
def create_quiz_attempt(payload: QuizAttemptInput, db: Session = Depends(get_db), user: User = D
    ↳ epends(current_user)):
    if len(payload.question_ids) != len(set(payload.question_ids)):
        raise HTTPException(422, "题目列表不能包含重复项")
    questions = db.scalars(select(QuizQuestion).where(
        QuizQuestion.id.in_(payload.question_ids),
        (QuizQuestion.owner_id.is_(None)) | (QuizQuestion.owner_id == user.id),
    )).all()
    by_id = {question.id: question for question in questions}
    if len(by_id) != len(set(payload.question_ids)):
        raise HTTPException(422, "题目不存在或已下架")
    objective = [by_id[qid] for qid in payload.question_ids if by_id[qid].type != "简答题"]
    correct = 0
    results = []
    for question in objective:
        selected = payload.answers.get(str(question.id))
        is_correct = (isinstance(selected, list) and sorted(selected) == sorted(question.answer)
```

```
// 文件：backend/app/api/learning.py
    ) if isinstance(question.answer, list) else selected == question.answer
        if is_correct:
            correct += 1
        results.append({"questionId": question.id, "correct": is_correct, "selectedAnswer": sele
    cted,
                        "answer": question.answer, "explanation": question.explanation,
                        "misconception": question.misconceptions.get(str(selected), "") if isins
    tance(question.misconceptions, dict) and isinstance(selected, str) else "",
                        "conceptId": question.concept_id, "courseId": question.course_id})
        if question.concept_id and db.get(KnowledgeNode, question.concept_id):
            mastery_row = db.scalar(select(KnowledgeMastery).where(KnowledgeMastery.user_id == u
    ser.id, KnowledgeMastery.node_id == question.concept_id))
            if mastery_row is None:
                mastery_row = KnowledgeMastery(user_id=user.id, node_id=question.concept_id, mas
    tery=0)
                db.add(mastery_row)
            mastery_row.mastery = min(100, mastery_row.mastery + 15) if is_correct else max(0, m
    astery_row.mastery - 20)
    score = round(correct / len(objective) * 100) if objective else 0
    course_id = payload.course_id or (by_id[payload.question_ids[0]].course_id if payload.questi
    on_ids else "")
    if any(question.course_id != course_id for question in by_id.values()):
        raise HTTPException(422, "一次测验只能提交同一课程的题目")
    last_node_id = next((by_id[qid].concept_id for qid in reversed(payload.question_ids)
                         if by_id[qid].concept_id and db.get(KnowledgeNode, by_id[qid].concept_i
    d)), "")
    if course_id:
        _sync_course_progress(db, user.id, course_id, last_node_id)
    attempt = QuizAttempt(user_id=user.id, course_id=course_id, score=score,
                          question_count=len(objective), answers=payload.answers, results=result
    s,
                          duration_seconds=payload.duration_seconds, at=utcnow())
    db.add(attempt)
    profile = db.get(UserProfile, user.id)
    if profile is not None and profile.learning_reminders:
        db.add(UserNotification(user_id=user.id, title="测验已完成", body=f"本次得分 {score} 分
    ，共完成 {len(objective)} 道客观题。"))
    db.commit()
    db.refresh(attempt)
    return {"id": attempt.id, "at": attempt.at.isoformat(), "score": attempt.score,
            "questionCount": attempt.question_count, "courseId": attempt.course_id,
            "durationSeconds": attempt.duration_seconds, "results": results}
@router.get("/quiz-attempts")
def list_quiz_attempts(db: Session = Depends(get_db), user: User = Depends(current_user)):
    attempts = db.scalars(select(QuizAttempt).where(QuizAttempt.user_id == user.id).order_by(Qui
    zAttempt.at.desc())).all()
    courses = {course.id: course.name for course in db.scalars(select(LearningCourse)).all()}
    return [{"id": item.id, "at": item.at.isoformat(), "score": item.score, "questionCount": ite
    m.question_count,
             "courseId": item.course_id, "course": courses.get(item.course_id, item.course_id)}
```

```
// 文件：backend/app/api/learning.py
    ↳ for item in attempts]
@router.get("/quiz-attempts/{attempt_id}")
def get_quiz_attempt(attempt_id: int, db: Session = Depends(get_db), user: User = Depends(curren
    ↳ t_user)):
    attempt = db.scalar(select(QuizAttempt).where(QuizAttempt.id == attempt_id, QuizAttempt.user
    ↳ _id == user.id))
    if not attempt:
        raise HTTPException(404, "测验记录不存在")
    return {"id": attempt.id, "at": attempt.at.isoformat(), "score": attempt.score,
            "questionCount": attempt.question_count, "courseId": attempt.course_id,
            "durationSeconds": attempt.duration_seconds, "results": attempt.results or []}
@router.get("/mistakes")
def list_mistakes(db: Session = Depends(get_db), user: User = Depends(current_user)):
    rows = db.scalars(select(UserMistake).where(UserMistake.user_id == user.id).order_by(UserMis
    ↳ take.updated_at.desc())).all()
    return [row.payload for row in rows]
@router.put("/mistakes")
def replace_mistakes(payload: MistakeListInput, db: Session = Depends(get_db), user: User = Depe
    ↳ nds(current_user)):
    ids = {item.question_id for item in payload.items}
    if len(ids) != len(payload.items):
        raise HTTPException(422, "错题列表不能包含重复题目")
    if ids:
        unknown = db.scalars(select(QuizQuestion.id).where(QuizQuestion.id.in_(ids),
            or_(QuizQuestion.owner_id.is_(None), QuizQuestion.owner_id == user.id))).all()
        if set(unknown) != ids:
            raise HTTPException(422, "错题包含不存在的题目")
    existing = {row.question_id: row for row in db.scalars(select(UserMistake).where(UserMistake
    ↳ .user_id == user.id)).all()}
    for question_id, row in existing.items():
        if question_id not in ids:
            db.delete(row)
    for item in payload.items:
        row = existing.get(item.question_id)
        if row:
            row.payload = item.payload
            row.updated_at = utcnow()
        else:
            db.add(UserMistake(user_id=user.id, question_id=item.question_id, payload=item.paylo
    ↳ ad))
    db.commit()
    return [item.payload for item in payload.items]
def _task_dict(task: StudyTaskRecord) -> dict:
    return {"id": task.id, "title": task.title, "course": task.course, "date": task.date,
            "priority": task.priority, "done": task.done}
@router.get("/study-plans")
def list_study_tasks(db: Session = Depends(get_db), user: User = Depends(current_user)):
    tasks = db.scalars(select(StudyTaskRecord).where(StudyTaskRecord.user_id == user.id).order_b
    ↳ y(StudyTaskRecord.id)).all()
    return [_task_dict(task) for task in tasks]
```

```
// 文件：backend/app/api/learning.py
@router.put("/study-plans")
def save_study_tasks(payload: StudyPlanInput, db: Session = Depends(get_db), user: User = Depend
    ↳ s(current_user)):
    provided_ids = [task.id for task in payload.tasks if task.id is not None]
    if len(provided_ids) != len(set(provided_ids)):
        raise HTTPException(422, "学习计划不能包含重复任务")
    existing = {task.id: task for task in db.scalars(select(StudyTaskRecord).where(StudyTaskReco
    ↳ rd.user_id == user.id)).all()}
    retained: set[int] = set()
    saved: list[StudyTaskRecord] = []
    for item in payload.tasks:
        values = item.model_dump(exclude={"id"})
        row = existing.get(item.id) if item.id is not None else None
        if row:
            for key, value in values.items():
                setattr(row, key, value)
            retained.add(row.id)
        else:
            row = StudyTaskRecord(user_id=user.id, **values)
            db.add(row)
        saved.append(row)
    for task_id, row in existing.items():
        if task_id not in retained:
            db.delete(row)
    db.flush()
    result = [_task_dict(task) for task in saved]
    db.commit()
    return result
@router.get("/dashboard")
def learning_dashboard(db: Session = Depends(get_db), user: User = Depends(current_user)):
    now = datetime.now(timezone.utc)
    start = (now.date() - timedelta(days=6))
    attempts = db.scalars(select(QuizAttempt).where(QuizAttempt.user_id == user.id, QuizAttempt.
    ↳ at >= datetime.combine(start, datetime.min.time(), timezone.utc)).order_by(QuizAttempt.at)).all(
    ↳ )
    all_attempts = db.scalars(select(QuizAttempt).where(QuizAttempt.user_id == user.id).order_by
    ↳ (QuizAttempt.at.desc())).all()
    tasks = db.scalars(select(StudyTaskRecord).where(StudyTaskRecord.user_id == user.id)).all()
    mistakes = db.scalars(select(UserMistake).where(UserMistake.user_id == user.id)).all()
    sessions = db.scalars(select(StudySession).where(StudySession.user_id == user.id, StudySessi
    ↳ on.started_at >= datetime.combine(start, datetime.min.time(), timezone.utc))).all()
    day_scores: dict[str, list[int]] = {}
    active_days = {item.at.date() for item in all_attempts}
    for attempt in attempts:
        day_scores.setdefault(attempt.at.date().isoformat(), []).append(attempt.score)
    week_scores = [round(sum(day_scores[d.isoformat()]) / len(day_scores[d.isoformat()])) if d.i
    ↳ soformat() in day_scores else None
                   for d in (start + timedelta(days=i) for i in range(7))]
    cursor = now.date()
    if cursor not in active_days:
```

```
// 文件：backend/app/api/learning.py
        cursor -= timedelta(days=1)
    streak = 0
    while cursor in active_days:
        streak += 1
        cursor -= timedelta(days=1)
    course_list = list_courses(db, user)
    week_seconds = [0] * 7
    today_seconds = 0
    for session in sessions:
        started_at = session.started_at if session.started_at.tzinfo else session.started_at.rep
    ↳ lace(tzinfo=timezone.utc)
        seconds = session.duration_seconds if session.state == "completed" else max(0, min(12 *
    ↳ 3600, int((now - started_at).total_seconds())))
        index = (session.started_at.date() - start).days
        if 0 <= index < 7:
            week_seconds[index] += seconds
        if session.started_at.date() == now.date():
            today_seconds += seconds
    mastered = db.scalar(select(func.count(KnowledgeMastery.id)).where(KnowledgeMastery.user_id
    ↳ == user.id, KnowledgeMastery.mastery >= 80)) or 0
    return {"hoursToday": round(today_seconds / 3600, 1), "mastered": mastered,
            "weeklyTasks": sum(item.done for item in tasks), "streak": streak, "weekHours": [rou
    ↳ nd(seconds / 3600, 1) for seconds in week_seconds],
            "weekScores": week_scores, "courses": course_list, "tasks": [_task_dict(item) for it
    ↳ em in tasks]}
def seed_learning_catalog(db: Session) -> None:
    catalog_path = Path(__file__).resolve().parents[1] / "learning_catalog.json"
    catalog = json.loads(catalog_path.read_text(encoding="utf-8"))
    for item in catalog["courses"]:
        if db.get(LearningCourse, item["id"]) is None:
            db.add(LearningCourse(id=item["id"], name=item["name"], category=item["category"], p
    ↳ oints=item["points"],
                                  chapter=item["chapter"], color=item["color"], icon=item["icon"
    ↳ ]))
    db.flush()
    for item in catalog["graphNodes"]:
        if db.get(KnowledgeNode, item["id"]) is None:
            db.add(KnowledgeNode(id=item["id"], course_id="ds", name=item["name"], type=item["ty
    ↳ pe"], description=item["description"]))
    for course_id, items in DEFAULT_CURRICULA.items():
        if db.get(KnowledgeNode, course_id) is None:
            course = db.get(LearningCourse, course_id)
            if course:
                db.add(KnowledgeNode(id=course_id, course_id=course_id, name=course.name, type="
    ↳ course", description=f"{course.name}课程的核心概念与先修关系。"))
        for node_id, name, node_type, description, _parent in items:
            if db.get(KnowledgeNode, node_id) is None:
                db.add(KnowledgeNode(id=node_id, course_id=course_id, name=name, type=node_type,
    ↳  description=description))
    db.flush()
```

```
// 文件: backend/app/api/learning.py
    existing_edges = {(edge.course_id, edge.source, edge.target) for edge in db.scalars(select(K
    ↳ nowledgeEdge)).all()}
    for item in catalog["graphLinks"]:
        key = ("ds", item["source"], item["target"])
        if key not in existing_edges:
            db.add(KnowledgeEdge(course_id=key[0], source=key[1], target=key[2]))
            existing_edges.add(key)
    for course_id, items in DEFAULT_CURRICULA.items():
        for node_id, _name, _node_type, _description, parent_id in items:
            key = (course_id, parent_id, node_id)
            if key not in existing_edges and db.get(KnowledgeNode, parent_id) and db.get(Knowled
    ↳ geNode, node_id):
                db.add(KnowledgeEdge(course_id=course_id, source=parent_id, target=node_id))
                existing_edges.add(key)
    db.flush()
    for item in catalog["questions"]:
        if db.get(QuizQuestion, item["id"]) is None:
            db.add(QuizQuestion(id=item["id"], course_id=next((c["id"] for c in catalog["courses
    ↳ "] if c["name"] == item.get("course")), "ds"),
                                type=item["type"], difficulty=item["difficulty"], topic=item["to
    ↳ pic"], concept_id=item.get("conceptId", ""),
                                title=item["title"], options=item.get("options", []), answer=ite
    ↳ m.get("answer", ""),
                                explanation=item["explanation"], misconceptions=item.get("miscon
    ↳ ceptions", {}),
                                transfer_question_ids=item.get("transferQuestionIds", []), evide
    ↳ nce=item.get("evidence")))
    db.flush()
    for node in db.scalars(select(KnowledgeNode).where(KnowledgeNode.type.in_(["concept", "appli
    ↳ cation"]), KnowledgeNode.owner_id.is_(None))).all():
        ensure_default_question_for_node(db, node)
    db.commit()
// 文件: backend/app/api/learning_extended.py
import json
import logging
import re
import uuid
from difflib import SequenceMatcher
from datetime import datetime, timedelta, timezone
from urllib.parse import urlparse
from fastapi import APIRouter, Depends, HTTPException
from openai import APIConnectionError
from pydantic import BaseModel
from sqlalchemy import select, or_
from sqlalchemy.orm import Session
from app.core.db import get_db
from app.core.security import current_user
from app.models import (CourseProgress, KnowledgeBase, KnowledgeEdge, KnowledgeMastery, Knowledg
    ↳ eNode, LearningCourse,
                        LearningPathPreference, MistakeVariantSet, QuizQuestion, StudySession, U
    ↳ ser, UserCourseBinding, UserMistake, UserNotification, UserProfile, utcnow)
```

```
// 文件：backend/app/api/learning_extended.py
from app.schemas import CourseBindingInput, CourseProgressInput, MasteryInput, MistakeReviewInpu
    ↳ t, StudySessionStartInput
from app.rag.pipeline import _client
from app.core.config import get_settings
router = APIRouter(prefix="/api", tags=["learning-activity"])
logger = logging.getLogger(__name__)
class VariantSubmitInput(BaseModel):
    answers: dict[str, str]
@router.post("/mistakes/{question_id}/variants")
async def generate_mistake_variants(question_id: int, db: Session = Depends(get_db), user: User
    ↳ = Depends(current_user)):
    mistake = db.scalar(select(UserMistake).where(UserMistake.user_id == user.id, UserMistake.qu
    ↳ estion_id == question_id))
    question = db.get(QuizQuestion, question_id)
    if mistake is None or question is None or (question.owner_id is not None and question.owner_
    ↳ id != user.id):
        raise HTTPException(404, "错题不存在")
    node = db.get(KnowledgeNode, question.concept_id) if question.concept_id else None
    course = db.get(LearningCourse, question.course_id)
    concept = node.name if node else question.topic or "该题考查的知识点"
    concept_description = node.description if node else question.explanation
    evidence = question.evidence or {}
    source_title = question.title.strip()
    source_normalized = re.sub(r"[\W_]+", "", source_title).lower()
    source_options = sorted(re.sub(r"[\W_]+", "", str(option)).lower() for option in (question.o
    ↳ ptions or []))
    source_answer = re.sub(r"[\W_]+", "", str(question.answer)).lower()
    prompt_data = {
        "course": course.name if course else "",
        "concept": concept,
        "concept_description": concept_description,
        "source_question": source_title,
        "source_options": question.options or [],
        "source_answer": question.answer,
        "source_explanation": question.explanation,
        "evidence": evidence,
        "difficulty": question.difficulty,
    }
    system_prompt = (
        "你是大学课程的命题教师。请围绕给定考点设计恰好 3 道全新的单选题，用于举一反三。"
        "三题必须考查同一核心知识，但分别采用不同情境、条件或推理路径；不能照抄、换词复述或仅替
    ↳ 换数字后复现原题。"
        "每题提供 4 个互不重复的选项，answer 必须与其中一个选项逐字相同，explanation 简洁说明依
    ↳ 据。"
        "只使用给定的课程、知识点和解析信息，不补充无关事实。若信息不足以可靠命题，返回 {\"quest
    ↳ ions\": []}。"
        "只返回 JSON：{\"questions\":[{\"title\":\"...\",\"options\":[\"...\",\"...\",\"...\",\"
    ↳ ...\"],\"answer\":\"...\",\"explanation\":\"...\"}]}"
    )
    try:
```

```
// 文件：backend/app/api/learning_extended.py
        settings = get_settings()
        generation_options = {}
        llm_host = (urlparse(settings.llm_base_url).hostname or "").lower()
        if llm_host.endswith("aliyuncs.com") and settings.llm_model.lower().startswith("deepseek
    ↳ -v4"):
            # DeepSeek V4 enables lengthy chain-of-thought by default. These
            # questions are simple transfer exercises, so return the requested
            # JSON directly and keep the response bounded.
            generation_options = {
                "extra_body": {"enable_thinking": False},
                "max_tokens": 2048,
            }
        response = await _client().chat.completions.create(
            model=get_settings().llm_model,
            messages=[
                {"role": "system", "content": system_prompt},
                {"role": "user", "content": json.dumps(prompt_data, ensure_ascii=False)},
            ],
            temperature=0.7,
            response_format={"type": "json_object"},
            **generation_options,
        )
        content = (response.choices[0].message.content or "").strip()
        content = re.sub(r"^```(?:json)?\s*|\s*```$", "", content, flags=re.IGNORECASE)
        generated = json.loads(content).get("questions", [])
    except Exception as error:
        if isinstance(error, HTTPException):
            raise
        if isinstance(error, APIConnectionError):
            logger.warning("LLM connection failed while generating variants for question %s", qu
    ↳ estion_id, exc_info=True)
            raise HTTPException(502, "大模型服务连接不稳定，系统已自动重试一次，请稍后点击重新生
    ↳ 成") from None
        logger.exception("Failed to generate variants for question %s", question_id)
        raise HTTPException(502, f"变式题生成失败：{type(error).__name__}") from None
    if not isinstance(generated, list) or len(generated) != 3:
        raise HTTPException(502, "暂时无法生成 3 道可靠的变式题，请稍后重试")
    variants = []
    seen_titles = []
    for item in generated:
        if not isinstance(item, dict):
            raise HTTPException(502, "变式题格式无效，请稍后重试")
        title = str(item.get("title", "")).strip()
        title_normalized = re.sub(r"[\W_]+", "", title).lower()
        options = item.get("options")
        answer = item.get("answer")
        explanation = str(item.get("explanation", "")).strip()
        normalized_options = sorted(re.sub(r"[\W_]+", "", option).lower() for option in options)
    ↳  if isinstance(options, list) and all(isinstance(option, str) for option in options) else []
        repeats_source_content = title_normalized == source_normalized and normalized_options ==
```

```
// 文件：backend/app/api/learning_extended.py
    ↳  source_options and re.sub(r"[\W_]+", "", str(answer)).lower() == source_answer
        if (not title or not title_normalized or title_normalized == source_normalized
                or SequenceMatcher(None, title_normalized, source_normalized).ratio() >= 0.9
                or any(SequenceMatcher(None, title_normalized, previous).ratio() >= 0.9 for prev
    ↳ ious in seen_titles)
                or repeats_source_content
                or not isinstance(options, list) or len(options) != 4
                or any(not isinstance(option, str) or not option.strip() for option in options)
                or len(set(options)) != 4 or answer not in options or not explanation):
            raise HTTPException(502, "生成题目与原题重复或格式不完整，请重新生成")
        seen_titles.append(title_normalized)
        variants.append({"id": len(variants) + 1, "title": title, "options": options, "answer":
    ↳ answer, "explanation": explanation})
    now = utcnow()
    db.query(MistakeVariantSet).filter(MistakeVariantSet.expires_at < now).delete(synchronize_se
    ↳ ssion=False)
    variant_set = MistakeVariantSet(id=str(uuid.uuid4()), user_id=user.id, source_question_id=qu
    ↳ estion_id,
                                    variants=variants, expires_at=now + timedelta(minutes=30), c
    ↳ reated_at=now)
    db.add(variant_set)
    db.commit()
    return {"setId": variant_set.id, "course": course.name if course else "课程练习", "topic": c
    ↳ oncept, "sourceTitle": source_title,
            "conceptDescription": concept_description, "difficulty": question.difficulty,
            "variants": [{key: value for key, value in item.items() if key not in {"answer", "ex
    ↳ planation"}} for item in variants]}
def _record_mistake_review(row: UserMistake, correct: bool, selected_answer: str, note: str) ->
    ↳ dict:
    data = dict(row.payload)
    now = utcnow()
    attempts = list(data.get("attempts", []))
    attempts.append({"at": now.isoformat(), "correct": correct, "selectedAnswer": selected_answe
    ↳ r,
                     "feedback": note or ("复习回答正确" if correct else "复习回答错误")})
    data["attempts"] = attempts[-100:]
    if correct:
        data["correctStreak"] = int(data.get("correctStreak", 0)) + 1
        data["lastCorrectAt"] = now.isoformat()
        data["mastered"] = data["correctStreak"] >= 2
        data["nextReviewAt"] = "" if data["mastered"] else (now + timedelta(days=3)).isoformat()
    else:
        data["wrong"] = int(data.get("wrong", 0)) + 1
        data["correctStreak"] = 0
        data["mastered"] = False
        data["misconception"] = note or "本次复习未答对，建议回看关联知识点。"
        data["lastWrongAt"] = now.isoformat()
        data["nextReviewAt"] = (now + timedelta(days=1)).isoformat()
    row.payload, row.updated_at = data, now
    return data
```

```
// 文件：backend/app/api/learning_extended.py
@router.post("/mistakes/{question_id}/variants/{set_id}/submit")
def submit_mistake_variants(question_id: int, set_id: str, payload: VariantSubmitInput,
                            db: Session = Depends(get_db), user: User = Depends(current_user)):
    row = db.scalar(select(UserMistake).where(UserMistake.user_id == user.id, UserMistake.questi
    ↳ on_id == question_id))
    variant_set = db.scalar(select(MistakeVariantSet).where(
        MistakeVariantSet.id == set_id, MistakeVariantSet.user_id == user.id,
        MistakeVariantSet.source_question_id == question_id, MistakeVariantSet.expires_at > utcn
    ↳ ow(),
    ))
    if row is None or variant_set is None:
        raise HTTPException(404, "练习已失效，请重新生成")
    variants = variant_set.variants or []
    expected_ids = {str(item["id"]) for item in variants}
    if len(variants) != 3 or set(payload.answers) != expected_ids:
        raise HTTPException(422, "请提交三道题的完整答案")
    results = []
    for item in variants:
        selected = payload.answers[str(item["id"])]
        if selected not in item["options"]:
            raise HTTPException(422, "答案选项无效")
        results.append({"id": item["id"], "correct": selected == item["answer"], "answer": item[
    ↳ "answer"],
                        "explanation": item["explanation"]})
    correct_count = sum(result["correct"] for result in results)
    missed = "、".join(f"第 {index + 1} 题" for index, result in enumerate(results) if not resul
    ↳ t["correct"])
    note = f"举一反三练习答对 {correct_count}/3。{f'需要巩固：{missed}。' if missed else '三道变
    ↳ 式题全部答对。'}"
    updated_mistake = _record_mistake_review(row, correct_count == 3, f"{correct_count}/3", note
    ↳ )
    db.delete(variant_set)
    db.commit()
    return {"correctCount": correct_count, "total": 3, "results": results, "mistake": updated_mi
    ↳ stake}
@router.post("/study-sessions", status_code=201)
def start_study_session(payload: StudySessionStartInput, db: Session = Depends(get_db), user: Us
    ↳ er = Depends(current_user)):
    if payload.course_id and not db.get(LearningCourse, payload.course_id):
        raise HTTPException(404, "课程不存在")
    if db.scalar(select(StudySession.id).where(StudySession.user_id == user.id, StudySession.sta
    ↳ te == "active")):
        raise HTTPException(409, "已有进行中的学习计时，请先结束或恢复计时")
    row = StudySession(id=str(uuid.uuid4()), user_id=user.id, course_id=payload.course_id, start
    ↳ ed_at=utcnow(), state="active")
    db.add(row)
    db.commit()
    return {"id": row.id, "courseId": row.course_id, "startedAt": row.started_at.isoformat(), "s
    ↳ tate": row.state}
@router.get("/study-sessions/active")
```

```
// 文件: backend/app/api/learning_extended.py
def get_active_study_session(db: Session = Depends(get_db), user: User = Depends(current_user)):
    row = db.scalar(select(StudySession).where(StudySession.user_id == user.id, StudySession.sta
    ↳ te == "active").order_by(StudySession.started_at.desc()))
    return None if row is None else {"id": row.id, "courseId": row.course_id, "startedAt": row.s
    ↳ tarted_at.isoformat(), "state": row.state}
@router.post("/study-sessions/{session_id}/finish")
def finish_study_session(session_id: str, db: Session = Depends(get_db), user: User = Depends(cu
    ↳ rrent_user)):
    row = db.scalar(select(StudySession).where(StudySession.id == session_id, StudySession.user_
    ↳ id == user.id))
    if row is None:
        raise HTTPException(404, "学习计时不存在")
    if row.state == "active":
        row.ended_at = utcnow()
        started_at = row.started_at if row.started_at.tzinfo else row.started_at.replace(tzinfo=
    ↳ timezone.utc)
        row.duration_seconds = max(1, min(43200, int((row.ended_at - started_at).total_seconds()
    ↳ )))
        row.state = "completed"
        profile = db.get(UserProfile, user.id)
        if row.duration_seconds >= 300 and profile is not None and profile.learning_reminders:
            minutes = max(1, round(row.duration_seconds / 60))
            db.add(UserNotification(user_id=user.id, title="专注学习完成", body=f"本次专注学习 {
    ↳ minutes} 分钟，记录已保存到账户。"))
        db.commit()
    return {"id": row.id, "courseId": row.course_id, "startedAt": row.started_at.isoformat(),
            "endedAt": row.ended_at.isoformat() if row.ended_at else None, "durationSeconds": ro
    ↳ w.duration_seconds, "state": row.state}
@router.get("/analytics")
def get_learning_analytics(days: int = 7, db: Session = Depends(get_db), user: User = Depends(cu
    ↳ rrent_user)):
    days = min(90, max(1, days))
    start = datetime.now(timezone.utc).date() - timedelta(days=days - 1)
    rows = db.scalars(select(StudySession).where(StudySession.user_id == user.id, StudySession.s
    ↳ tarted_at >= datetime.combine(start, datetime.min.time(), timezone.utc))).all()
    buckets = {(start + timedelta(days=i)).isoformat(): 0 for i in range(days)}
    for row in rows:
        started_at = row.started_at if row.started_at.tzinfo else row.started_at.replace(tzinfo=
    ↳ timezone.utc)
        seconds = row.duration_seconds if row.state == "completed" else max(0, min(43200, int((u
    ↳ tcnow() - started_at).total_seconds())))
        key = row.started_at.date().isoformat()
        if key in buckets:
            buckets[key] += seconds
    return {"days": [{"date": day, "seconds": seconds, "hours": round(seconds / 3600, 2)} for da
    ↳ y, seconds in buckets.items()],
            "totalSeconds": sum(buckets.values()), "totalHours": round(sum(buckets.values()) / 3
    ↳ 600, 2)}
@router.get("/me/courses/{course_id}/progress")
def get_course_progress(course_id: str, db: Session = Depends(get_db), user: User = Depends(curr
```

```
// 文件：backend/app/api/learning_extended.py
    ↳ ent_user)):
    if not db.get(LearningCourse, course_id):
        raise HTTPException(404, "课程不存在")
    row = db.scalar(select(CourseProgress).where(CourseProgress.user_id == user.id, CourseProgre
    ↳ ss.course_id == course_id))
    return {"courseId": course_id, "progress": row.progress if row else 0, "lastNodeId": row.las
    ↳ t_node_id if row else ""}
@router.patch("/me/courses/{course_id}/progress")
def update_course_progress(course_id: str, payload: CourseProgressInput, db: Session = Depends(g
    ↳ et_db), user: User = Depends(current_user)):
    if not db.get(LearningCourse, course_id):
        raise HTTPException(404, "课程不存在")
    if payload.last_node_id and db.scalar(select(KnowledgeNode.id).where(KnowledgeNode.id == pay
    ↳ load.last_node_id, KnowledgeNode.course_id == course_id,
            or_(KnowledgeNode.owner_id.is_(None), KnowledgeNode.owner_id == user.id))) is None:
        raise HTTPException(422, "知识点不属于该课程")
    row = db.scalar(select(CourseProgress).where(CourseProgress.user_id == user.id, CourseProgre
    ↳ ss.course_id == course_id))
    if row is None:
        row = CourseProgress(user_id=user.id, course_id=course_id)
        db.add(row)
    row.progress, row.last_node_id = payload.progress, payload.last_node_id
    db.commit()
    return {"courseId": course_id, "progress": row.progress, "lastNodeId": row.last_node_id, "up
    ↳ datedAt": row.updated_at.isoformat()}
@router.get("/me/courses/{course_id}/learning-path")
def get_personalized_learning_path(course_id: str, db: Session = Depends(get_db), user: User = D
    ↳ epends(current_user)):
    """Recommend weak concepts with their graph prerequisites in prerequisite order."""
    course = db.get(LearningCourse, course_id)
    if course is None:
        raise HTTPException(404, "课程不存在")
    nodes = db.scalars(select(KnowledgeNode).where(KnowledgeNode.course_id == course_id,
        or_(KnowledgeNode.owner_id.is_(None), KnowledgeNode.owner_id == user.id))).all()
    if not nodes:
        return {"courseId": course_id, "courseName": course.name, "steps": [], "progress": 0}
    node_by_id = {node.id: node for node in nodes}
    edges = db.scalars(select(KnowledgeEdge).where(KnowledgeEdge.course_id == course_id,
        or_(KnowledgeEdge.owner_id.is_(None), KnowledgeEdge.owner_id == user.id))).all()
    prerequisites: dict[str, set[str]] = {node_id: set() for node_id in node_by_id}
    for edge in edges:
        if edge.source in node_by_id and edge.target in node_by_id:
            prerequisites[edge.target].add(edge.source)
    mastery = {
        row.node_id: row.mastery
        for row in db.scalars(select(KnowledgeMastery).where(
            KnowledgeMastery.user_id == user.id,
            KnowledgeMastery.node_id.in_(list(node_by_id)),
        )).all()
    }
```

```
// 文件：src/views/RagChatView.vue
    ↳ 请先创建或选择一个知识库';return}try{const session=await ragRequest<ChatSession>('/rag/sessions'
    ↳ ,{method:'POST',body:JSON.stringify({knowledge_base_id:answerMode.value==='library'?selectedKb.v
    ↳ alue:'',answer_mode:answerMode.value,title:'新建学习对话'})});activeSession.value=session.id;mes
    ↳ sages.value=[];await loadSessions()}catch(e){error.value=e instanceof Error?e.message:'新建对话
    ↳ 失败'}}
async function changeAnswerMode(mode:AnswerMode){if(busy.value||mode===answerMode.value)return;a
    ↳ nswerMode.value=mode;activeSession.value='';messages.value=[];error.value='';await loadSessions(
    ↳ )}
async function deleteSession(session:ChatSession){if(!window.confirm(`删除对话“${session.title}
    ↳ ”？`))return;try{await ragRequest(`/rag/sessions/${session.id}`,{method:'DELETE'});sessions.val
    ↳ ue=sessions.value.filter(item=>item.id!==session.id);if(activeSession.value===session.id){active
    ↳ Session.value='';messages.value=[]}}catch(e){error.value=e instanceof Error?e.message:'删除失败'
    ↳ }}
async function scrollDown(){await nextTick();chatBox.value?.scrollTo({top:chatBox.value.scrollHe
    ↳ ight,behavior:'smooth'})}
function readSseBlock(block:string,onEvent:(name:string,data:any)=>void){let eventName='message'
    ↳ ;const data:string[]=[];for(const line of block.split(/\r?\n/)){if(line.startsWith('event:'))eve
    ↳ ntName=line.slice(6).trim();else if(line.startsWith('data:'))data.push(line.slice(5).trim())}if(
    ↳ !data.length)return;const value=data.join('\n');try{onEvent(eventName,JSON.parse(value))}catch{o
    ↳ nEvent(eventName,value)}}
async function sendQuestion(raw=input.value, retryIndex?:number){const question=raw.trim();if(!q
    ↳ uestion||busy.value)return;error.value='';if(!activeSession.value)await createSession();if(!acti
    ↳ veSession.value)return
  let assistantIndex:number
  if(retryIndex!==undefined){assistantIndex=retryIndex;messages.value[assistantIndex]={role:'ass
    ↳ istant',content:'',citations:[],loading:true}}
  else{messages.value.push({role:'user',content:question});messages.value.push({role:'assistant'
    ↳ ,content:'',citations:[],loading:true});assistantIndex=messages.value.length-1;input.value=''}
  busy.value=true;aborter=new AbortController();await scrollDown()
  try{
    const token=authState.token;const response=await fetch(ragPath('/rag/chat/stream'),{method:'
    ↳ POST',headers:{'Content-Type':'application/json',Authorization:`Bearer ${token}`},body:JSON.stri
    ↳ ngify({session_id:activeSession.value,question}),signal:aborter.signal})
    if(response.status===401){logout();throw new Error('登录状态已失效，请重新登录')}
    if(!response.ok){let detail=`请求失败 (${response.status})`;try{detail=(await response.json(
    ↳ )).detail||detail}catch{};throw new Error(detail)}
    if(!response.body)throw new Error('浏览器不支持流式响应')
    const reader=response.body.getReader();const decoder=new TextDecoder();let buffer='';let com
    ↳ pleted=false
    while(true){const {value,done}=await reader.read();if(done)break;buffer+=decoder.decode(valu
    ↳ e,{stream:true});let boundary:number;while((boundary=buffer.indexOf('\n\n'))>=0){const block=buf
    ↳ fer.slice(0,boundary);buffer=buffer.slice(boundary+2);readSseBlock(block,(name,data)=>{const tar
    ↳ get=messages.value[assistantIndex];if(!target)return;if(name==='citations')target.citations=data
    ↳ ;if(name==='delta')target.content+=data;if(name==='done'){target.id=data.message_id;target.conte
    ↳ nt=typeof data.answer==='string'?data.answer:'';target.loading=false;target.error='';completed=t
    ↳ rue}if(name==='error'){const message=typeof data.message==='string'&&data.message?data.message:'
    ↳ 回答生成失败';target.loading=false;target.error=message;error.value=message}});void scrollDown()
    ↳ }}
    if(!completed&&!messages.value[assistantIndex].error)messages.value[assistantIndex].error='
    ↳ 连接中断，服务端未保存本轮回答。可重试。'
```

```
// 文件：src/views/RagChatView.vue
    await loadSessions()
  }catch(e){const target=messages.value[assistantIndex];target.loading=false;if((e as Error).nam
    ↳ e==='AbortError')target.error='已中断生成';else{target.error=e instanceof Error?e.message:'问答
    ↳ 请求失败';error.value=target.error}}
  finally{busy.value=false;aborter=undefined;await scrollDown()}}
function stopGeneration(){aborter?.abort()}
function retry(messageIndex:number){const question=messages.value[messageIndex-1]?.content;if(qu
    ↳ estion)void sendQuestion(question,messageIndex)}
async function rateAnswer(message:ChatMessage,rating:'helpful'|'unhelpful'){if(!message.id)retur
    ↳ n;try{await ragRequest(`/rag/messages/${message.id}/feedback`,{method:'PUT',body:JSON.stringify(
    ↳ {rating})});message.feedback=rating}catch(e){error.value=e instanceof Error?e.message:'评价提交
    ↳ 失败'}}
async function openCitation(citation:Citation){preview.value=null;previewBusy.value=true;try{pre
    ↳ view.value=await ragRequest(`/documents/${citation.document_id}/preview?chunk_id=${encodeURIComp
    ↳ onent(citation.chunk_id)}`)}catch(e){error.value=e instanceof Error?e.message:'无法打开原文'}fin
    ↳ ally{previewBusy.value=false}}
watch(selectedKb,async()=>{if(answerMode.value!=='library')return;activeSession.value='';message
    ↳ s.value=[];await loadSessions()})
onMounted(async()=>{window.addEventListener('studypath:auth-expired',expireAuth);if(authState.to
    ↳ ken){try{await ragRequest('/auth/me');authenticated.value=true;await loadKnowledgeBases()}catch{
    ↳ clearCredentials()}finally{validatingAuth.value=false}}else validatingAuth.value=false})
onBeforeUnmount(()=>{aborter?.abort();window.clearInterval(pollTimer);window.removeEventListener
    ↳ ('studypath:auth-expired',expireAuth)})
</script>
<template>
  <div v-if="validatingAuth" class="rag-auth-wrap"><div class="rag-library-loading"><LoaderCircl
    ↳ e class="spin"/>正在验证登录状态…</div></div>
  <div v-else-if="!authenticated" class="rag-auth-wrap"><RagAuthGate @authenticated="authDone"/>
    ↳ </div>
  <section v-else class="rag-chat-page">
    <aside class="rag-session-rail">
      <div class="rag-rail-heading"><div><span class="rag-eyebrow">AI STUDY ASSISTANT</span><h2>
    ↳ 智能答疑</h2></div><button class="rag-icon-button" title="退出登录" @click="logout"><X :size="16
    ↳ "/></button></div>
      <button class="rag-button primary wide" :disabled="answerMode==='library'&&!selectedKb" @c
    ↳ lick="createSession"><MessageSquarePlus :size="16"/>新建对话</button>
      <div class="rag-mode-field">
        <span class="rag-mode-label">回答方式</span>
        <div class="rag-mode-options" role="group" aria-label="选择回答方式">
          <button type="button" class="rag-mode-option" :class="{selected:answerMode==='library'
    ↳ }" :aria-pressed="answerMode==='library'" @click="changeAnswerMode('library')"><BookOpen :size="
    ↳ 14"/>本地资料库</button>
          <button type="button" class="rag-mode-option" :class="{selected:answerMode==='model'}"
    ↳  :aria-pressed="answerMode==='model'" @click="changeAnswerMode('model')"><Sparkles :size="14"/>
    ↳ 大模型</button>
        </div>
      </div>
      <label v-if="answerMode==='library'" class="rag-field rag-kb-select"><span>检索知识库</spa
    ↳ n><div><BookOpen :size="15"/><select v-model="selectedKb"><option value="" disabled>选择知识库</
    ↳ option><option v-for="kb in knowledgeBases" :key="kb.id" :value="kb.id">{{kb.name}}{{kb.course_n
```

```
// 文件：src/views/RagChatView.vue
    ↳ ame?` · ${kb.course_name}`:''}}</option></select><ChevronDown :size="14"/></div></label>
      <div class="rag-session-title">最近对话 <span>{{sessions.length}}</span></div>
      <div class="rag-session-list"><button v-for="session in sessions" :key="session.id" class=
    ↳ "rag-session-item" :class="{selected:activeSession===session.id}" @click="selectSession(session.
    ↳ id)"><MessageSquarePlus :size="15"/><span>{{session.title}}</span><span class="rag-session-delet
    ↳ e" title="删除对话" @click.stop="deleteSession(session)"><Trash2 :size="14"/></span></button><di
    ↳ v v-if="!sessions.length" class="rag-empty-note">这里会显示你保存的学习对话。</div></div>
      <div class="rag-private-note"><LockKeyhole :size="14"/><span>此账号的资料与历史对话和其他
    ↳ 账号隔离。</span></div>
    </aside>
    <main class="rag-chat-main">
      <header class="rag-chat-header"><div><span class="rag-eyebrow">{{answerMode==='library'?(s
    ↳ electedKnowledgeBase?.course_name||'个人知识库'):'通用知识问答'}}</span><h1>{{sessions.find(s=>s
    ↳ .id===activeSession)?.title||(answerMode==='library'?'课程资料问答':'大模型问答')}}</h1><p>{{ans
    ↳ werMode==='library'?'回答依据当前知识库检索到的原文，来源可打开核对。':'由大模型根据已有知识直接
    ↳ 回答，不会检索本地资料。'}}</p></div><a v-if="answerMode==='library'" class="rag-button subtle"
    ↳ href="/library"><FileText :size="15"/>管理资料</a></header>
      <div v-if="answerMode==='library'&&!knowledgeBases.length" class="rag-empty-state"><div cl
    ↳ ass="rag-empty-icon"><BookOpen/></div><h3>先创建知识库并上传课程资料</h3><p>资料完成解析和向量索
    ↳ 引后，才能基于原文进行问答。</p><a class="rag-button primary" href="/library">前往学习资料库 <Ar
    ↳ rowRight :size="15"/></a></div>
      <div v-else-if="answerMode==='library'&&selectedKnowledgeBase?.document_count===0" class="
    ↳ rag-inline-alert"><AlertCircle :size="17"/><span>当前知识库还没有可用资料。请先上传并等待索引状
    ↳ 态变为“已就绪”。</span><a href="/library">前往上传</a></div>
      <div v-else ref="chatBox" class="rag-message-scroll">
        <div v-if="!messages.length" class="rag-welcome"><span class="rag-welcome-spark"><Sparkl
    ↳ es :size="20"/></span><h2>{{answerMode==='library'?'从课程资料出发，弄懂每一个问题':'直接向大模
    ↳ 型提问，探索更多知识'}}</h2><p>{{answerMode==='library'?'答案会附上真实文档片段与页码。试着问：'
    ↳ :'大模型将依据已有知识回答，不会引用本地资料。试着问：'}}</p><div class="rag-prompt-grid"><butto
    ↳ n v-for="prompt in prompts" :key="prompt" @click="input=prompt">{{prompt}} <ArrowRight :size="14
    ↳ "/></button></div></div>
        <article v-for="(message,index) in messages" :key="message.id||`local-${index}`" class="
    ↳ rag-message" :class="message.role">
          <div class="rag-avatar"><span v-if="message.role==='user'">我</span><Sparkles v-else :
    ↳ size="15"/></div><div class="rag-message-body"><div class="rag-message-label">{{message.role==='
    ↳ user'?'你':'智学研途'}}</div><div v-if="message.role==='user'" class="rag-user-bubble">{{message
    ↳ .content}}</div><template v-else><div class="rag-assistant-content" v-html="renderMarkdown(messa
    ↳ ge.content)"/><div v-if="message.loading" class="rag-generating"><LoaderCircle :size="14" class=
    ↳ "spin"/>{{answerMode==='library'?'正在依据资料组织回答…':'正在由大模型生成回答…'}}</div><div v
    ↳ -if="message.id&&!message.loading&&!message.error" class="rag-message-feedback"><span>回答有帮助
    ↳ 吗？</span><button :class="{active:message.feedback==='helpful'}" @click="rateAnswer(message,'he
    ↳ lpful')">有帮助</button><button :class="{active:message.feedback==='unhelpful'}" @click="rateAns
    ↳ wer(message,'unhelpful')">需改进</button></div><div v-if="message.citations?.length" class="rag-
    ↳ message-citations"><button v-for="(citation,citeIndex) in message.citations" :key="citation.cita
    ↳ tion_id" @click="openCitation(citation)"><span>{{citeIndex+1}}</span>{{citation.document_name}}<
    ↳ small>{{citation.page_start?`第 ${citation.page_start} 页`:'片段'}}</small></button></div><div v
    ↳ -if="message.error" class="rag-message-error"><AlertCircle :size="14"/>{{message.error}}<button
    ↳ @click="retry(index)">重试</button></div></template></div>
        </article>
      </div>
```

```
// 文件：src/views/RagChatView.vue
      <div v-if="error" class="rag-global-error"><AlertCircle :size="15"/>{{error}}<button @clic
    ↳ k="error=''">关闭</button></div>
      <form v-if="answerMode==='model'||knowledgeBases.length" class="rag-composer" @submit.prev
    ↳ ent="sendQuestion()"><textarea v-model="input" rows="2" maxlength="5000" :placeholder="answerMod
    ↳ e==='library'?'输入课程问题，按 Enter 发送，Shift + Enter 换行':'输入问题，按 Enter 发送，Shift
    ↳ + Enter 换行'" :disabled="busy||(answerMode==='library'&&selectedKnowledgeBase?.document_count==
    ↳ =0)" @keydown.enter.exact.prevent="sendQuestion()"/><div class="rag-composer-bottom"><span>{{ans
    ↳ werMode==='library'?'检索片段将发送至已配置的大模型服务；请勿上传未获授权的敏感资料。':'此模式直
    ↳ 接使用已配置的大模型服务，不检索本地资料。'}}</span><button v-if="busy" type="button" class="rag
    ↳ -button stop" @click="stopGeneration"><Square :size="13"/>中断</button><button v-else class="rag
    ↳ -button primary send" :disabled="!input.trim()||(answerMode==='library'&&selectedKnowledgeBase?.
    ↳ document_count===0)"><Send :size="15"/>发送</button></div></form>
    </main>
    <aside class="rag-evidence-rail"><div class="rag-evidence-head"><div><span class="rag-eyebro
    ↳ w">{{answerMode==='library'?'GROUNDED ANSWERS':'MODEL ANSWERS'}}</span><h3>{{answerMode==='libra
    ↳ ry'?'参考来源':'回答模式'}}</h3></div><span class="rag-source-count">{{latestCitations.length}}<
    ↳ /span></div><p class="rag-evidence-hint">{{answerMode==='library'?'来源来自本轮真实检索结果。匹
    ↳ 配分是检索分数，不代表答案正确概率。':'大模型依据已有知识回答，不读取本地资料库，因此不会显示文
    ↳ 档引用。'}}</p><div v-if="latestCitations.length" class="rag-evidence-list"><button v-for="(cita
    ↳ tion,index) in latestCitations" :key="citation.citation_id" class="rag-evidence-card" @click="op
    ↳ enCitation(citation)"><span class="rag-evidence-index">{{String(index+1).padStart(2,'0')}}</span
    ↳ ><b>{{citation.document_name}}</b><small>{{citation.page_start?`第 ${citation.page_start}${citat
    ↳ ion.page_end&&citation.page_end!==citation.page_start?`–${citation.page_end}`:''} 页`:citation.
    ↳ section_title||'页码未标注'}}</small><span v-if="citation.section_title" class="rag-evidence-sec
    ↳ tion">{{citation.section_title}}</span><p>{{citation.snippet}}</p><span class="rag-evidence-scor
    ↳ e">检索分数 {{citation.retrieval_score.toFixed(3)}} · 点击查看原文</span></button></div><div v-
    ↳ else class="rag-evidence-empty"><PanelRightClose :size="20"/><p>{{answerMode==='library'?'发送问
    ↳ 题后，这里会列出检索到的原文和页码。':'切换到本地资料库模式后，可查看回答引用的原文。'}}</p></di
    ↳ v><div class="rag-evidence-foot"><Check :size="14"/>{{answerMode==='library'?'引用由服务端文档片
    ↳ 段生成':'当前回答未使用本地资料'}}</div></aside>
    <div v-if="preview||previewBusy" class="rag-modal-backdrop" @click.self="preview=null"><sect
    ↳ ion class="rag-preview-modal"><header><div><span class="rag-eyebrow">原文定位</span><h3>{{previe
    ↳ w?.document_name||'正在打开资料'}}</h3><small v-if="preview?.page">第 {{preview.page}}{{preview.
    ↳ page_end&&preview.page_end!==preview.page?`–${preview.page_end}`:''}} 页 · {{preview.section_t
    ↳ itle||'未识别章节'}}</small></div><button class="rag-icon-button" @click="preview=null"><X :size
    ↳ ="17"/></button></header><div v-if="previewBusy" class="rag-preview-loading"><LoaderCircle class
    ↳ ="spin"/>正在读取保存的原文…</div><div v-else class="rag-preview-text"><div v-if="preview?.high
    ↳ light" class="rag-original-quote"><b>检索引用片段</b><blockquote>{{preview.highlight}}</blockquo
    ↳ te></div><div class="rag-preview-page-label">{{preview?.page?`文档第 ${preview.page} 页原文`:'解
    ↳ 析后的文档文本'}}</div><pre>{{preview?.text}}</pre><p class="rag-muted">页码来自原始文档提取；当
    ↳ 前显示提取文本，精确版面高亮暂不可用。</p></div></section></div>
  </section>
</template>
<style>
@import 'katex/dist/katex.min.css';
.rag-chat-page{display:grid;grid-template-columns:230px minmax(0,1fr) 292px;gap:16px;min-height:
    ↳ calc(100vh - 166px);height:calc(100vh - 166px);color:#28253a}.rag-session-rail,.rag-evidence-rai
    ↳ l,.rag-chat-main{background:#fff;border:1px solid #ece9f2;border-radius:18px;box-shadow:0 4px 18
    ↳ px #32245c08}.rag-session-rail{padding:18px 14px;display:flex;flex-direction:column;min-height:0
    ↳ }.rag-rail-heading,.rag-evidence-head,.rag-chat-header,.rag-preview-modal header{display:flex;al
```

```
// 文件：src/views/RagChatView.vue
    ↳ ign-items:center;justify-content:space-between;gap:12px}.rag-rail-heading{margin:2px 2px 18px}.r
    ↳ ag-rail-heading h2,.rag-chat-header h1,.rag-evidence-head h3,.rag-preview-modal h3{margin:4px 0
    ↳ 0;letter-spacing:-.025em}.rag-rail-heading h2{font-size:19px}.rag-eyebrow{font-size:10px;letter-
    ↳ spacing:.12em;color:#8c73d8;font-weight:700}.rag-button,.rag-icon-button,.rag-link-button{border
    ↳ :0;cursor:pointer;display:inline-flex;align-items:center;justify-content:center;gap:8px;font:inh
    ↳ erit;text-decoration:none}.rag-button{min-height:38px;border-radius:10px;padding:0 13px;font-siz
    ↳ e:12px;font-weight:650}.rag-button.primary{color:white;background:linear-gradient(120deg,#7048dc
    ↳ ,#8b68ef);box-shadow:0 5px 12px #7953df29}.rag-button.primary:disabled{opacity:.5;cursor:not-all
    ↳ owed}.rag-button.wide{width:100%}.rag-button.subtle{background:#f6f3fd;color:#7557ca}.rag-button
    ↳ .stop{background:#fff0f0;color:#b74958}.rag-button.send{margin-left:auto}.rag-icon-button{width:
    ↳ 32px;height:32px;border-radius:9px;background:#f6f4fa;color:#77718a}.rag-field{display:grid;gap:
    ↳ 7px;margin:18px 0 20px}.rag-field>span{font-size:11px;color:#888299}.rag-kb-select>div{display:f
    ↳ lex;align-items:center;gap:8px;border:1px solid #ece9f3;border-radius:10px;padding:0 10px;color:
    ↳ #8064cf}.rag-kb-select select{height:39px;min-width:0;flex:1;border:0;background:transparent;col
    ↳ or:#423c55;outline:0;font-size:12px}.rag-session-title{display:flex;justify-content:space-betwee
    ↳ n;padding:0 4px 8px;color:#8b8598;font-size:11px}.rag-session-title span,.rag-source-count{min-w
    ↳ idth:22px;text-align:center;border-radius:20px;padding:3px 7px;background:#f3effc;color:#7653d4;
    ↳ font-size:10px}.rag-session-list{overflow:auto;flex:1;display:flex;flex-direction:column;gap:4px
    ↳ }.rag-session-item{display:flex;align-items:center;gap:9px;text-align:left;padding:10px 8px;bord
    ↳ er:0;border-radius:9px;background:transparent;color:#69657a;font-size:11px;cursor:pointer}.rag-s
    ↳ ession-item.selected,.rag-session-item:hover{background:#f5f1ff;color:#6746c7}.rag-session-item>
    ↳ span:nth-child(2){flex:1;min-width:0;overflow:hidden;text-overflow:ellipsis;white-space:nowrap}.
    ↳ rag-session-delete{opacity:0}.rag-session-item:hover .rag-session-delete{opacity:1}.rag-empty-no
    ↳ te{color:#aaa4b6;font-size:11px;padding:10px 5px}.rag-private-note{display:flex;gap:8px;align-it
    ↳ ems:flex-start;color:#928ba2;font-size:10px;line-height:1.6;padding:12px 6px 0;border-top:1px so
    ↳ lid #f0edf4}.rag-private-note svg{flex:none;color:#8f77d4}.rag-chat-main{display:flex;flex-direc
    ↳ tion:column;min-width:0;overflow:hidden}.rag-chat-header{padding:20px 24px 16px;border-bottom:1p
    ↳ x solid #f0edf4}.rag-chat-header h1{font-size:18px}.rag-chat-header p{font-size:11px;color:#9993
    ↳ a4;margin:5px 0 0}.rag-message-scroll{padding:20px 24px;flex:1;min-height:0;overflow-y:auto;scro
    ↳ llbar-width:thin;scrollbar-color:#ddd5f0 transparent}.rag-welcome{max-width:520px;margin:7vh aut
    ↳ o 20px;text-align:center}.rag-welcome-spark,.rag-empty-icon{display:grid;place-items:center;marg
    ↳ in:auto;width:48px;height:48px;border-radius:16px;color:#7956d5;background:#f1ebff}.rag-welcome
    ↳ h2{font-size:20px;letter-spacing:-.035em;margin:16px 0 7px}.rag-welcome>p,.rag-empty-state>p{fon
    ↳ t-size:12px;color:#9993a5}.rag-prompt-grid{display:grid;grid-template-columns:1fr 1fr;gap:8px;ma
    ↳ rgin-top:22px}.rag-prompt-grid button{border:1px solid #eeeaf5;background:#fff;border-radius:11p
    ↳ x;padding:12px;color:#5d576e;font-size:11px;text-align:left;display:flex;justify-content:space-b
    ↳ etween;gap:8px;cursor:pointer}.rag-prompt-grid button:hover{border-color:#d2c4f5;color:#7150ca}.
    ↳ rag-message{display:flex;align-items:flex-start;gap:10px;margin:0 0 24px}.rag-message.user{flex-
    ↳ direction:row-reverse}.rag-avatar{flex:none;width:29px;height:29px;border-radius:10px;display:gr
    ↳ id;place-items:center;background:#f1ebff;color:#7855d4;font-size:11px;font-weight:700}.rag-messa
    ↳ ge.user .rag-avatar{background:#ecebf2;color:#666172}.rag-message-body{max-width:min(90%,710px);
    ↳ min-width:80px}.rag-message.user .rag-message-body{text-align:right}.rag-message-label{font-size
    ↳ :10px;color:#9892a4;margin:0 2px 7px}.rag-user-bubble{display:inline-block;text-align:left;paddi
    ↳ ng:11px 14px;background:#7652d9;color:#fff;border-radius:14px 4px 14px 14px;font-size:13px;line-
    ↳ height:1.7;white-space:pre-wrap}.rag-assistant-content{font-size:13px;color:#484457;line-height:
    ↳ 1.8;overflow-wrap:anywhere}.rag-assistant-content>*:first-child{margin-top:0}.rag-assistant-cont
    ↳ ent h1,.rag-assistant-content h2,.rag-assistant-content h3{font-size:15px;color:#302b41}.rag-ass
    ↳ istant-content pre{padding:12px;background:#25213a;color:#eee9ff;overflow:auto;border-radius:10p
    ↳ x;font-size:11px}.rag-assistant-content :not(pre)>code{padding:2px 5px;border-radius:4px;backgro
    ↳ und:#f3effa;color:#6e4ccc}.rag-assistant-content blockquote{margin:10px 0;padding:2px 12px;borde
```

// 文件：src/views/RagChatView.vue

```
↳ r-left:3px solid #b69eea;color:#797387}.rag-assistant-content table{border-collapse:collapse;fon
↳ t-size:11px}.rag-assistant-content th,.rag-assistant-content td{border:1px solid #e9e5ef;padding
↳ :6px 9px}.rag-generating{display:flex;align-items:center;gap:7px;color:#8d829f;font-size:10px}.s
↳ pin{animation:rag-spin .9s linear infinite}@keyframes rag-spin{to{transform:rotate(360deg)}}.rag
↳ -message-feedback{display:flex;align-items:center;gap:6px;margin-top:9px;color:#9a94a6;font-size
↳ :10px}.rag-message-feedback button{border:1px solid #ece8f2;background:#fff;border-radius:6px;pa
↳ dding:4px 8px;color:#766d87;font-size:10px;cursor:pointer}.rag-message-feedback button.active{bo
↳ rder-color:#bda9ef;color:#704bd0;background:#f7f3ff}.rag-message-citations{display:flex;flex-wra
↳ p:wrap;gap:6px;margin-top:10px}.rag-message-citations button{display:flex;align-items:center;gap
↳ :5px;padding:5px 8px;border:1px solid #eee9f6;background:#fff;border-radius:7px;color:#6855a1;fo
↳ nt-size:9px;cursor:pointer;max-width:240px;overflow:hidden;text-overflow:ellipsis;white-space:no
↳ wrap}.rag-message-citations button span{display:grid;place-items:center;width:16px;height:16px;b
↳ order-radius:5px;background:#efe8ff;color:#704bd0}.rag-message-citations small{color:#999}.rag-m
↳ essage-error,.rag-global-error{display:flex;align-items:center;gap:7px;color:#ac4857;font-size:1
↳ 0px;margin-top:8px}.rag-message-error button,.rag-global-error button{border:0;background:none;c
↳ olor:#704bd0;cursor:pointer;font:inherit}.rag-composer{margin:0 20px 18px;padding:11px 12px 9px;
↳ border:1px solid #e8e3f0;border-radius:14px;box-shadow:0 5px 16px #3a2b6010}.rag-composer textar
↳ ea{border:0;outline:0;resize:vertical;width:100%;min-height:48px;color:#413c4e;font:inherit;font
↳ -size:12px;line-height:1.6}.rag-composer textarea::placeholder{color:#aaa4b5}.rag-composer-botto
↳ m{display:flex;align-items:center;gap:8px}.rag-composer-bottom>span{font-size:9px;color:#a29bac;
↳ line-height:1.5}.rag-evidence-rail{padding:19px 15px;display:flex;flex-direction:column;min-heig
↳ ht:0;overflow:hidden}.rag-evidence-head h3{font-size:15px}.rag-evidence-hint{margin:10px 0 13px;
↳ font-size:10px;color:#9a94a6;line-height:1.6}.rag-evidence-list{overflow:auto;display:flex;flex-
↳ direction:column;gap:9px}.rag-evidence-card{position:relative;text-align:left;border:1px solid #
↳ eeeaf4;background:#fff;border-radius:12px;padding:12px;cursor:pointer;color:#393448}.rag-evidenc
↳ e-card:hover{border-color:#cdbdf2;box-shadow:0 5px 15px #7352cf10}.rag-evidence-index{float:righ
↳ t;color:#a38adf;font-size:10px}.rag-evidence-card>b{display:block;font-size:11px;padding-right:2
↳ 8px}.rag-evidence-card>small{display:block;margin-top:5px;color:#827a92;font-size:10px}.rag-evid
↳ ence-section{display:inline-block;margin-top:6px;color:#7959c9;background:#f6f2ff;border-radius:
↳ 5px;padding:3px 5px;font-size:9px}.rag-evidence-card>p{font-size:10px;line-height:1.65;color:#71
↳ 6b7d;display:-webkit-box;-webkit-line-clamp:5;-webkit-box-orient:vertical;overflow:hidden;margin
↳ :8px 0}.rag-evidence-score{display:block;padding-top:7px;border-top:1px solid #f1eef5;color:#9d9
↳ 6a7;font-size:9px}.rag-evidence-empty{margin:auto;text-align:center;color:#b0a9b9;max-width:190p
↳ x}.rag-evidence-empty p{font-size:11px;line-height:1.7}.rag-evidence-foot{display:flex;align-ite
↳ ms:center;gap:6px;margin-top:auto;padding-top:12px;color:#9189a0;font-size:9px}.rag-evidence-foo
↳ t svg{color:#53ae95}.rag-inline-alert,.rag-global-error{margin:10px 22px;padding:10px 12px;borde
↳ r-radius:9px;background:#fff8e9;color:#9d7138;font-size:11px}.rag-inline-alert{display:flex;alig
↳ n-items:center;gap:8px}.rag-inline-alert a{margin-left:auto;color:#7954d6;text-decoration:none}.
↳ rag-empty-state{margin:auto;text-align:center;padding:30px 15px}.rag-empty-state h3{font-size:16
↳ px}.rag-empty-state>p{margin-bottom:20px}.rag-global-error{background:#fff0f0}.rag-auth-wrap{dis
↳ play:grid;place-items:center;min-height:calc(100vh - 180px)}.rag-auth-card{width:min(410px,100%)
↳ ;background:#fff;border:1px solid #eeeaf4;border-radius:20px;padding:32px;box-shadow:0 12px 40px
↳  #32245c0d;text-align:center}.rag-auth-mark{width:48px;height:48px;border-radius:15px;display:gr
↳ id;place-items:center;margin:0 auto 18px;background:linear-gradient(140deg,#8861e9,#6d46d3);colo
↳ r:white}.rag-auth-card h2{font-size:20px;margin:7px 0}.rag-muted,.rag-auth-card>small{color:#999
↳ 3a5;font-size:11px}.rag-auth-form{text-align:left;display:grid;gap:14px;margin-top:22px}.rag-aut
↳ h-form label>span{display:block;color:#777185;font-size:11px;margin-bottom:6px}.rag-auth-form la
↳ bel>div{height:42px;border:1px solid #e9e5ef;border-radius:9px;display:flex;align-items:center;g
↳ ap:9px;padding:0 11px;color:#958aa9}.rag-auth-form input{border:0;outline:0;width:100%;font:inhe
↳ rit;font-size:12px;color:#3c3748}.rag-error{font-size:11px;color:#bf4351;margin:0}.rag-link-butt
```

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// 文件：src/views/RagChatView.vue
    ↳ on{margin:14px auto;color:#7855d4;background:none;font-size:11px}.rag-auth-card>small{display:bl
    ↳ ock;margin-top:7px}.rag-modal-backdrop{position:fixed;inset:0;z-index:30;background:#241a3d4f;di
    ↳ splay:grid;place-items:center;padding:24px}.rag-preview-modal{width:min(740px,100%);max-height:m
    ↳ in(82vh,900px);overflow:hidden;display:flex;flex-direction:column;background:#fff;border-radius:
    ↳ 16px;box-shadow:0 24px 70px #21133d35}.rag-preview-modal header{padding:20px 23px;border-bottom:
    ↳ 1px solid #eeeaf3}.rag-preview-modal h3{font-size:16px}.rag-preview-modal small{display:block;co
    ↳ lor:#8b8498;margin-top:5px;font-size:10px}.rag-preview-text{overflow:auto;padding:20px 24px}.rag
    ↳ -preview-text pre{font:12px/1.85 inherit;color:#555064;white-space:pre-wrap;overflow-wrap:anywhe
    ↳ re;margin:0}.rag-preview-page-label{font-size:10px;color:#90899c;margin:0 0 9px}.rag-original-qu
    ↳ ote{padding:12px 14px;background:#faf8ff;border:1px solid #eee8f8;border-radius:10px;margin-bott
    ↳ om:16px;color:#6247ae;font-size:10px}.rag-original-quote blockquote{margin:6px 0 0;color:#5f5a6e
    ↳ ;line-height:1.8;font-size:11px}.rag-preview-loading{padding:70px;text-align:center;color:#8d849
    ↳ e;font-size:12px}.rag-preview-loading svg{vertical-align:middle}.rag-global-error{display:flex;a
    ↳ lign-items:center}
.rag-mode-field{display:grid;gap:7px;margin:17px 0 0}.rag-mode-label{font-size:11px;color:#88829
    ↳ 9}.rag-mode-options{display:grid;grid-template-columns:1fr 1fr;gap:6px}.rag-mode-option{display:
    ↳ flex;align-items:center;justify-content:center;gap:5px;min-width:0;min-height:36px;padding:0 5px
    ↳ ;border:1px solid #ece9f3;border-radius:9px;background:#fff;color:#777185;font:inherit;font-size
    ↳ :10px;cursor:pointer;white-space:nowrap}.rag-mode-option:hover{border-color:#cdbdf2;color:#7150c
    ↳ a}.rag-mode-option.selected{border-color:#c9b8f2;background:#f6f2ff;color:#704bd0;box-shadow:ins
    ↳ et 0 0 0 1px #e7ddfb}
@media(max-width:1180px){.rag-chat-page{grid-template-columns:210px minmax(0,1fr);height:auto;mi
    ↳ n-height:calc(100vh - 166px)}.rag-evidence-rail{grid-column:2;min-height:160px;max-height:330px}
    ↳ .rag-evidence-list{display:grid;grid-template-columns:repeat(2,minmax(0,1fr))}.rag-evidence-foot
    ↳ {margin-top:12px}}
@media(max-width:760px){.rag-chat-page{display:flex;flex-direction:column;height:auto}.rag-sessi
    ↳ on-rail{min-height:auto}.rag-session-list{max-height:190px}.rag-chat-main{min-height:75vh}.rag-e
    ↳ vidence-rail{min-height:200px}.rag-message-scroll{min-height:360px;padding:16px}.rag-chat-header
    ↳ {padding:16px}.rag-chat-header h1{font-size:16px}.rag-prompt-grid{grid-template-columns:1fr}.rag
    ↳ -evidence-list{grid-template-columns:1fr}.rag-composer{margin:0 12px 12px}.rag-composer-bottom{a
    ↳ lign-items:flex-end}.rag-composer-bottom>span{flex:1}.rag-auth-card{padding:26px 20px}.rag-modal
    ↳ -backdrop{padding:10px}.rag-preview-modal{max-height:90vh}}
</style>
// 文件：src/views/RagLibraryView.vue
<script setup lang="ts">
import { computed, nextTick, onBeforeUnmount, onMounted, ref, watch } from 'vue'
import { AlertCircle, BookOpen, CheckCircle2, ChevronDown, Eye, FileText, FolderPlus, LoaderCirc
    ↳ le, Pencil, Plus, RotateCw, Trash2, Upload, X } from 'lucide-vue-next'
import RagAuthGate from '../components/RagAuthGate.vue'
import { authState, clearCredentials, ragRequest } from '../rag/api'
type KB={id:string;name:string;course_name:string;document_count:number;created_at:string}
type Doc={id:string;knowledge_base_id:string;file_name:string;file_type:string;file_size:number;
    ↳ status:string;error_message:string;chunk_count:number;created_at:string}
type Preview={document_id:string;document_name:string;page:number|null;page_end?:number|null;sec
    ↳ tion_title?:string;text:string;highlight?:string}
const authenticated=ref(false);const validatingAuth=ref(!!authState.token);const bases=ref<KB[]>
    ↳ ([]);const selected=ref('');const docs=ref<Doc[]>([]);const loading=ref(false);const busy=ref(fa
    ↳ lse);const error=ref('');const notice=ref('');const kbName=ref('');const courseName=ref('');cons
    ↳ t showCreate=ref(false);const nameInput=ref<HTMLInputElement>();const fileInput=ref<HTMLInputEle
    ↳ ment>();const preview=ref<Preview|null>(null);const previewBusy=ref(false);const page=ref(1);let
    ↳  pollTimer:number|undefined
```

```
// 文件：src/views/RagLibraryView.vue
const current=computed(()=>bases.value.find(k=>k.id===selected.value));const processing=computed
    ↳ (()=>docs.value.some(doc=>['UPLOADED','PARSING','CHUNKING','EMBEDDING','INDEXING'].includes(doc.
    ↳ status)))
function authDone(){authenticated.value=true;void loadBases()}
function logout(){clearCredentials();authenticated.value=false;bases.value=[];docs.value=[]}
function expireAuth(){logout();error.value='登录状态已过期，请重新登录'}
function explain(e:unknown){error.value=e instanceof Error?e.message:'服务请求失败'}
async function loadBases(){try{bases.value=await ragRequest<KB[]>('/knowledge-bases');if(!select
    ↳ ed.value||!bases.value.some(k=>k.id===selected.value))selected.value=bases.value[0]?.id||''}catc
    ↳ h(e){explain(e)}}
async function loadDocs(){if(!selected.value){docs.value=[];return}loading.value=true;try{docs.v
    ↳ alue=await ragRequest<Doc[]>(`/knowledge-bases/${selected.value}/documents`);await loadBases()}c
    ↳ atch(e){explain(e)}finally{loading.value=false}}
async function openCreate(){showCreate.value=true;error.value='';notice.value='';await nextTick(
    ↳ );nameInput.value?.focus()}
async function createBase(){const name=kbName.value.trim();if(!name){error.value='请填写知识库名
    ↳ 称后再创建';nameInput.value?.focus();return}error.value='';notice.value='';busy.value=true;try{c
    ↳ onst result=await ragRequest<KB>('/knowledge-bases',{method:'POST',body:JSON.stringify({name,cou
    ↳ rse_name:courseName.value.trim()})});bases.value.unshift(result);selected.value=result.id;kbName
    ↳ .value='';courseName.value='';showCreate.value=false;notice.value='知识库已创建'}catch(e){explai
    ↳ n(e)}finally{busy.value=false}}
async function renameBase(){if(!current.value)return;const name=window.prompt('修改知识库名称',c
    ↳ urrent.value.name);if(!name?.trim())return;try{await ragRequest(`/knowledge-bases/${current.valu
    ↳ e.id}`,{method:'PATCH',body:JSON.stringify({name:name.trim(),course_name:current.value.course_na
    ↳ me})});await loadBases()}catch(e){explain(e)}}
async function deleteBase(){if(!current.value||!window.confirm(`删除知识库“${current.value.name
    ↳ }”及其中所有文档、索引和对话？此操作不可撤销。`))return;try{await ragRequest(`/knowledge-bases/
    ↳ ${current.value.id}`,{method:'DELETE'});bases.value=bases.value.filter(k=>k.id!==selected.value)
    ↳ ;selected.value=bases.value[0]?.id||'';await loadDocs();notice.value='知识库及其数据已删除'}catc
    ↳ h(e){explain(e)}}
function chooseFiles(){fileInput.value?.click()}
async function uploadFiles(event:Event){const target=event.target as HTMLInputElement;const file
    ↳ s=Array.from(target.files||[]);target.value='';if(!selected.value||!files.length)return;busy.val
    ↳ ue=true;error.value='';for(const file of files){const body=new FormData();body.append('file',fil
    ↳ e);try{const item=await ragRequest<Doc>(`/knowledge-bases/${selected.value}/documents`,{method:'
    ↳ POST',body});docs.value.unshift(item);notice.value=`${file.name} 已保存，正在真实解析并建立索引`
    ↳ }catch(e){error.value=`${file.name}：${e instanceof Error?e.message:'上传失败'}`}}busy.value=fal
    ↳ se;await loadDocs()}
async function reindex(doc:Doc){try{const result=await ragRequest<Doc>(`/documents/${doc.id}/rei
    ↳ ndex`,{method:'POST'});Object.assign(doc,result);notice.value=`已开始重建 ${doc.file_name} 的索
    ↳ 引`;void loadDocs()}catch(e){explain(e)}}
async function removeDoc(doc:Doc){if(!window.confirm(`删除“${doc.file_name}”及其向量索引？`))r
    ↳ eturn;try{await ragRequest(`/documents/${doc.id}`,{method:'DELETE'});docs.value=docs.value.filte
    ↳ r(item=>item.id!==doc.id);notice.value='文档与向量索引已删除';await loadBases()}catch(e){explain
    ↳ (e)}}
async function openPreview(doc:Doc){preview.value=null;previewBusy.value=true;page.value=1;try{p
    ↳ review.value=await ragRequest<Preview>(`/documents/${doc.id}/preview`)}catch(e){explain(e)}final
    ↳ ly{previewBusy.value=false}}
async function loadPage(){if(!preview.value)return;previewBusy.value=true;try{preview.value=awai
    ↳ t ragRequest<Preview>(`/documents/${preview.value.document_id}/preview?page=${Math.max(1,page.va
```

```
// 文件：src/views/RagLibraryView.vue
    ↳ lue)}`)}catch(e){explain(e)}finally{previewBusy.value=false}}
function statusLabel(status:string){return ({UPLOADED:'已上传',PARSING:'解析中',CHUNKING:'分块中
    ↳ ',EMBEDDING:'向量化中',INDEXING:'写入向量库',READY:'已就绪',FAILED:'处理失败'} as Record<string,
    ↳ string>)[status]||status}
function statusClass(status:string){return status==='READY'?'ready':status==='FAILED'?'failed':[
    ↳ 'UPLOADED','PARSING','CHUNKING','EMBEDDING','INDEXING'].includes(status)?'processing':'uploaded'
    ↳ }
function size(bytes:number){return bytes<1024*1024?`${Math.max(1,Math.round(bytes/1024))} KB`:`$
    ↳ {(bytes/1024/1024).toFixed(1)} MB`}
watch(selected,()=>{void loadDocs()})
onMounted(async()=>{window.addEventListener('studypath:auth-expired',expireAuth);if(authState.to
    ↳ ken){try{await ragRequest('/auth/me');authenticated.value=true;await loadBases()}catch{clearCred
    ↳ entials()}finally{validatingAuth.value=false}}else validatingAuth.value=false;pollTimer=window.s
    ↳ etInterval(()=>{if(processing.value)void loadDocs()},1800)})
onBeforeUnmount(()=>{window.clearInterval(pollTimer);window.removeEventListener('studypath:auth-
    ↳ expired',expireAuth)})
</script>
<template>
  <div v-if="validatingAuth" class="rag-auth-wrap"><div class="rag-library-loading"><LoaderCircl
    ↳ e class="spin"/>正在验证登录状态…</div></div>
  <div v-else-if="!authenticated" class="rag-auth-wrap"><RagAuthGate @authenticated="authDone"/>
    ↳ </div>
  <section v-else class="rag-library-page">
    <header class="rag-library-heading"><div><span class="rag-eyebrow">YOUR COURSE MATERIALS</sp
    ↳ an><h1>学习资料库</h1><p>文档会在后端完成解析、分块、Embedding 与向量索引；只有“已就绪”的资料
    ↳ 可用于问答。</p></div><button type="button" class="rag-button primary" @click="showCreate ? show
    ↳ Create=false : openCreate()"><FolderPlus :size="16"/>{{showCreate?'收起创建表单':'新建知识库'}}<
    ↳ /button></header>
    <div class="rag-cloud-notice"><AlertCircle :size="16"/><span>资料存储在本地后端账号空间。提
    ↳ 问时检索片段会发送给你配置的大模型服务；请勿上传未获授权的敏感资料。</span></div>
    <form v-if="showCreate" class="rag-create-base" @submit.prevent="createBase"><label>知识库名
    ↳ 称 <span class="rag-required">必填</span><input ref="nameInput" v-model="kbName" maxlength="120"
    ↳  placeholder="例如：操作系统原理" :aria-invalid="!!error" :aria-describedby="error ? 'kb-create-
    ↳ hint' : undefined" @input="error=''"/></label><label>所属课程（可选）<input v-model="courseName"
    ↳  maxlength="120" placeholder="例如：计算机科学"/></label><button type="submit" class="rag-button
    ↳  primary" :disabled="busy"><LoaderCircle v-if="busy" :size="14" class="spin"/><Plus v-else :size
    ↳ ="14"/>{{busy?'正在创建…':'创建知识库'}}</button><button type="button" class="rag-icon-button"
    ↳ aria-label="关闭创建表单" @click="showCreate=false;error=''"><X :size="16"/></button></form>
    <div v-if="error" id="kb-create-hint" class="rag-library-error" role="alert"><AlertCircle :s
    ↳ ize="15"/>{{error}}<button type="button" aria-label="关闭错误提示" @click="error=''"><X :size="1
    ↳ 4"/></button></div><div v-if="notice" class="rag-library-notice" role="status"><CheckCircle2 :si
    ↳ ze="15"/>{{notice}}<button type="button" aria-label="关闭提示" @click="notice=''"><X :size="14"/
    ↳ ></button></div>
    <div class="rag-library-layout"><aside class="rag-kb-rail"><div class="rag-list-label">我的
    ↳ 知识库 <span>{{bases.length}}</span></div><button v-for="kb in bases" :key="kb.id" class="rag-kb
    ↳ -row" :class="{active:selected===kb.id}" @click="selected=kb.id"><BookOpen :size="16"/><span><b>
    ↳ {{kb.name}}</b><small>{{kb.course_name||'未关联课程'}} · {{kb.document_count}} 份资料</small></
    ↳ span><ChevronDown :size="14"/></button><div v-if="!bases.length" class="rag-kb-empty">创建知识库
    ↳ 后即可上传第一份课程资料。</div><div v-if="current" class="rag-kb-actions"><button @click="renam
    ↳ eBase"><Pencil :size="13"/>重命名</button><button @click="deleteBase"><Trash2 :size="13"/>删除知
```

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// 文件：src/views/RagLibraryView.vue
    ↳ 识库</button></div></aside>
      <main class="rag-library-main"><template v-if="current"><div class="rag-material-toolbar">
    ↳ <div><span class="rag-eyebrow">{{current.course_name||'PERSONAL LIBRARY'}}</span><h2>{{current.n
    ↳ ame}}</h2><p>{{docs.length}} 份资料 · 只对当前账号开放</p></div><button class="rag-button prima
    ↳ ry" :disabled="busy" @click="chooseFiles"><Upload :size="15"/>上传资料</button><input ref="fileI
    ↳ nput" type="file" accept=".pdf,.docx,.txt,.md" multiple hidden @change="uploadFiles"/></div>
          <div class="rag-upload-drop" @click="chooseFiles"><div><span class="rag-upload-icon"><
    ↳ Upload :size="18"/></span><b>拖入或选择课程资料</b><p>支持 PDF、DOCX、TXT、Markdown · 单文件最
    ↳ 大 20 MB</p></div><button class="rag-button subtle" :disabled="busy">{{busy?'上传中…':'浏览文件
    ↳ '}}</button></div>
          <div class="rag-docs-heading"><div><h3>文档与索引</h3><span>后端状态会自动刷新</span><
    ↳ /div><button class="rag-icon-button" title="刷新" @click="loadDocs"><RotateCw :size="15"/></butt
    ↳ on></div>
          <div v-if="loading&&!docs.length" class="rag-library-loading"><LoaderCircle class="spi
    ↳ n"/>读取文档列表…</div>
          <div v-else-if="!docs.length" class="rag-library-empty"><FileText :size="23"/><b>这个
    ↳ 知识库还没有文档</b><span>上传课程材料后，系统会生成真实向量索引。</span></div>
          <div v-else class="rag-document-list"><article v-for="doc in docs" :key="doc.id" class
    ↳ ="rag-document-row"><div class="rag-doc-icon" :class="doc.file_type.toLowerCase()">{{doc.file_ty
    ↳ pe}}</div><div class="rag-doc-main"><div class="rag-doc-name">{{doc.file_name}}<small>{{size(doc
    ↳ .file_size)}}</small></div><div class="rag-doc-meta"><span>{{doc.chunk_count}} 个文本块</span><s
    ↳ pan>·</span><span>{{new Date(doc.created_at).toLocaleDateString('zh-CN')}}</span><span v-if="do
    ↳ c.status==='READY'">· 已写入向量库</span></div><p v-if="doc.error_message" class="rag-doc-error
    ↳ ">{{doc.error_message}}</p></div><span class="rag-status-pill" :class="statusClass(doc.status)">
    ↳ <LoaderCircle v-if="statusClass(doc.status)==='processing'" :size="12" class="spin"/><CheckCircl
    ↳ e2 v-else-if="doc.status==='READY'" :size="12"/><AlertCircle v-else-if="doc.status==='FAILED'" :
    ↳ size="12"/>{{statusLabel(doc.status)}}</span><div class="rag-doc-actions"><button v-if="doc.stat
    ↳ us==='READY'" title="预览原文" @click="openPreview(doc)"><Eye :size="15"/></button><button v-if=
    ↳ "doc.status==='FAILED'" title="重建索引" @click="reindex(doc)"><RotateCw :size="15"/></button><b
    ↳ utton title="删除文档" @click="removeDoc(doc)"><Trash2 :size="15"/></button></div></article></di
    ↳ v>
        </template><div v-else class="rag-library-empty"><FolderPlus :size="25"/><b>先创建一个知
    ↳ 识库</b><span>按课程或主题整理资料，再让答疑助手检索其中的原文。</span><button type="button" cla
    ↳ ss="rag-button primary" @click="openCreate"><Plus :size="14"/>新建知识库</button></div></main></
    ↳ div>
    <div v-if="preview||previewBusy" class="rag-modal-backdrop" @click.self="preview=null"><sect
    ↳ ion class="rag-preview-modal"><header><div><span class="rag-eyebrow">文档预览</span><h3>{{previe
    ↳ w?.document_name||'正在读取'}}</h3><small>{{preview?.page?`第 ${preview.page} 页原文`: '解析后的
    ↳ 文本'}}</small></div><button class="rag-icon-button" @click="preview=null"><X :size="17"/></butt
    ↳ on></header><div v-if="preview?.document_name && preview.document_name.toLowerCase().endsWith('.
    ↳ pdf')" class="rag-page-controls"><button class="rag-button subtle" @click="page=Math.max(1,page-
    ↳ 1);loadPage()">上一页</button><label>页码 <input v-model.number="page" type="number" min="1" @ch
    ↳ ange="loadPage"/></label><button class="rag-button subtle" @click="page++;loadPage()">下一页</bu
    ↳ tton></div><div v-if="previewBusy" class="rag-preview-loading"><LoaderCircle class="spin"/>正在
    ↳ 读取原文…</div><div v-else class="rag-preview-text"><div v-if="preview?.highlight" class="rag-o
    ↳ riginal-quote"><b>检索到的原文片段</b><blockquote>{{preview.highlight}}</blockquote></div><pre>{
    ↳ {preview?.text}}</pre><p class="rag-muted">{{preview?.page?'页码来自原始 PDF 提取；文本定位来自
    ↳ 真实页面内容。':'精确版面定位暂不可用，当前显示后端提取的文档文本。'}}</p></div></section></div>
  </section>
</template>
```

```
// 文件：src/views/RagLibraryView.vue
<style>
.rag-library-page{color:#302c3f}.rag-library-heading{display:flex;align-items:flex-end;justify-c
    ↳ ontent:space-between;gap:20px;margin:2px 2px 17px}.rag-library-heading h1{font-size:22px;margin:
    ↳ 5px 0;letter-spacing:-.03em}.rag-library-heading p{font-size:11px;color:#9690a2;margin:0;line-he
    ↳ ight:1.6}.rag-cloud-notice{display:flex;align-items:flex-start;gap:9px;padding:11px 13px;border:
    ↳ 1px solid #eee8fb;border-radius:10px;background:#faf8ff;color:#786a9a;font-size:10px;line-height
    ↳ :1.6;margin-bottom:16px}.rag-cloud-notice svg{flex:none;color:#9678db}.rag-create-base{display:f
    ↳ lex;align-items:flex-end;gap:12px;padding:15px;margin:0 0 15px;border:1px solid #ece7f5;backgrou
    ↳ nd:white;border-radius:13px}.rag-create-base label{display:grid;gap:6px;flex:1;color:#7f788e;fon
    ↳ t-size:10px}.rag-required{color:#b34b5c;font-size:9px}.rag-create-base input[aria-invalid="true"
    ↳ ]{border-color:#c45c68;box-shadow:0 0 0 3px #c45c6818}.rag-create-base input{height:37px;border:
    ↳ 1px solid #e8e4ed;border-radius:8px;padding:0 10px;font:inherit;font-size:11px;outline:0}.rag-cr
    ↳ eate-base input:focus{border-color:#a38be3}.rag-library-layout{display:grid;grid-template-column
    ↳ s:235px minmax(0,1fr);gap:14px;align-items:start}.rag-kb-rail,.rag-library-main{background:white
    ↳ ;border:1px solid #ece9f2;border-radius:15px;box-shadow:0 4px 18px #32245c08}.rag-kb-rail{paddin
    ↳ g:13px;min-height:300px}.rag-list-label{display:flex;justify-content:space-between;padding:4px 5
    ↳ px 11px;color:#8c8599;font-size:10px}.rag-list-label span{color:#9b8bbf}.rag-kb-row{width:100%;d
    ↳ isplay:flex;align-items:center;gap:9px;padding:10px 8px;border:0;background:transparent;color:#8
    ↳ 0768e;border-radius:9px;text-align:left;cursor:pointer}.rag-kb-row.active,.rag-kb-row:hover{colo
    ↳ r:#7654ce;background:#f6f2ff}.rag-kb-row>span{flex:1;min-width:0}.rag-kb-row b,.rag-kb-row small
    ↳ {display:block;overflow:hidden;text-overflow:ellipsis;white-space:nowrap}.rag-kb-row b{font-size
    ↳ :11px;font-weight:600}.rag-kb-row small{font-size:9px;color:#a09aab;margin-top:4px}.rag-kb-empty
    ↳ {padding:12px 5px;color:#a39cac;font-size:10px;line-height:1.7}.rag-kb-actions{display:flex;gap:
    ↳ 6px;margin-top:14px;padding-top:10px;border-top:1px solid #f0edf4}.rag-kb-actions button{display
    ↳ :flex;align-items:center;gap:4px;border:0;background:none;color:#8d8799;font:inherit;font-size:9
    ↳ px;cursor:pointer}.rag-kb-actions button:last-child{color:#b95c68}.rag-library-main{padding:19px
    ↳ ;min-height:420px}.rag-material-toolbar{display:flex;align-items:center;justify-content:space-be
    ↳ tween;gap:12px;margin-bottom:14px}.rag-material-toolbar h2{font-size:16px;margin:4px 0}.rag-mate
    ↳ rial-toolbar p{color:#9b95a6;font-size:10px;margin:0}.rag-upload-drop{display:flex;align-items:c
    ↳ enter;justify-content:space-between;gap:15px;padding:17px 18px;border:1px dashed #cfc5e7;border-
    ↳ radius:12px;background:linear-gradient(110deg,#fcfbff,#faf8ff);cursor:pointer}.rag-upload-drop>d
    ↳ iv{display:grid;grid-template-columns:34px auto;column-gap:10px;align-items:center}.rag-upload-i
    ↳ con{grid-row:span 2;width:34px;height:34px;display:grid;place-items:center;border-radius:10px;ba
    ↳ ckground:#efe9ff;color:#805fd2}.rag-upload-drop b{font-size:11px;color:#514a64}.rag-upload-drop
    ↳ p{font-size:9px;color:#9b95a5;margin:4px 0 0}.rag-library-notice,.rag-library-error{display:flex
    ↳ ;align-items:center;gap:7px;margin-top:12px;padding:9px 11px;border-radius:8px;background:#f0fbf
    ↳ 6;color:#468c73;font-size:10px}.rag-library-notice button,.rag-library-error button{margin-left:
    ↳ auto;border:0;background:none;color:inherit;cursor:pointer}.rag-library-error{background:#fff1f1
    ↳ ;color:#af4d59}.rag-docs-heading{display:flex;align-items:center;justify-content:space-between;m
    ↳ argin:22px 0 9px}.rag-docs-heading h3{font-size:12px;margin:0}.rag-docs-heading span{display:blo
    ↳ ck;color:#a19bab;font-size:9px;margin-top:4px}.rag-library-loading,.rag-library-empty{display:gr
    ↳ id;place-items:center;gap:8px;padding:50px 16px;color:#a39cad;text-align:center;font-size:10px}.
    ↳ rag-library-empty>b{font-size:12px;color:#696276}.rag-library-empty>span{line-height:1.6}.rag-li
    ↳ brary-empty>.rag-button{margin-top:7px}.rag-document-list{display:flex;flex-direction:column}.ra
    ↳ g-document-row{display:flex;align-items:center;gap:11px;padding:12px 5px;border-top:1px solid #f
    ↳ 1eef4}.rag-doc-icon{flex:none;width:36px;height:40px;display:grid;place-items:center;border-radi
    ↳ us:8px;background:#f5efff;color:#835eda;font-size:8px;font-weight:700}.rag-doc-icon.pdf{backgrou
    ↳ nd:#fff0f0;color:#ca5966}.rag-doc-icon.docx{background:#eef4ff;color:#5078c5}.rag-doc-icon.md,.r
    ↳ ag-doc-icon.txt{background:#f0f8f5;color:#54a184}.rag-doc-main{flex:1;min-width:0}.rag-doc-name{
    ↳ font-size:11px;font-weight:600;color:#514b60;display:flex;align-items:center;gap:8px;min-width:0
```

```
// 文件：src/views/RagLibraryView.vue
    ↳ }.rag-doc-name:first-line{overflow-wrap:anywhere}.rag-doc-name small{flex:none;color:#a39dad;fon
    ↳ t-size:9px;font-weight:400}.rag-doc-meta{display:flex;gap:5px;margin-top:5px;color:#a09aaa;font-
    ↳ size:9px}.rag-doc-error{font-size:9px;color:#bc5860;margin:6px 0 0;line-height:1.5;overflow-wrap
    ↳ :anywhere}.rag-status-pill{flex:none;display:flex;align-items:center;gap:5px;padding:5px 7px;bor
    ↳ der-radius:20px;font-size:9px}.rag-status-pill.ready{color:#498e74;background:#edf8f2}.rag-statu
    ↳ s-pill.failed{color:#bd5864;background:#fff1f1}.rag-status-pill.processing{color:#8b6bcf;backgro
    ↳ und:#f4efff}.rag-status-pill.uploaded{color:#827b90;background:#f3f1f6}.rag-doc-actions{display:
    ↳ flex;gap:3px}.rag-doc-actions button{display:grid;place-items:center;width:29px;height:29px;bord
    ↳ er:0;border-radius:7px;background:transparent;color:#9992a3;cursor:pointer}.rag-doc-actions butt
    ↳ on:hover{background:#f4f0fb;color:#7251ce}.rag-page-controls{display:flex;align-items:center;gap
    ↳ :8px;padding:10px 23px;border-bottom:1px solid #f0edf3}.rag-page-controls label{font-size:10px;c
    ↳ olor:#8d8798}.rag-page-controls input{width:65px;padding:7px;border:1px solid #eae6ef;border-rad
    ↳ ius:6px;font:inherit}.rag-library-page .rag-modal-backdrop{z-index:35}
@media(max-width:980px){.rag-library-layout{grid-template-columns:190px minmax(0,1fr)}}
@media(max-width:720px){.rag-library-layout{grid-template-columns:1fr}.rag-kb-rail{min-height:au
    ↳ to}.rag-kb-actions{margin-top:8px}.rag-library-heading{align-items:flex-start}.rag-library-headi
    ↳ ng h1{font-size:19px}.rag-library-main{padding:14px}.rag-doc-name{display:block;line-height:1.5}
    ↳ .rag-doc-name small{display:block}.rag-status-pill{font-size:8px;padding:4px 5px}.rag-doc-action
    ↳ s button{width:25px}.rag-create-base{flex-wrap:wrap}.rag-create-base label{min-width:150px}.rag-
    ↳ upload-drop{padding:13px}.rag-upload-drop p{max-width:220px}}
</style>
// 文件：src/views/WorkspaceView.vue
<script setup lang="ts">
import { computed, nextTick, onMounted, onBeforeUnmount, reactive, ref, watch } from 'vue'
import { useRoute, useRouter } from 'vue-router'
import MarkdownIt from 'markdown-it'
import type { EChartsInstance, EChartsOption } from '../utils/echarts'
import { ArrowLeft, ArrowRight, ArrowUpRight, BookOpen, Check, CheckCheck, ChevronDown, ChevronL
    ↳ eft, ChevronRight, CircleHelp, Clock3, Download, FileText, Filter, Flame, FolderOpen, Info, Ligh
    ↳ tbulb, ListFilter, Maximize2, MessageSquare, MoreHorizontal, Network, Plus, RotateCcw, Search, S
    ↳ end, Settings2, Sparkles, Star, Target, ThumbsDown, ThumbsUp, Trash2, Upload, X, ZoomIn, ZoomOut
    ↳ , Copy as CopyIcon, Bell as BellIcon, LockKeyhole as LockIcon } from 'lucide-vue-next'
import ChartPanel from '../components/ChartPanel.vue'
import { studyApi } from '../api'
import { useStudyStore } from '../stores/study'
import { profileState, saveProfile, type StudyProfile } from '../stores/profile'
import { preferences, savePreferences } from '../stores/preferences'
import type { Course, KnowledgeNode, Question, Material } from '../types'
import '../styles/learning-loop.css'
type Source = { document: string; section: string; excerpt: string; score: string }
type Message = { role: 'user'|'assistant'; text: string; sources?: Source[]; loading?: boolean;
    ↳ feedback?: string }
const route = useRoute(); const router = useRouter(); const store = useStudyStore(); const md =
    ↳ new MarkdownIt({ html: false, linkify: true, breaks: true })
const section = computed(() => String(route.meta.section || '')); const pageTitle = computed(()
    ↳ => String(route.meta.title || '学习空间'))
const q = ref(''); const selectedCourse = ref('全部课程'); const courseCategoryFilter=ref('全部
    ↳ 课程'); const drawer = ref(false); const toast = ref(''); let toastTimer: number | undefined
function notify(message: string) { toast.value=message; window.clearTimeout(toastTimer); toastTi
    ↳ mer=window.setTimeout(()=>toast.value='',2600) }
function markdown(text:string) { return md.render(text) }
```

```
// 文件：src/views/WorkspaceView.vue
// Learning graph: force layout, search, category toggles and node inspector.
const graphHost = ref<HTMLElement>(); const graphFull = ref(false); const graphNodes = ref<Knowl
    ↳ edgeNode[]>([]); const graphLinks = ref<Array<{source:string;target:string}>>([]); const courses
    ↳  = ref<Course[]>([]); const graphCourseId = ref(String(route.query.course || '')); const graphCo
    ↳ urseName = computed(()=>courses.value.find(course=>course.id===graphCourseId.value)?.name||'当前
    ↳ 课程'); const selectedNode = ref<KnowledgeNode>({id:'',name:'',type:'concept',mastery:0,descript
    ↳ ion:''}); const selectedTypes = ref(['course','chapter','concept','application']); let graphChar
    ↳ t: EChartsInstance | undefined; let graphObserver: ResizeObserver | undefined; let graphEcharts:
    ↳  typeof import('../utils/echarts') | undefined
const typeMeta: Record<string,{label:string;color:string}> = { course:{label:'课程',color:'#7054
    ↳ d9'}, chapter:{label:'章节',color:'#54b7a3'}, concept:{label:'知识点',color:'#8e80e8'}, applicat
    ↳ ion:{label:'综合应用',color:'#e9a45a'} }
const graphTextCanvas = document.createElement('canvas')
const graphTextContext = graphTextCanvas.getContext('2d')
function graphNodeSize(node: KnowledgeNode, compact: boolean, fontSize: number) {
  const baseSize = compact
    ? node.id === graphCourseId.value ? 56 : node.type === 'chapter' ? 44 : node.type === 'appli
    ↳ cation' ? 34 : 30
    : node.id === graphCourseId.value ? 76 : node.type === 'chapter' ? 55 : node.type === 'appli
    ↳ cation' ? 37 : 34
  if (!graphTextContext) return baseSize
  const fontWeight = node.id === graphCourseId.value ? 700 : 600
  graphTextContext.font = `${fontWeight} ${fontSize}px sans-serif`
  const textWidth = graphTextContext.measureText(node.name).width
  // Leave room for the node border and a little breathing space around the label.
  return Math.max(baseSize, Math.ceil(textWidth + 16))
}
function graphOption(): EChartsOption { const compact=window.innerWidth<=680; const visibles=gra
    ↳ phNodes.value.filter(n=>selectedTypes.value.includes(n.type)); const ids=new Set(visibles.map(n=
    ↳ >n.id)); return { animationDuration:500, tooltip:{trigger:'item',formatter:(p:any)=>p.data?.name
    ↳ ||''}, legend:{show:false}, series:[{type:'graph',layout:'force',roam:true,draggable:true,scaleL
    ↳ imit:{min:.55,max:2.8},force:{repulsion:compact?75:260,edgeLength:compact?[38,72]:[58,140],gravi
    ↳ ty:.08},data:visibles.map(n=>{const fontSize=compact?(n.id===graphCourseId.value?12:9):(n.id===g
    ↳ raphCourseId.value?14:11);return {id:n.id,name:n.name,symbolSize:graphNodeSize(n,compact,fontSiz
    ↳ e),itemStyle:{color:typeMeta[n.type].color,borderColor:'#fff',borderWidth:n.id===graphCourseId.v
    ↳ alue?5:3,shadowBlur:n.id===graphCourseId.value?28:12,shadowColor:n.id===graphCourseId.value?'rgb
    ↳ a(112,84,217,.32)':'rgba(89,76,140,.15)'},label:{show:true,color:'#fff',fontWeight:n.id===graphC
    ↳ ourseId.value?700:600,fontSize},emphasis:{scale:1.2,itemStyle:{borderColor:'#fff',borderWidth:4}
    ↳ }}}),links:graphLinks.value.filter(l=>ids.has(l.source)&&ids.has(l.target)).map(l=>({...l, symbo
    ↳ l:['none','arrow'],symbolSize:7,lineStyle:{color:'#c8c2dc',width:1.5,curveness:.06}})),edgeSymbo
    ↳ l:['none','arrow'],edgeSymbolSize:6,emphasis:{focus:'adjacency',lineStyle:{width:3,color:'#8c75d
    ↳ c'}},selectedMode:'single'}] } }
async function loadGraph(courseId=graphCourseId.value) { if(!courseId){graphNodes.value=[];graph
    ↳ Links.value=[];return} graphCourseId.value=courseId; const graph=await studyApi.graph(courseId);
    ↳  graphNodes.value=graph.nodes; graphLinks.value=graph.links; selectedNode.value=graph.nodes.find
    ↳ ((node:KnowledgeNode)=>node.id===String(route.query.node||''))||graph.nodes[0]||{id:'',name:'',t
    ↳ ype:'concept',mastery:0,description:''}; if(section.value==='graph'){await nextTick();await draw
    ↳ Graph()} }
async function drawGraph() { if(!graphHost.value)return; graphEcharts ||= await import('../utils
    ↳ /echarts'); if(!graphHost.value||section.value!=='graph')return; if(!graphChart) graphChart=grap
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ hEcharts.echarts.init(graphHost.value); const instance=graphChart; instance.setOption(graphOptio
    ↳ n(),true); instance.off('click'); instance.on('click',(p:any)=>{if(p.dataType==='node'){const it
    ↳ em=graphNodes.value.find(n=>n.id===p.data.id);if(item){selectedNode.value=item;drawer.value=true
    ↳ }}}); graphObserver?.disconnect();graphObserver=new ResizeObserver(()=>graphChart?.resize());gra
    ↳ phObserver.observe(graphHost.value) }
function focusNode(id:string) { const n=graphNodes.value.find(x=>x.id===id);if(n){selectedNode.v
    ↳ alue=n;drawer.value=true;graphChart?.dispatchAction({type:'focusNodeAdjacency',seriesIndex:0,dat
    ↳ aIndex:graphNodes.value.filter(x=>selectedTypes.value.includes(x.type)).findIndex(x=>x.id===id)}
    ↳ );graphChart?.dispatchAction({type:'highlight',seriesIndex:0,dataIndex:graphNodes.value.filter(x
    ↳ =>selectedTypes.value.includes(x.type)).findIndex(x=>x.id===id)})} }
function zoomGraph(delta:number){graphChart?.dispatchAction({type:delta>0?'dataZoom':'dataZoom',
    ↳ delta})}
watch([selectedTypes,section,graphFull],async()=>{if(section.value==='graph'){await nextTick();d
    ↳ rawGraph()}},{deep:true})
watch(section,async(nextSection)=>{try{if(nextSection==='graph')await loadGraph(String(route.que
    ↳ ry.course||graphCourseId.value));if(nextSection==='path')await loadPersonalizedPath(String(route
    ↳ .query.course||graphCourseId.value));if(nextSection==='quiz'&&(localStorage.getItem('studypath.r
    ↳ eviewQuestionId')||localStorage.getItem('studypath.quizConceptId')))await beginQuiz()}catch(erro
    ↳ r){notify(error instanceof Error?error.message:'学习内容加载失败')}})
watch(()=>route.query.course,async(value)=>{if(typeof value!=='string'||!value)return;graphCours
    ↳ eId.value=value;try{if(section.value==='graph')await loadGraph(value);if(section.value==='path')
    ↳ await loadPersonalizedPath(value)}catch(error){notify(error instanceof Error?error.message:'课程
    ↳ 数据加载失败')}})
onMounted(async()=>{await store.loadTasks();try{courses.value=await studyApi.courses();if(!quizC
    ↳ ourseId.value&&courses.value.length)quizCourseId.value=courses.value[0].id;if(!graphCourseId.val
    ↳ ue&&courses.value.length)graphCourseId.value=courses.value[0].id;quizTopic.value=courses.value.f
    ↳ ind(course=>course.id===quizCourseId.value)?.name||'';mistakeList.value=await studyApi.mistakes(
    ↳ );if(!mistakeList.value.length){try{const legacy=JSON.parse(localStorage.getItem('studypath.lear
    ↳ ning-loop.mistakes')||'[]') as MistakeRecord[];if(legacy.length){mistakeList.value=await studyAp
    ↳ i.saveMistakes(legacy.map(item=>({question_id:item.questionId,payload:item as unknown as Record<
    ↳ string,unknown>})));localStorage.removeItem('studypath.learning-loop.mistakes')}}catch{}quizHis
    ↳ tory.value=await studyApi.quizAttempts();studyAnalytics.value=await studyApi.analytics(30)}catch
    ↳ (error){notify(error instanceof Error?error.message:'学习数据加载失败')}if(section.value==='grap
    ↳ h')await loadGraph(String(route.query.course||graphCourseId.value));if(section.value==='path')aw
    ↳ ait loadPersonalizedPath(String(route.query.course||graphCourseId.value));if(section.value==='qu
    ↳ iz')await loadGraph(quizCourseId.value);if(section.value==='quiz'&&(localStorage.getItem('studyp
    ↳ ath.reviewQuestionId')||localStorage.getItem('studypath.quizConceptId')))await beginQuiz() })
onBeforeUnmount(()=>{graphObserver?.disconnect();graphChart?.dispose();window.clearTimeout(toast
    ↳ Timer)})
const prerequisites = computed(()=>graphLinks.value.filter(edge=>edge.target===selectedNode.valu
    ↳ e.id).map(edge=>edge.source))
const related = computed(()=>graphLinks.value.filter(l=>l.source===selectedNode.value.id||l.targ
    ↳ et===selectedNode.value.id).map(l=>graphNodes.value.find(n=>n.id===(l.source===selectedNode.valu
    ↳ e.id?l.target:l.source))).filter(Boolean))
// AI Q&A and evidence drawer. The mock label travels with each response.
const chats=ref([{id:1,title:'进程与线程有什么区别？',course:'操作系统',time:'今天 10:42'},{id:2
    ↳ ,title:'二叉搜索树的删除操作',course:'数据结构',time:'昨天 20:18'},{id:3,title:'TCP 拥塞窗口如何
    ↳ 变化',course:'计算机网络',time:'周一 16:30'}]); const activeChat=ref(1); const messages=ref<Mess
    ↳ age[]>([]); const chatInput=ref(''); const chatBusy=ref(false); const chatScroll=ref<HTMLElement
    ↳ >(); const sourceSelected=ref<Source|null>(null); const sourcesOpen=ref(true)
```

```
// 文件：src/views/WorkspaceView.vue
const promptExamples=['解释操作系统中的进程与线程','二叉搜索树与平衡二叉树的区别','TCP 三次握手
    ↳ 为什么不能改为两次','动态规划和贪心算法如何选择']
watch(activeChat,()=>{messages.value=[]})
async function sendQuestion(value: string | Event = chatInput.value){const text=(typeof value===
    ↳ 'string'?value:chatInput.value).trim();if(!text||chatBusy.value)return;messages.value.push({role
    ↳ :'user',text});chatInput.value='';chatBusy.value=true;messages.value.push({role:'assistant',text
    ↳ :'',loading:true});await nextTick();chatScroll.value?.scrollTo({top:chatScroll.value.scrollHeigh
    ↳ t,behavior:'smooth'});const result=await studyApi.ask(text);messages.value[messages.value.length
    ↳ -1]={role:'assistant',text:result.answer,sources:result.sources};chatBusy.value=false;await next
    ↳ Tick();chatScroll.value?.scrollTo({top:chatScroll.value.scrollHeight,behavior:'smooth'})}
async function copyAnswer(text:string){await navigator.clipboard?.writeText(text);notify('回答已
    ↳ 复制')}
function newChat(){activeChat.value=Date.now();chats.value.unshift({id:activeChat.value,title:'
    ↳ 新建学习对话',course:selectedCourse.value==='全部课程'?'未选择课程':selectedCourse.value,time:'
    ↳ 刚刚'});messages.value=[]}
// Library uploads, validation, local demonstration only.
const statusFilter=ref('全部状态'); const materials=ref<Material[]>(JSON.parse(localStorage.getI
    ↳ tem('studypath.materials')||'null')||[{id:1,name:'数据结构课程讲义.pdf',type:'PDF',size:'8.4 MB'
    ↳ ,status:'已索引',course:'数据结构',date:'2026/10/16'},{id:2,name:'操作系统复习提纲.docx',type:'D
    ↳ OCX',size:'1.2 MB',status:'已索引',course:'操作系统',date:'2026/10/14'},{id:3,name:'TCP 拥塞控制
    ↳ 学习笔记.md',type:'MD',size:'18 KB',status:'处理中',course:'计算机网络',date:'2026/10/17'},{id:4
    ↳ ,name:'算法设计课堂记录.txt',type:'TXT',size:'42 KB',status:'待处理',course:'数据结构',date:'202
    ↳ 6/10/18'}] as Material[]); const fileInput=ref<HTMLInputElement>(); const materialDetail=ref<Mat
    ↳ erial|null>(null)
const visibleMaterials=computed(()=>materials.value.filter(m=>m.name.toLowerCase().includes(q.va
    ↳ lue.toLowerCase())&&(selectedCourse.value==='全部课程'||m.course===selectedCourse.value)&&(statu
    ↳ sFilter.value==='全部状态'||m.status===statusFilter.value)))
function saveMaterials(){localStorage.setItem('studypath.materials',JSON.stringify(materials.val
    ↳ ue))}
function uploadFiles(files:FileList|null){if(!files)return;for(const file of Array.from(files)){
    ↳ const ext=file.name.split('.').pop()?.toLowerCase()||'';if(!['pdf','docx','txt','md'].includes(e
    ↳ xt)){notify(`${file.name}：仅支持 PDF、DOCX、TXT、Markdown`);continue}if(file.size>20*1024*1024)
    ↳ {notify(`${file.name}：文件需小于 20 MB`);continue}const item:Material={id:Date.now()+Math.rando
    ↳ m(),name:file.name,type:ext.toUpperCase(),size:file.size>1024*1024?`${(file.size/1024/1024).toFi
    ↳ xed(1)} MB`:`${Math.max(1,Math.round(file.size/1024))} KB`,status:'待处理',course:selectedCourse
    ↳ .value==='全部课程'?'数据结构':selectedCourse.value,date:new Date().toLocaleDateString('zh-CN')}
    ↳ ;materials.value.unshift(item);saveMaterials();window.setTimeout(()=>{const current=materials.va
    ↳ lue.find(m=>m.id===item.id);if(current){current.status='失败';saveMaterials()}},2200);notify('已
    ↳ 加入账户数据队列；尚未解析或写入向量数据库。')} }
function deleteMaterial(id:number){materials.value=materials.value.filter(m=>m.id!==id);saveMate
    ↳ rials();notify('资料已从本地列表移除')}
// Quiz evaluates only objective answers from the selected mock question set.
type LearningAttempt = { at: string; correct: boolean; selectedAnswer: string; feedback: string
    ↳ }
type MistakeRecord = { id: number; questionId: number; conceptId?: string; title: string; topic:
    ↳  string; course: string; wrong: number; difficulty: string; saved: boolean; misconception: strin
    ↳ g; lastWrongAt: string; nextReviewAt: string; correctStreak: number; lastCorrectAt: string | nul
    ↳ l; mastered: boolean; attempts: LearningAttempt[]; evidence?: Question['evidence'] }
const mistakeList=ref<MistakeRecord[]>([])
const studyAnalytics=ref<{days:Array<{date:string;seconds:number;hours:number}>;totalSeconds:num
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ ber;totalHours:number}>({days:[],totalSeconds:0,totalHours:0})
const quizQuestions=ref<Question[]>([]); const quizStarted=ref(false); const quizAnswers=ref<Rec
    ↳ ord<number,string|string[]>>({}); const quizSubmitted=ref(false); const currentQuestion=ref(0);
    ↳ const quizCount=ref('3'); const quizCourseId=ref(String(route.query.course||'')); const quizTopi
    ↳ c=ref(''); const reviewMode=ref(false)
const reviewQuestionId=ref<number|null>(null); const quizStartedAt=ref(0)
const score=computed(()=>{const objective=quizQuestions.value.filter(x=>x.type!=='简答题');const
    ↳  correct=objective.filter(x=>{const a=quizAnswers.value[x.id];return Array.isArray(x.answer)?Arr
    ↳ ay.isArray(a)&&x.answer.length===a.length&&x.answer.every(v=>a.includes(v)):a===x.answer}).lengt
    ↳ h;return Math.round(correct/Math.max(1,objective.length)*100)})
function saveMistakes(){return studyApi.saveMistakes(mistakeList.value.map(item=>({question_id:i
    ↳ tem.questionId,payload:item as unknown as Record<string,unknown>})))}
function scheduleIn(days:number){const date=new Date();date.setDate(date.getDate()+days);return
    ↳ date.toISOString()}
const dueMistakes=computed(()=>mistakeList.value.filter(item=>!item.mastered&&new Date(item.next
    ↳ ReviewAt)<=new Date()).length)
const adaptiveMistake=computed(()=>mistakeList.value.filter(item=>!item.mastered).sort((a,b)=>Nu
    ↳ mber(new Date(a.nextReviewAt)>new Date())-Number(new Date(b.nextReviewAt)>new Date())||b.wrong-a
    ↳ .wrong)[0])
const weakMistakes=computed(()=>mistakeList.value.filter(item=>!item.mastered).sort((a,b)=>b.wro
    ↳ ng-a.wrong).slice(0,3))
type LearningPathStep={nodeId:string;name:string;type:string;mastery:number;wrongCount:number;du
    ↳ e:boolean;prerequisites:string[];state:'current'|'next';reason:string}
type PathViewStep=LearningPathStep&{id:string;tag:string;time:string;done:boolean}
const generatedPath=ref<LearningPathStep[]>([]);const pathProgress=ref(0);const pathLoading=ref(
    ↳ false);const pathError=ref('');const pathCustomized=ref(false);const pathDirty=ref(false)
async function loadPersonalizedPath(courseId=graphCourseId.value){if(!courseId){generatedPath.va
    ↳ lue=[];pathProgress.value=0;pathError.value='当前账号没有可用课程';return}graphCourseId.value=co
    ↳ urseId;pathLoading.value=true;pathError.value='';try{const result=await studyApi.learningPath(co
    ↳ urseId);generatedPath.value=result.steps;pathProgress.value=result.progress;pathCustomized.value
    ↳ =Boolean(result.customized);pathDirty.value=false}catch(error){pathError.value=error instanceof
    ↳ Error?error.message:'个性化学习路径加载失败';throw error}finally{pathLoading.value=false}}
const pathSteps=computed<PathViewStep[]>(()=>generatedPath.value.map((step,index)=>({ ...step,id
    ↳ :step.nodeId,tag:step.due?'到期复习':step.wrongCount?`错题 ${step.wrongCount} 次`:step.mastery<4
    ↳ 0?'重点巩固':'能力提升',time:`约 ${Math.max(15,preferences.dailyFocusHours*15)} 分钟`,done:step.
    ↳ mastery>=80,state:index===0?'current':'next' })))
function movePathStep(index:number,delta:number){const target=index+delta;if(target<0||target>=g
    ↳ eneratedPath.value.length)return;const reordered=[...generatedPath.value];const [step]=reordered
    ↳ .splice(index,1);reordered.splice(target,0,step);generatedPath.value=reordered;pathDirty.value=t
    ↳ rue}
async function savePathOrder(){try{await studyApi.saveLearningPath(graphCourseId.value,generated
    ↳ Path.value.map(step=>step.nodeId));pathCustomized.value=true;pathDirty.value=false;notify('自定
    ↳ 义学习顺序已保存到账户')}catch(error){notify(error instanceof Error?error.message:'学习路径保存
    ↳ 失败')}}
async function resetPathOrder(){try{await studyApi.saveLearningPath(graphCourseId.value,[]);path
    ↳ Customized.value=false;pathDirty.value=false;await loadPersonalizedPath();notify('已恢复系统推荐
    ↳ 顺序')}catch(error){notify(error instanceof Error?error.message:'恢复推荐顺序失败')}}
function rebuildPath(){void loadPersonalizedPath().then(()=>notify('已根据课程先修关系、掌握度和
    ↳ 错题重新规划')).catch(error=>notify(error instanceof Error?error.message:'学习路径更新失败'))}
async function addPlanTask(title:string,course=selectedCourse.value==='全部课程'?(courses.value[
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ 0]?.name||'学习空间'):selectedCourse.value){try{await store.addTask(title,course,`${selectedPlan
    ↳ Day.value} · 09:00`);notify('学习任务已保存到账户')}catch(error){notify(error instanceof Error?
    ↳ error.message:'任务保存失败，请重试')}}
async function togglePlanTask(id:number){try{await store.toggleTask(id)}catch(error){notify(erro
    ↳ r instanceof Error?error.message:'任务状态保存失败，请重试')}}
async function addCurrentToPlan(){const step=pathSteps.value.find(item=>!item.done);if(!step)ret
    ↳ urn;await addPlanTask(`学习「${step.name}」`,graphCourseName.value)}
async function startPathStep(step:LearningPathStep){localStorage.setItem('studypath.quizConceptI
    ↳ d',step.nodeId);localStorage.setItem('studypath.quizCourseId',graphCourseId.value);await router.
    ↳ push({path:'/quiz',query:{course:graphCourseId.value}})}
async function beginConceptPractice(nodeId:string,courseId:string){localStorage.setItem('studypa
    ↳ th.quizConceptId',nodeId);localStorage.setItem('studypath.quizCourseId',courseId);quizCourseId.v
    ↳ alue=courseId;await router.push({path:'/quiz',query:{course:courseId}})}
async function toggleCurrentMastery(){try{const mastery=selectedNode.value.mastery>=80?40:100;aw
    ↳ ait studyApi.saveMastery(selectedNode.value.id,mastery);selectedNode.value.mastery=mastery;const
    ↳  node=graphNodes.value.find(item=>item.id===selectedNode.value.id);if(node)node.mastery=mastery;
    ↳ const learnable=graphNodes.value.filter(item=>item.type==='concept'||item.type==='application');
    ↳ const progress=Math.round(learnable.filter(item=>item.mastery>=80).length/Math.max(1,learnable.l
    ↳ ength)*100);await studyApi.saveCourseProgress(graphCourseId.value,progress,selectedNode.value.id
    ↳ );notify(mastery>=80?'已记录为掌握':'已更新掌握度')}catch(error){notify(error instanceof Error?e
    ↳ rror.message:'掌握度保存失败')}}
function diagnosisFor(question:Question){const answer=quizAnswers.value[question.id];if(answer==
    ↳ =undefined||answer===''')return '未作答；建议先阅读知识卡，再尝试举一反三练习。';if(isCorrect(que
    ↳ stion))return '回答正确，继续通过间隔复测确认稳定掌握。';const choices=Array.isArray(answer)?ans
    ↳ wer:[answer];const distractor=choices.find(choice=>!(Array.isArray(question.answer)?question.ans
    ↳ wer.includes(choice):choice===question.answer));if(distractor)return question.misconceptions?.[d
    ↳ istractor]||'所选答案与目标概念不符，先核对知识卡再做举一反三练习。';return '你选对了部分内容，
    ↳ 但还遗漏了必要条件；对照正确选项和知识卡逐项检查。'}
const quizConceptFilter=ref('')
const quizConceptOptions=computed(()=>graphNodes.value.filter(node=>node.type==='concept'||node.
    ↳ type==='application'))
watch(quizCourseId,async(courseId)=>{quizConceptFilter.value='';if(courseId){graphCourseId.value
    ↳ =courseId;try{await loadGraph(courseId)}catch(error){notify(error instanceof Error?error.message
    ↳ :'课程知识点加载失败')}}})
async function beginQuiz(){
  let bank:Question[]
  try{bank=await studyApi.questions()}catch(error){notify(error instanceof Error?error.message:'
    ↳ 题库加载失败');return}
  const requested=Number(localStorage.getItem('studypath.reviewQuestionId')||'')
  const conceptId=localStorage.getItem('studypath.quizConceptId')||quizConceptFilter.value
  const courseId=localStorage.getItem('studypath.quizCourseId')||String(route.query.course||quiz
    ↳ CourseId.value)
  const hasReview=Number.isFinite(requested)&&requested>0
  let selected:Question[]=[]
  reviewMode.value=hasReview
  reviewQuestionId.value=hasReview?requested:null
  if(hasReview){
    const target=bank.find(question=>question.id===requested)
    const transfer=target?.transferQuestionIds?.map((id:number)=>bank.find(question=>question.id
    ↳ ===id)).find(item=>item&&item.courseId===target.courseId)
```

```
// 文件：src/views/WorkspaceView.vue
    selected=[target,transfer].filter((item):item is Question=>Boolean(item))
    if(target){quizCourseId.value=target.courseId||courseId;quizTopic.value=target.course||quizT
    ↳ opic.value}
  }else if(conceptId){
    quizConceptFilter.value=conceptId
    selected=bank.filter(question=>question.conceptId===conceptId&&(!courseId||question.courseId
    ↳ ===courseId)&&question.type!=='简答题').slice(0,Number(quizCount.value))
  }else{
    selected=bank.filter(question=>question.courseId===courseId&&question.type!=='简答题').slice
    ↳ (0,Number(quizCount.value))
  }
  localStorage.removeItem('studypath.reviewQuestionId')
  localStorage.removeItem('studypath.quizConceptId')
  localStorage.removeItem('studypath.quizCourseId')
  if(!selected.length){reviewMode.value=false;quizQuestions.value=[];notify('当前课程或知识点暂
    ↳ 时没有可用客观题。请更换课程或联系管理员补充题目。');return}
  quizCourseId.value=selected[0].courseId||courseId||courses.value[0]?.id||''
  quizTopic.value=selected[0].course||courses.value.find(course=>course.id===selected[0].courseI
    ↳ d)?.name||''
  quizQuestions.value=selected
  quizStarted.value=true;quizSubmitted.value=false;quizAnswers.value={};currentQuestion.value=0;
    ↳ quizStartedAt.value=Date.now()
}
function toggleAnswer(question:Question,option:string){if(question.type==='多选题'){const arr=Ar
    ↳ ray.isArray(quizAnswers.value[question.id])?[...quizAnswers.value[question.id] as string[]]:[];q
    ↳ uizAnswers.value[question.id]=arr.includes(option)?arr.filter(x=>x!==option):[...arr,option]}els
    ↳ e quizAnswers.value[question.id]=option}
async function submitQuiz(){let stored:any;try{stored=await studyApi.saveQuizAttempt({course_id:
    ↳ quizQuestions.value[0]?.courseId||quizCourseId.value,question_ids:quizQuestions.value.map(item=>
    ↳ item.id),answers:Object.fromEntries(Object.entries(quizAnswers.value).map(([id,value])=>[id,valu
    ↳ e])),duration_seconds:Math.max(0,Math.round((Date.now()-quizStartedAt.value)/1000))});for(const
    ↳ result of stored.results||[]){const question=quizQuestions.value.find(item=>item.id===result.que
    ↳ stionId);if(question){question.answer=result.answer;question.explanation=result.explanation;if(r
    ↳ esult.misconception&&typeof result.selectedAnswer==='string')question.misconceptions={[result.se
    ↳ lectedAnswer]:result.misconception}}}quizHistory.value.unshift({...stored,course:quizQuestions.v
    ↳ alue[0]?.course||quizTopic.value});quizSubmitted.value=true}catch(error){notify(error instanceof
    ↳  Error?error.message:'测验记录保存失败');return}const now=new Date();const concepts=new Map<stri
    ↳ ng,Question[]>();for(const question of quizQuestions.value){if(question.type==='简答题')continue
    ↳ ;const key=question.conceptId||String(question.id);concepts.set(key,[...(concepts.get(key)||[]),
    ↳ question])}for(const group of concepts.values()){const question=group.find(item=>!isCorrect(item
    ↳ ))||group[0];const prior=mistakeList.value.find(item=>item.questionId===question.id)||mistakeLis
    ↳ t.value.find(item=>group.some(candidate=>candidate.id===item.questionId));const selected=quizAns
    ↳ wers.value[question.id];const attempt:LearningAttempt={at:now.toISOString(),correct:isCorrect(qu
    ↳ estion),selectedAnswer:Array.isArray(selected)?selected.join('、'):selected||'未作答',feedback:d
    ↳ iagnosisFor(question)};if(reviewMode&&prior){const updated=await studyApi.reviewMistake(prior.qu
    ↳ estionId,{correct:attempt.correct,selected_answer:attempt.selectedAnswer,note:attempt.feedback})
    ↳ ;Object.assign(prior,updated);continue}if(!isCorrect(question)){if(prior){prior.wrong+=1;prior.c
    ↳ orrectStreak=0;prior.mastered=false;prior.misconception=attempt.feedback;prior.lastWrongAt=now.t
    ↳ oISOString();prior.nextReviewAt=scheduleIn(1);prior.evidence=question.evidence||prior.evidence;p
    ↳ rior.attempts.push(attempt)}else{mistakeList.value.unshift({id:question.id,questionId:question.i
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ d,conceptId:question.conceptId,title:question.title,topic:question.topic,course:question.course|
    ↳ |courses.value.find(course=>course.id===question.courseId)?.name||'学习空间',wrong:1,difficulty:
    ↳ question.difficulty,saved:false,misconception:attempt.feedback,lastWrongAt:now.toISOString(),nex
    ↳ tReviewAt:scheduleIn(1),correctStreak:0,lastCorrectAt:null,mastered:false,attempts:[attempt],evi
    ↳ dence:question.evidence})}}else if(prior&&!prior.mastered){prior.attempts.push(attempt);const la
    ↳ st=prior.lastCorrectAt?new Date(prior.lastCorrectAt):null;const separated=Boolean(last&&now.getT
    ↳ ime()-last.getTime()>=24*60*60*1000);if(separated){prior.correctStreak+=1;prior.lastCorrectAt=no
    ↳ w.toISOString();if(prior.correctStreak>=2){prior.mastered=true;prior.nextReviewAt=''}else prior.
    ↳ nextReviewAt=scheduleIn(3)}else{prior.lastCorrectAt=now.toISOString();prior.correctStreak=Math.m
    ↳ ax(1,prior.correctStreak);prior.nextReviewAt=scheduleIn(1)}}}await saveMistakes();notify('测验与
    ↳ 错题记录已保存到账户')}
const selectedAnswer=(question:Question,option:string)=>Array.isArray(quizAnswers.value[question
    ↳ .id])?quizAnswers.value[question.id].includes(option):quizAnswers.value[question.id]===option
function isCorrect(question:Question){const answer=quizAnswers.value[question.id];return Array.i
    ↳ sArray(question.answer)?Array.isArray(answer)&&question.answer.length===answer.length&&question.
    ↳ answer.every(x=>answer.includes(x)):answer===question.answer}
const filteredCourses=computed(()=>courses.value.filter(c=>c.name.toLowerCase().includes(q.value
    ↳ .toLowerCase())&&(courseCategoryFilter.value===''全部课程''||c.category===courseCategoryFilter.val
    ↳ ue)))
const courseDetail=ref<string|null>(null)
const courseDetailNodes=ref<KnowledgeNode[]>([]);const courseDetailLoading=ref(false);const cour
    ↳ seDetailError=ref('')
const courseCategories=computed(()=>Array.from(new Set(courses.value.map(course=>course.category
    ↳ ).filter(Boolean))))
async function openCourse(courseId:string){courseDetail.value=courseId;courseDetailNodes.value=[
    ↳ ];courseDetailError.value='';courseDetailLoading.value=true;try{const graph=await studyApi.graph
    ↳ (courseId);courseDetailNodes.value=graph.nodes}catch(error){courseDetailError.value=error instan
    ↳ ceof Error?error.message:'课程目录读取失败'}finally{courseDetailLoading.value=false}}
function openCourseNode(courseId:string,nodeId:string){courseDetail.value=null;void router.push(
    ↳ {path:'/graph',query:{course:courseId,node:nodeId}})}
function viewCourseGraph(courseId:string){courseDetail.value=null;void router.push({path:'/graph
    ↳ ',query:{course:courseId}})}
const mistakeDifficulty=ref('全部难度'); const sortMistakesDescending=ref(true)
const visibleMistakes=computed(()=>mistakeList.value.filter(item=>!item.mastered&&(selectedCours
    ↳ e.value===''全部课程''||item.course===selectedCourse.value)&&(mistakeDifficulty.value===''全部难度''
    ↳ ||item.difficulty===mistakeDifficulty.value)).sort((a,b)=>sortMistakesDescending.value?b.wrong-a
    ↳ .wrong:a.wrong-b.wrong))
function openMistakePractice(questionId:number){void router.push(`/mistake-practice/${questionId
    ↳ }`)}
async function toggleMistakeSaved(item:MistakeRecord){const next=!item.saved;try{await studyApi.
    ↳ updateMistake(item.questionId,{saved:next});item.saved=next}catch(error){notify(error instanceof
    ↳  Error?error.message:'收藏状态保存失败')}}
const goal=ref(preferences.weeklyGoalHours); const daily=ref(preferences.dailyFocusHours); const
    ↳  notifications=ref(preferences.learningReminders); const displayName=computed(()=>profileState.n
    ↳ ame); const profileDraft=reactive<StudyProfile>({...profileState})
watch([goal,daily,notifications],([weekly,duration,reminders])=>{preferences.weeklyGoalHours=wee
    ↳ kly;preferences.dailyFocusHours=duration;preferences.learningReminders=reminders;savePreferences
    ↳ ()})
async function saveProfileChanges(){if(!profileDraft.name.trim()){notify('姓名不能为空');return}
    ↳ try{await saveProfile(profileDraft);notify('个人资料已保存到账户')}catch(error){notify(error ins
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ tanceof Error?error.message:'资料保存失败')}}
type QuizRecord={at:string;score:number;questionCount:number;course:string}
const quizHistory=ref<QuizRecord[]>([])
const analyticsRange=ref(7)
const rangeRecords=computed(()=>quizHistory.value.filter(record=>Date.now()-new Date(record.at).
    ↳ getTime()<analyticsRange.value*86400000))
const averageScore=computed(()=>rangeRecords.value.length?Math.round(rangeRecords.value.reduce((
    ↳ sum,item)=>sum+item.score,0)/rangeRecords.value.length):null)
const analyticsPeriod=computed(()=>{const end=new Date();const start=new Date();start.setDate(en
    ↳ d.getDate()-analyticsRange.value+1);return `${start.toLocaleDateString('zh-CN',{month:'long',day
    ↳ :'numeric',timeZone:'Asia/Shanghai'})} — ${end.toLocaleDateString('zh-CN',{month:'long',day:'nu
    ↳ meric',timeZone:'Asia/Shanghai'})}`})
const trendOption=computed(()=>{const days=Array.from({length:analyticsRange.value},(_,index)=>{
    ↳ const date=new Date();date.setHours(0,0,0,0);date.setDate(date.getDate()-(analyticsRange.value-1
    ↳ -index));return date});const labels=days.map(date=>`${date.getMonth()+1}/${date.getDate()}`);con
    ↳ st values=days.map(date=>{const key=date.toDateString();const attempts=rangeRecords.value.filter
    ↳ (item=>new Date(item.at).toDateString()===key);return attempts.length?Math.round(attempts.reduce
    ↳ ((sum,item)=>sum+item.score,0)/attempts.length):null});return {tooltip:{trigger:'axis'},grid:{le
    ↳ ft:8,right:16,top:16,bottom:24,containLabel:true},xAxis:{type:'category',boundaryGap:false,data:
    ↳ labels,axisLine:{show:false},axisTick:{show:false},axisLabel:{color:'#8b8998'}},yAxis:{type:'val
    ↳ ue',min:0,max:100,splitLine:{lineStyle:{color:'#efedf3',type:'dashed'}},axisLabel:{color:'#aaa8b
    ↳ 5',formatter:'{value}%'}},series:[{type:'line',smooth:true,data:values,lineStyle:{color:'#7958df
    ↳ ',width:3},areaStyle:{color:'rgba(121,88,223,.12)'},symbolSize:7,itemStyle:{color:'#7958df'}}]}
    ↳ as EChartsOption})
const distributionOption=computed(()=>({tooltip:{trigger:'item'},legend:{bottom:0,textStyle:{col
    ↳ or:'#777486'}},series:[{type:'pie',radius:['48%','72%'],data:[{name:'已掌握错题',value:mistakeLi
    ↳ st.value.filter(item=>item.mastered).length,itemStyle:{color:'#7958df'}},{name:'待复测错题',valu
    ↳ e:mistakeList.value.filter(item=>!item.mastered&&new Date(item.nextReviewAt)>new Date()).length,
    ↳ itemStyle:{color:'#b7a7ed'}},{name:'已到期错题',value:dueMistakes.value,itemStyle:{color:'#e8e5e
    ↳ f'}}],label:{show:false}}]} as EChartsOption))
const todayTasks=computed(()=>store.tasks.filter(task=>taskDate(task.date)===selectedPlanDay.val
    ↳ ue))
const nextPlanDay=computed(()=>{const date=new Date(`${selectedPlanDay.value}T12:00:00`);date.se
    ↳ tDate(date.getDate()+1);return date.toISOString().slice(0,10)})
const nextDayTasks=computed(()=>store.tasks.filter(task=>taskDate(task.date)===nextPlanDay.value
    ↳ ))
const calendarDays=computed(()=>Array.from({length:7},(_,index)=>{const date=new Date(`${selecte
    ↳ dPlanDay.value}T12:00:00`);date.setDate(date.getDate()+index);const value=date.toISOString().sli
    ↳ ce(0,10);return {value,label:new Intl.DateTimeFormat('zh-CN',{weekday:'short',day:'numeric'}).fo
    ↳ rmat(date),count:store.tasks.filter(task=>taskDate(task.date)===value).length}}))
const selectedPlanDay=ref(new Date().toISOString().slice(0,10)); const planView=ref<'list'|'cale
    ↳ ndar'>('list')
const planDate=computed(()=>new Intl.DateTimeFormat('zh-CN',{year:'numeric',month:'long',day:'nu
    ↳ meric',weekday:'long',timeZone:'Asia/Shanghai'}).format(new Date(`${selectedPlanDay.value}T12:00
    ↳ :00`)))
const tomorrowDate=computed(()=>new Intl.DateTimeFormat('zh-CN',{month:'long',day:'numeric',week
    ↳ day:'short',timeZone:'Asia/Shanghai'}).format(new Date(Date.now()+86400000)))
function taskDate(value:string){if(value.includes('明天'))return new Date(Date.now()+86400000).t
    ↳ oLocaleDateString('sv-SE');if(value.includes('今天'))return new Date().toLocaleDateString('sv-SE
    ↳ ');const prefix=value.match(/^\d{4}-\d{2}-\d{2}/)?.[0];if(prefix)return prefix;const parsed=new
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ Date(value);return Number.isNaN(parsed.getTime())?new Date().toLocaleDateString('sv-SE'):parsed.
    ↳ toLocaleDateString('sv-SE')}
function shiftPlanDate(days:number){const date=new Date(`${selectedPlanDay.value}T12:00:00`);dat
    ↳ e.setDate(date.getDate()+days);selectedPlanDay.value=date.toISOString().slice(0,10)}
function goToday(){selectedPlanDay.value=new Date().toISOString().slice(0,10)}
function scrollSettings(id:string){document.getElementById(id)?.scrollIntoView({behavior:'smooth
    ↳ ',block:'start'})}
function exportAnalytics(){const payload={generatedAt:new Date().toISOString(),rangeDays:analyti
    ↳ csRange.value,quizAttempts:rangeRecords.value,tasks:store.tasks,mistakes:mistakeList.value};cons
    ↳ t blob=new Blob([JSON.stringify(payload,null,2)],{type:'application/json'});const url=URL.create
    ↳ ObjectURL(blob);const link=document.createElement('a');link.href=url;link.download=`studypath-re
    ↳ port-${selectedPlanDay.value}.json`;link.click();URL.revokeObjectURL(url)}
async function editTask(task: {id:number;title:string}){const title=window.prompt('编辑学习任务'
    ↳ ,task.title);if(title?.trim()){const current=store.tasks.find(item=>item.id===task.id);if(curren
    ↳ t){const previous=current.title;current.title=title.trim();try{await store.persist();notify('任
    ↳ 务已保存到账户')}catch(error){current.title=previous;notify(error instanceof Error?error.message
    ↳ :'任务保存失败，请重试')}}}}
</script>
<template>
  <div class="page-content workspace-page">
    <section class="page-intro"><div><div class="page-context"><span class="context-dot"></span>
    ↳ 智学研途 <span>/</span> {{ pageTitle }}</div><h1>{{ section==='graph'?'课程知识图谱':section==='
    ↳ path'?'你的专属学习路径':section==='chat'?'AI 智能答疑':section==='analytics'?'学习数据分析':pag
    ↳ eTitle }}</h1><p>{{ section==='path'?'根据当前课程图谱、个人掌握度与错题复习状态生成学习顺序。':
    ↳ section==='graph'?'查看课程知识点与先修关系；节点不代表个人掌握程度。':section==='chat'?'结合课
    ↳ 程资料与可追溯证据，让每个问题都有依据。':section==='library'?'集中整理课程资料，为检索问答与复
    ↳ 习建立知识底座。':section==='quiz'?'用一组短测验检验理解，把薄弱知识点及时找出来。':section==='c
    ↳ ourses'?'课程进度与知识点目录来自当前账号和课程图谱。':'把学习记录转化为清晰的下一步。' }}</p></
    ↳ div><div class="intro-actions"><span v-if="section==='chat'||section==='library'" class="demo-ta
    ↳ g"><Sparkles :size="12" /> 本地演示</span><button v-if="section==='chat'" class="btn btn-primary
    ↳ " @click="newChat"><Plus :size="15" />新建对话</button><button v-if="section==='path'" class="bt
    ↳ n btn-primary" @click="rebuildPath"><RotateCcw :size="15" />重新生成路径</button><button v-if="s
    ↳ ection==='path'" class="btn btn-secondary" @click="addCurrentToPlan"><Plus :size="15" />加入学习
    ↳ 计划</button><button v-if="section==='quiz'&&!quizStarted" class="btn btn-primary" @click="begin
    ↳ Quiz"><Sparkles :size="15" />开始测验</button><button v-if="section==='library'" class="btn btn-
    ↳ primary" @click="fileInput?.click()"><Upload :size="15" />上传资料</button><input ref="fileInput
    ↳ " type="file" hidden multiple accept=".pdf,.docx,.txt,.md" @change="uploadFiles(($event.target a
    ↳ s HTMLInputElement).files)" /></div></section>
    <!-- AI learning chat -->
    <section v-if="section==='chat'" class="chat-layout panel">
      <aside class="chat-sidebar"><div class="chat-side-header"><div class="chat-side-title">对
    ↳ 话记录 <span>{{ chats.length }}</span></div><button class="icon-button" title="新建对话" @click=
    ↳ "newChat"><Plus :size="16" /></button></div><label class="field-search"><Search :size="15" /><in
    ↳ put v-model="q" placeholder="搜索历史对话" /></label><select v-model="selectedCourse" class="cha
    ↳ t-course-select"><option>全部课程</option><option v-for="c in courses" :key="c.id">{{ c.name }}<
    ↳ /option></select><div class="chat-history-label">最近对话</div><div class="chat-history-list"><b
    ↳ utton v-for="chat in chats.filter(x=>x.title.includes(q))" :key="chat.id" class="chat-history-it
    ↳ em" :class="{current:activeChat===chat.id}" @click="activeChat=chat.id"><MessageSquare :size="15
    ↳ " /><span><b>{{ chat.title }}</b><small>{{ chat.course }} · {{ chat.time }}</small></span><More
    ↳ Horizontal :size="15" /></button><div v-if="!chats.filter(x=>x.title.includes(q)).length" class=
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ "empty-small">没有找到相关对话</div></div><div class="chat-side-foot"><span class="chat-status">
    ↳ <i></i>演示问答已就绪</span><small>回答基于本地 Mock 数据</small></div></aside>
      <div class="chat-main">
        <div v-if="messages.length" ref="chatScroll" class="chat-thread">
          <div class="thread-date">今天 · {{ chats.find(x => x.id === activeChat)?.course || '
    ↳ 课程问答' }}</div>
          <article v-for="(message, i) in messages" :key="i" class="message" :class="message.rol
    ↳ e">
            <div v-if="message.role === 'assistant'" class="assistant-avatar"><Sparkles :size="1
    ↳ 5" /></div>
            <div class="message-body">
              <div class="message-author">{{ message.role === 'user' ? '我' : '智学研途学习助手'
    ↳  }} <span v-if="message.role === 'assistant'" class="mock-label">MOCK</span></div>
              <div v-if="message.loading" class="typing"><i></i><i></i><i></i><span>正在整理课程
    ↳ 资料与答案…</span></div>
              <div v-else class="markdown-body" v-html="markdown(message.text)"></div>
              <div v-if="message.role === 'assistant' && !message.loading" class="message-tools"
    ↳ >
                <button @click="copyAnswer(message.text)"><CopyIcon :size="13" />复制</button>
                <button :class="{positive: message.feedback === 'up'}" @click="message.feedback
    ↳ = message.feedback === 'up' ? '' : 'up'; notify('感谢你的反馈')"><ThumbsUp :size="13" />有帮助</
    ↳ button>
                <button :class="{positive: message.feedback === 'down'}" @click="message.feedbac
    ↳ k = message.feedback === 'down' ? '' : 'down'; notify('已记录反馈')"><ThumbsDown :size="13" />反
    ↳ 馈</button>
              </div>
              <div v-if="message.sources?.length" class="evidence-box">
                <button class="evidence-toggle" @click="sourcesOpen = !sourcesOpen"><span><BookO
    ↳ pen :size="15" />参考来源 <b>{{ message.sources.length }}</b></span><ChevronDown :size="15" :cla
    ↳ ss="{rotate: !sourcesOpen}" /></button>
                <div v-if="sourcesOpen" class="source-list">
                  <button v-for="source in message.sources" :key="source.document" class="source
    ↳ -card" @click="sourceSelected = source">
                    <span class="source-file"><FileText :size="15" /></span>
                    <span class="source-copy"><b>{{ source.document }}</b><small>{{ source.secti
    ↳ on }}</small><span>“{{ source.excerpt }}”</span><em><Check :size="11" /> {{ source.score }} ·
    ↳  本地模拟匹配</em></span>
                    <ArrowUpRight :size="14" />
                  </button>
                </div>
                <small class="evidence-note"><Info :size="12" />匹配分数为演示数据，非真实检索结
    ↳ 果。</small>
              </div>
              <div v-else-if="message.role === 'assistant' && !message.loading" class="no-eviden
    ↳ ce"><Info :size="14" />未检索到足够可靠的课程资料，回答仅供参考。</div>
            </div>
          </article>
          <div v-if="!chatBusy" class="followup-row"><span>你还可以问</span><button @click="send
    ↳ Question('能举一个具体的例子吗？')">能举个例子吗？ <ArrowUpRight :size="12" /></button><button @
    ↳ click="sendQuestion('请用更简单的话解释一遍')">换个方式解释 <ArrowUpRight :size="12" /></button>
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ </div>
        </div>
        <div v-else class="chat-welcome">
          <div class="welcome-spark"><Sparkles :size="20" /></div>
          <h2>你好，{{ displayName }}<br /><span>让我们一起把问题弄明白。</span></h2>
          <p>你好，我是智学研途学习助手。你可以向我提问课程概念、算法原理，或者上传学习资料后进
    ↳ 行针对性问答。</p>
          <div class="chat-course-picker"><span>参考课程资料</span><select v-model="selectedCour
    ↳ se"><option>全部课程</option><option v-for="c in courses" :key="c.id">{{ c.name }}</option></sel
    ↳ ect><span class="mock-label">本地 Mock</span></div>
          <div class="prompt-grid"><button v-for="(prompt, i) in promptExamples" :key="prompt" @
    ↳ click="sendQuestion(prompt)"><span class="prompt-number">0{{ i + 1 }}</span>{{ prompt }}<ArrowUp
    ↳ Right :size="14" /></button></div>
          <div class="mock-notice"><Info :size="14" />当前为本地演示回复，尚未连接真实大模型或向
    ↳ 量检索服务。</div>
        </div>
        <div class="chat-composer">
          <div class="composer-tools"><button @click="notify('资料选择功能请前往学习资料库上传')
    ↳ "><Plus :size="15" />添加资料</button><span><i></i>{{ selectedCourse === '全部课程' ? '所有课程'
    ↳  : '检索 ' + selectedCourse }}</span></div>
          <div class="composer-input"><textarea v-model="chatInput" rows="2" placeholder="输入你
    ↳ 的问题…（Enter 发送，Shift + Enter 换行）" @keydown.enter.exact.prevent="sendQuestion()"></text
    ↳ area><button class="send-button" :disabled="!chatInput.trim() || chatBusy" aria-label="发送" @cl
    ↳ ick="sendQuestion()"><Send :size="16" /></button></div>
          <div class="composer-foot"><span>AI 生成内容请结合课程资料核对</span><span>⌘ ↵ 发送</s
    ↳ pan></div>
        </div>
      </div>    </section>
    <!-- Interactive course graph -->
    <section v-else-if="section==='graph'" class="graph-page panel" :class="{'graph-full':graphF
    ↳ ull}"><div class="graph-toolbar"><div class="graph-controls"><select v-model="graphCourseId" ari
    ↳ a-label="选择课程" @change="loadGraph(graphCourseId)"><option v-for="course in courses" :key="co
    ↳ urse.id" :value="course.id">{{course.name}}</option></select><label class="field-search"><Search
    ↳  :size="14" /><input v-model="q" placeholder="搜索知识点" @keydown.enter="focusNode(graphNodes.f
    ↳ ind(x=>x.name.includes(q))?.id||graphCourseId)" /><button v-if="q" @click="focusNode(graphNodes.
    ↳ find(x=>x.name.includes(q))?.id||graphCourseId)"><ArrowRight :size="14" /></button></label><butt
    ↳ on class="btn btn-soft" @click="selectedTypes=['course','chapter','concept','application'];selec
    ↳ tedNode=graphNodes[0];drawGraph()"><RotateCcw :size="14" />重置布局</button></div><div class="gr
    ↳ aph-actions"><button class="icon-button" title="全屏" @click="graphFull=!graphFull"><X v-if="gra
    ↳ phFull" :size="17"/><Maximize2 v-else :size="17" /></button><button class="btn btn-primary btn-c
    ↳ ompact" @click="router.push('/path')"><Sparkles :size="14"/>生成学习路径</button></div></div><di
    ↳ v class="graph-body"><div class="graph-legend-panel"><div class="graph-summary"><span>{{graphCou
    ↳ rseName}}</span><strong>{{graphNodes.length}}</strong><small>个知识点</small></div><label class=
    ↳ "field-search"><Search :size="14"/><input v-model="q" placeholder="查找知识点" /></label><div cl
    ↳ ass="legend-title">节点类型</div><button v-for="(meta,type) in typeMeta" :key="type" class="lege
    ↳ nd-item" @click="selectedTypes.includes(type)?selectedTypes=selectedTypes.filter(x=>x!==type):se
    ↳ lectedTypes.push(type);drawGraph()"><i :style="{background:meta.color}"/><span>{{ meta.label }}<
    ↳ /span><Check v-if="selectedTypes.includes(type)" :size="14" /></button><div class="legend-divide
    ↳ r"></div><div class="graph-data-note">节点与先修关系来自服务端课程目录；掌握度按当前账号记录显示
    ↳ 。</div><div class="graph-hint"><Network :size="15"/><span>拖动节点调整布局<br/>滚轮缩放 · 点击
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ 查看详情</span></div></div><div class="graph-canvas"><div ref="graphHost" class="graph-echart"><
    ↳ /div><div class="graph-canvas-caption"><span><i class="canvas-pulse"></i>当前展示 <b>{{graphNode
    ↳ s.filter(x=>selectedTypes.includes(x.type)).length}}</b> 个知识点</span><span>箭头表示知识依赖关
    ↳ 系</span></div><div class="graph-zoom"><button title="缩小" @click="graphChart?.dispatchAction({
    ↳ type:'graphRoam',seriesIndex:0,zoom:0.85})"><ZoomOut :size="15"/></button><span>可拖动探索</span
    ↳ ><button title="放大" @click="graphChart?.dispatchAction({type:'graphRoam',seriesIndex:0,zoom:1.
    ↳ 15})"><ZoomIn :size="15"/></button></div></div></div><Transition name="drawer"><aside v-if="draw
    ↳ er" class="node-drawer"><div class="drawer-top"><span class="node-type-pill" :style="{'--node-co
    ↳ lor':typeMeta[selectedNode.type].color}"><i></i>{{ typeMeta[selectedNode.type].label }}</span><b
    ↳ utton class="icon-button" aria-label="关闭详情" @click="drawer=false"><X :size="17"/></button></
    ↳ div><div class="node-heading"><h2>{{selectedNode.name}}</h2><span>{{graphCourseName}} · {{selec
    ↳ tedNode.type==='chapter'?'章节':'知识点'}}</span></div><div class="mastery-editor"><span>个人掌
    ↳ 握度 <b>{{selectedNode.mastery}}%</b></span><button class="btn btn-soft" @click="toggleCurrentMa
    ↳ stery">{{selectedNode.mastery>=80?'调整为待巩固':'标记已掌握'}}</button></div><div class="drawer
    ↳ -section"><h4>知识点解释</h4><p>{{selectedNode.description}}</p></div><div class="drawer-section
    ↳ "><h4>前置知识 <span>{{prerequisites.length}}</span></h4><div class="chip-list"><button v-for="i
    ↳ d in prerequisites" :key="id" @click="focusNode(id)">{{graphNodes.find(x=>x.id===id)?.name}} <Ar
    ↳ rowUpRight :size="11"/></button><small v-if="!prerequisites.length">这是当前课程的根节点</small>
    ↳ </div></div><div class="drawer-section"><h4>相关知识点 <span>{{related.length}}</span></h4><div
    ↳ class="related-list"><button v-for="item in related.slice(0,4)" :key="item?.id" @click="item&&fo
    ↳ cusNode(item.id)"><span class="related-dot" :style="{background:typeMeta[item?.type||'concept'].
    ↳ color}"/>{{item?.name}}<ArrowRight :size="13"/></button></div></div><div class="drawer-section">
    ↳ <h4>学习资料</h4><p>资料上传并完成索引后，真实来源会显示在 AI 答疑引用中。</p><button class="tex
    ↳ t-link" @click="router.push('/library')">前往资料库 <ArrowUpRight :size="13"/></button></div><di
    ↳ v class="drawer-actions"><button class="btn btn-primary" @click="router.push('/chat');chatInput=
    ↳ `请讲解${selectedNode.name}`"><Sparkles :size="14"/>AI 讲解</button><button class="btn btn-secon
    ↳ dary" @click="addCurrentToPlan"><Plus :size="14"/>加入计划</button><button class="btn btn-soft"
    ↳ @click="beginConceptPractice(selectedNode.id,graphCourseId)"><Target :size="14"/>练习</button></
    ↳ div></aside></Transition></section>
    <!-- Courses -->
    <template v-else-if="section==='courses'"><div class="content-toolbar"><label class="field-s
    ↳ earch wide"><Search :size="15"/><input v-model="q" placeholder="搜索课程名称" /></label><select
    ↳ v-model="courseCategoryFilter" class="filter-select"><option>全部课程</option><option v-for="cat
    ↳ egory in courseCategories" :key="category">{{category}}</option></select><button class="btn btn-
    ↳ soft" @click="router.push('/admin/curriculum')"><Settings2 :size="14"/>课程内容维护</button></di
    ↳ v><div class="course-grid"><article v-for="course in filteredCourses" :key="course.id" class="co
    ↳ urse-card panel"><div class="course-card-head"><span class="course-icon-large" :style="{'--cours
    ↳ e-color':course.color}">{{course.icon}}</span><button class="icon-button" title="课程选项" @clic
    ↳ k="openCourse(course.id)"><MoreHorizontal :size="17"/></button></div><span class="course-categor
    ↳ y">{{course.category}}</span><h3>{{course.name}}</h3><p>当前章节：{{course.chapter}}</p><div cla
    ↳ ss="course-progress-label"><span>{{course.points}} 个目录知识点</span><b>已学 {{course.progress}
    ↳ }%</b></div><button class="course-continue" @click="openCourse(course.id)"><span>查看课程目录 <A
    ↳ rrowRight :size="14"/></span><small>个人进度 {{course.progress}}%</small></button></article><div
    ↳  v-if="!filteredCourses.length" class="empty-state panel"><BookOpen :size="22"/><b>没有找到这门
    ↳ 课程</b><span>换个关键词试试</span></div></div><div v-if="courseDetail" class="overlay" @click.s
    ↳ elf="courseDetail=null"><div class="modal-card"><button class="icon-button modal-close" @click="
    ↳ courseDetail=null"><X :size="17"/></button><div class="modal-icon" :style="{'--course-color':cou
    ↳ rses.find(c=>c.id===courseDetail)?.color}">{{courses.find(c=>c.id===courseDetail)?.icon}}</div><
    ↳ h2>{{courses.find(c=>c.id===courseDetail)?.name}}</h2><p>{{courses.find(c=>c.id===courseDetail)?
    ↳ .points}} 个目录知识点 · 当前进度 {{courses.find(c=>c.id===courseDetail)?.progress||0}}%</p><h4
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ >课程知识点</h4><div v-if="courseDetailLoading" class="empty-state">正在读取课程图谱…</div><div
    ↳  v-else-if="courseDetailError" class="empty-state" role="alert">{{courseDetailError}}</div><div
    ↳ v-else-if="!courseDetailNodes.length" class="empty-state">这门课程暂时没有知识点，请先在课程内容
    ↳ 维护中添加。</div><button v-else v-for="(node,i) in courseDetailNodes" :key="node.id" class="cha
    ↳ pter-row" @click="openCourseNode(courseDetail!,node.id)"><span class="chapter-index">{{String(i+
    ↳ 1).padStart(2,'0')}}</span><span>{{node.name}}</span><span>{{node.type}} · 掌握 {{node.mastery}
    ↳ }%</span><ChevronRight :size="14"/></button><button class="btn btn-primary modal-primary" @click
    ↳ ="courseDetail=null;viewCourseGraph(courseDetail!)">打开课程知识图谱 <ArrowRight :size="14"/></b
    ↳ utton></div></div></template>
    <!-- Materials library -->
    <template v-else-if="section==='library'"><div class="library-notice"><Info :size="16"/><spa
    ↳ n><b>本地演示模式：</b>上传文件仅进入浏览器演示队列，尚未真实解析内容或写入向量数据库。</span><b
    ↳ utton @click="notify('正式环境可对接 POST /materials/upload 接口')">API 接入说明 <ArrowUpRight :
    ↳ size="12"/></button></div><div class="library-stats"><div><span>全部资料</span><b>{{materials.le
    ↳ ngth}}</b></div><div><span>已建立索引</span><b>{{materials.filter(m=>m.status==='已索引').length
    ↳ }}</b></div><div><span>待处理</span><b>{{materials.filter(m=>m.status==='待处理'||m.status==='处
    ↳ 理中').length}}</b></div><div><span>资料空间</span><b>6.8 <small>GB / 10 GB</small></b></div></d
    ↳ iv><div class="panel library-panel"><div class="library-toolbar"><label class="field-search wide
    ↳ "><Search :size="15"/><input v-model="q" placeholder="搜索文件名" /></label><select v-model="sel
    ↳ ectedCourse" class="filter-select"><option>全部课程</option><option v-for="c in courses" :key="c
    ↳ .id">{{c.name}}</option></select><select v-model="statusFilter" class="filter-select"><option>全
    ↳ 部状态</option><option>待处理</option><option>处理中</option><option>已索引</option><option>失败
    ↳ </option></select><button class="btn btn-secondary" @click="fileInput?.click()"><Upload :size="1
    ↳ 4"/>选择文件</button></div><div class="material-table"><div class="material-head"><span>文件名称
    ↳ </span><span>所属课程</span><span>文件大小</span><span>状态</span><span>更新时间</span><span></s
    ↳ pan></div><div v-for="mat in visibleMaterials" :key="mat.id" class="material-row"><span class="m
    ↳ aterial-name"><i class="file-type" :class="'file-'+mat.type.toLowerCase()">{{mat.type==='DOCX'?'
    ↳ W':mat.type==='PDF'?'P':mat.type==='MD'?'M':'T'}}</i><button @click="materialDetail=mat">{{mat.n
    ↳ ame}}</button></span><span>{{mat.course}}</span><span>{{mat.size}}</span><span><i class="status-
    ↳ chip" :class="'status-'+mat.status">{{mat.status}}</i></span><span>{{mat.date}}</span><span clas
    ↳ s="material-actions"><button title="预览详情" @click="materialDetail=mat"><ArrowUpRight :size="1
    ↳ 4"/></button><button title="删除资料" @click="deleteMaterial(mat.id)"><Trash2 :size="14"/></butt
    ↳ on></span></div><div v-if="!visibleMaterials.length" class="empty-state"><FolderOpen :size="23"/
    ↳ ><b>还没有匹配的资料</b><span>上传讲义或笔记，让课程知识更有迹可循。</span></div></div><div clas
    ↳ s="upload-drop" @click="fileInput?.click()" @dragover.prevent @drop.prevent="uploadFiles($event.
    ↳ dataTransfer?.files||null)"><span><Upload :size="18"/></span><b>拖放文件到此处，或 <u>浏览文件</
    ↳ u></b><small>PDF、DOCX、TXT、Markdown · 单个文件最大 20 MB</small></div></div><div v-if="materi
    ↳ alDetail" class="overlay" @click.self="materialDetail=null"><div class="modal-card material-moda
    ↳ l"><button class="icon-button modal-close" @click="materialDetail=null"><X :size="17"/></button>
    ↳ <span class="file-type file-pdf large-file">{{materialDetail.type.slice(0,1)}}</span><h2>{{mater
    ↳ ialDetail.name}}</h2><p>{{materialDetail.size}} · {{materialDetail.course}} · {{materialDetail
    ↳ .date}}</p><div class="preview-placeholder"><FileText :size="38"/><b>资料预览</b><span>本地演示
    ↳ 模式暂不解析文档正文。</span><small>上传接口对接后可展示原文与检索定位。</small></div><button cl
    ↳ ass="btn btn-primary modal-primary" @click="materialDetail=null">完成</button></div></div></temp
    ↳ late>
    <!-- Personalized path -->
    <section v-else-if="section==='path'" class="path-layout"><div class="path-main panel"><div
    ↳ class="path-topline"><span><Sparkles :size="14"/> {{ graphCourseName }} · 个性化学习路径</span>
    ↳ <div class="path-edit-actions"><button v-if="pathCustomized||pathDirty" class="btn btn-secondary
    ↳ " :disabled="pathLoading" @click="resetPathOrder">恢复推荐顺序</button><button v-if="pathDirty"
```

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// 文件：src/views/WorkspaceView.vue
    class="btn btn-primary" @click="savePathOrder">保存自定义顺序</button><button class="btn btn-sof
    t" :disabled="pathLoading" @click="rebuildPath"><RotateCcw :size="14"/>{{pathLoading?'正在规划':
    '重新规划'}}</button></div></div><div class="path-progress"><div><span>知识点掌握进度</span><b>{
    {pathProgress}}%</b></div><div class="progress-line"><i :style="{width:pathProgress+'%'}"/></div
    ></div><div v-if="pathLoading" class="adaptive-review-empty">正在结合课程先修关系、个人掌握度与
    错题记录生成学习顺序…</div><div v-else-if="pathError" class="adaptive-review-empty" role="alert
    ">{{pathError}} <button class="text-link" @click="rebuildPath">重试</button></div><div v-else-if
    ="!pathSteps.length" class="adaptive-review-empty"><CheckCheck :size="16"/>这门课程的知识点已达
    到当前掌握标准，继续练习可以巩固长期记忆。</div><div v-else class="timeline"><div v-for="(step,i
    ) in pathSteps" :key="step.nodeId" class="timeline-step" :class="'step-'+step.state"><div class=
    "timeline-rail"><span class="timeline-node"><Check v-if="step.done" :size="13"/><span v-else>{{S
    tring(i+1).padStart(2,'0')}}</span></span><i v-if="i<pathSteps.length-1"/></div><article class="
    step-card"><div class="step-head"><span class="step-tag" :class="step.due?'tag-current':step.wro
    ngCount?'tag-current':'tag-next'">{{step.tag}}</span><span class="step-time"><Clock3 :size="13"/
    >{{step.time}}</span><span v-if="step.state==='current'" class="now-tag">建议先学</span><span cl
    ass="path-order-controls"><button type="button" class="path-order-button" :disabled="i===0" :ari
    a-label="`将${step.name}上移`" title="上移" @click="movePathStep(i,-1)"><ChevronUp :size="13"/><
    /button><button type="button" class="path-order-button" :disabled="i===pathSteps.length-1" :aria
    -label="`将${step.name}下移`" title="下移" @click="movePathStep(i,1)"><ChevronDown :size="13"/><
    /button></span></div><h3>{{step.name}}</h3><p>{{step.reason}}</p><div class="step-mastery"><span
    >个人掌握度</span><b>{{step.mastery}}%</b></div><div class="progress-line"><i :style="{width:ste
    p.mastery+'%'}"/></div><div v-if="step.prerequisites.length" class="step-prereq"><span>前置知识
    : </span><b>{{step.prerequisites.map(id=>graphNodes.find(node=>node.id===id)?.name||id).join('、
    ')}}</b></div><button v-if="!step.done" class="step-cta" @click="startPathStep(step)">针对该知识
    点练习 <ArrowRight :size="13"/></button></article></div></div></div><aside class="path-aside"><a
    rticle class="panel goal-panel"><div class="side-panel-title"><Target :size="16"/><h3>本轮学习目
    标</h3></div><b class="goal-name">{{pathSteps.find(step=>!step.done)?.name||'巩固已掌握内容'}}</
    b><p>{{graphCourseName}}</p><div class="goal-ring"><div><strong>{{pathProgress}}%</strong><small
    >课程掌握</small></div></div><div class="goal-metrics"><span><b>{{pathSteps.length}}</b><small>
    推荐步骤</small></span><span><b>{{pathSteps.filter(step=>step.due).length}}</b><small>到期复习</
    small></span><span><b>{{pathSteps.filter(step=>step.wrongCount>0).length}}</b><small>错题关联</s
    mall></span></div></article><article class="panel weak-panel"><div class="side-panel-title"><Lig
    htbulb :size="16"/><h3>推荐依据</h3></div><p class="path-reason">优先处理已到期错题和重复错误，
    再按知识图谱的先修关系补齐前置知识。掌握度达到 80% 的节点会从本轮路线中移除。</p><button class="
    text-link" @click="router.push({path:'/graph',query:{course:graphCourseId}})">查看课程知识图谱 <
    ArrowRight :size="13"/></button></article><article class="path-tip"><Sparkles :size="16"/><div><
    b>个性化排序</b><p>路线读取当前账号掌握度与错题复习状态，并遵循课程知识图谱中的先修关系。</p><sm
    all>每次进入或重新规划时更新</small></div></article></aside></section>
    <!-- Quiz -->
    <section v-else-if="section==='quiz'" class="quiz-layout"><aside class="panel quiz-setup"><d
    iv class="quiz-setup-icon"><Target :size="20"/></div><h3>测验设置</h3><p>答错后会生成错因提示与
    间隔复测</p><label>当前课程<select v-model="quizCourseId"><option v-for="course in courses" :key
    ="course.id" :value="course.id">{{course.name}}</option></select></label><label>选择知识点（可选
    ）<select v-model="quizConceptFilter"><option value="">整门课程随机练习</option><option v-for="n
    ode in quizConceptOptions" :key="node.id" :value="node.id">{{node.name}}</option></select></labe
    l><label>题目数量<select v-model="quizCount"><option value="3">3 题</option><option value="5">5
    题</option></select></label><button class="btn btn-primary full-width" @click="beginQuiz"><Spark
    les :size="15"/>{{quizStarted&&!quizSubmitted?'重新开始':'开始测验'}}</button><small class="quiz
    -mock-note"><Info :size="12"/>题库按课程知识点生成；上传资料完成索引后也会补充对应练习</small><s
    mall v-if="dueMistakes" class="quiz-mock-note due-note"><Clock3 :size="12"/>有 {{dueMistakes}}
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ 道错题到期复习</small></aside><div class="quiz-main panel"><template v-if="!quizStarted"><div cl
    ↳ ass="quiz-empty"><div class="quiz-illustration"><div><BookOpen :size="26"/></div><span>✦</span><
    ↳ i></i></div><h2>{{dueMistakes?'先复测，再继续新内容':'来一次轻量知识检验'}}</h2><p>答题结果会进
    ↳ 入错题本；答错先看错因和知识卡，再用举一反三练习检验是否真正理解。</p><div class="quiz-facts"><s
    ↳ pan><b>{{quizCount}}</b><small>道练习题</small></span><span><b>次日</b><small>首次复测</small></
    ↳ span><span><b>2 次</b><small>间隔正确后掌握</small></span></div><button class="btn btn-primary"
    ↳ @click="beginQuiz"><Sparkles :size="15"/>开始测验 <ArrowRight :size="14"/></button></div></templ
    ↳ ate><template v-else-if="quizSubmitted"><div class="result-view"><span class="result-eyebrow"><C
    ↳ heckCheck :size="15"/>{{reviewMode?'复测已完成':'测验已完成'}}</span><h2>{{score>=80?'基础掌握得
    ↳ 不错！':score>=50?'再巩固一次会更好':'从解析开始稳步提升'}}</h2><p>测验结果已保存到账户，错题复
    ↳ 习记录会同步更新。{{reviewMode?'变式题答对后还需间隔复测，才会标记为掌握。':'错误题目已安排次日
    ↳ 复测。'}}</p><div class="score-circle" :style="{'--score':score+'%'}"><div><b>{{score}}<small>%<
    ↳ /small></b><span>正确率</span></div></div><div class="result-stats"><span><b>{{quizQuestions.fil
    ↳ ter(x=>x.type!=='简答题').length}}</b><small>客观题</small></span><span><b>{{Math.round(score)}}
    ↳ </b><small>正确率</small></span><span><b>{{dueMistakes}}</b><small>待复习</small></span></div><d
    ↳ iv class="review-list"><h3>答案解析 · 错因诊断与证据</h3><article v-for="(item,i) in quizQuesti
    ↳ ons" :key="item.id" class="review-item"><span class="review-mark" :class="{correct:isCorrect(ite
    ↳ m)}">{{i+1}}</span><div><b>{{item.title}}</b><p v-if="item.type === '简答题'">待人工评阅</p><p v
    ↳ -else>正确答案：{{Array.isArray(item.answer)?item.answer.join('、'):item.answer}}</p><small>{{is
    ↳ Correct(item)?item.explanation:diagnosisFor(item)}}</small><div v-if="item.evidence" class="quiz
    ↳ -evidence"><strong>{{item.evidence.title}}</strong><span>{{item.evidence.section}} · 内置示例知
    ↳ 识卡</span><p>{{item.evidence.excerpt}}</p></div><button v-if="!isCorrect(item)" class="text-lin
    ↳ k retry-variant" @click="openMistakePractice(item.id)"><RotateCcw :size="13"/>再做三道举一反三练
    ↳ 习 <ArrowRight :size="13"/></button></div></article></div><div class="result-actions"><button cl
    ↳ ass="btn btn-secondary" @click="router.push('/mistakes')"><BookOpen :size="14"/>查看错题闭环</bu
    ↳ tton><button class="btn btn-primary" @click="router.push('/analytics')">查看学习记录 <ArrowRight
    ↳  :size="14"/></button></div></div></template><template v-else><div class="quiz-progress-head"><s
    ↳ pan>{{quizQuestions[0]?.course||quizTopic}} · {{reviewMode?'间隔复测':'知识点自测'}}</span><spa
    ↳ n>{{reviewMode?'3 道新变式题':'内置示例题'}}</span></div><div class="quiz-progress-track"><i :st
    ↳ yle="{width:((currentQuestion+1)/quizQuestions.length*100)+'%'}"/></div><div class="question-met
    ↳ a"><span>第 {{currentQuestion+1}} 题 <small>/ {{quizQuestions.length}}</small></span><span class
    ↳ ="question-type">{{quizQuestions[currentQuestion]?.type}} · {{quizQuestions[currentQuestion]?.d
    ↳ ifficulty}}</span></div><h2 class="question-title">{{quizQuestions[currentQuestion]?.title}}</h2
    ↳ ><div class="option-list"><button v-for="(option, i) in quizQuestions[currentQuestion]?.options"
    ↳  :key="option" class="quiz-option" :class="{selected: selectedAnswer(quizQuestions[currentQuesti
    ↳ on], option)}" @click="toggleAnswer(quizQuestions[currentQuestion], option)"><span class="option
    ↳ -key">{{ ['A','B','C','D'][i] }}</span>{{ option }}<Check v-if="selectedAnswer(quizQuestions[cur
    ↳ rentQuestion], option)" :size="15" /></button></div>
<textarea v-if="quizQuestions[currentQuestion]?.type === '简答题'" class="short-answer" rows="4"
    ↳  placeholder="写下你的简要回答…" :value="quizAnswers[quizQuestions[currentQuestion].id]" @input
    ↳ ="quizAnswers[quizQuestions[currentQuestion].id] = ($event.target as HTMLTextAreaElement).value"
    ↳ ></textarea>
<div class="quiz-nav"><button class="btn btn-secondary" :disabled="currentQuestion === 0" @click
    ↳ ="currentQuestion--"><ArrowLeft :size="14" />上一题</button><div class="question-dots"><button v
    ↳ -for="(_, i) in quizQuestions" :key="i" :class="{active: i === currentQuestion, answered: quizAn
    ↳ swers[quizQuestions[i].id]}" @click="currentQuestion = i">{{ i + 1 }}</button></div><button v-if
    ↳ ="currentQuestion < quizQuestions.length - 1" class="btn btn-primary" @click="currentQuestion++"
    ↳ >下一题 <ArrowRight :size="14" /></button><button v-else class="btn btn-primary" @click="submitQ
    ↳ uiz"><CheckCheck :size="14" />提交测验</button></div>
<div class="quiz-bottom-note"><Info :size="13" />提交后系统将根据客观题答案计算分数；简答题需要
```

```
// 文件：src/views/WorkspaceView.vue
    ↳ 人工评阅。</div></template></div></section>
    <!-- Analytics -->
    <template v-else-if="section==='analytics'"><div class="analytics-controls"><div class="rang
    ↳ e-toggle"><button class="active" @click="analyticsRange=7" :class="{active:analyticsRange===7}">
    ↳ 近 7 天</button><button @click="analyticsRange=30" :class="{active:analyticsRange===30}">近 30
    ↳ 天</button></div><span><Clock3 :size="14"/>统计口径：当前账号的测验与学习记录</span><button clas
    ↳ s="btn btn-secondary" @click="exportAnalytics"><Download :size="14"/>导出报告</button></div><div
    ↳  class="analytics-stats"><article class="panel"><span>累计测验次数</span><b>{{quizHistory.length
    ↳ }} <small>次</small></b><em>仅统计已提交测验</em></article><article class="panel"><span>完成学习
    ↳ 任务</span><b>{{store.tasks.filter(t=>t.done).length}} <small>项</small></b><em>来自账户学习计划
    ↳ </em></article><article class="panel"><span>近 30 天学习时长</span><b>{{studyAnalytics.totalHour
    ↳ s}} <small>小时</small></b><em>由开始与结束专注计时累计</em></article><article class="panel"><sp
    ↳ an>近 {{analyticsRange}} 天平均正确率</span><b>{{averageScore ?? '—'}} <small v-if="averageScor
    ↳ e!==null">%</small></b><em>{{rangeRecords.length}} 次测验的平均正确率</em></article></div><div c
    ↳ lass="analytics-grid"><article class="panel analytics-chart"><div class="panel-heading"><div><h3
    ↳ >每日测验正确率</h3><p>单位：正确率（%）</p></div><span class="chart-period">{{analyticsPeriod}}
    ↳ </span></div><ChartPanel :option="trendOption" height="280px"/></article><article class="panel a
    ↳ nalytics-chart"><div class="panel-heading"><div><h3>错题复习状态</h3><p>按当前账号错题本掌握状态
    ↳ 统计</p></div></div><ChartPanel :option="distributionOption" height="285px"/></article></div><di
    ↳ v class="panel weak-table"><div class="panel-heading"><div><h3>需要关注的知识点</h3><p>按未掌握
    ↳ 错题次数排序</p></div><button class="text-link" @click="router.push('/mistakes')">进入错题本 <Ar
    ↳ rowRight :size="13"/></button></div><div v-for="item in mistakeList" :key="item.id" class="weak-
    ↳ table-row"><span class="weak-indicator"></span><span><b>{{item.topic}}</b><small>{{item.course}}
    ↳  · {{item.wrong}} 次错题</small></span><div class="weak-bar"><i :style="{width:(100-item.wrong*
    ↳ 15)+'%'}"/></div><b class="weak-score">{{item.wrong}} 次</b><button class="btn btn-soft" @click=
    ↳ "router.push('/mistakes')">去复习 <ArrowRight :size="12"/></button></div></div><div class="analy
    ↳ tics-footnote"><Info :size="13"/>测验、错题、学习计划和专注时长均来自当前登录账号。</div></templ
    ↳ ate>
    <!-- Mistakes -->
    <template v-else-if="section==='mistakes'"><div class="mistake-summary"><div class="panel"><
    ↳ span>待复习错题</span><b>{{mistakeList.filter(x=>!x.mastered).length}}</b><small>{{dueMistakes}}
    ↳  道已到复习时间</small></div><div class="panel"><span>已收藏</span><b>{{mistakeList.filter(x=>x.
    ↳ saved&&!x.mastered).length}}</b><small>优先练习重点题目</small></div><div class="panel"><span>已
    ↳ 验证掌握</span><b>{{mistakeList.filter(x=>x.mastered).length}}</b><small>至少两次间隔复测正确</s
    ↳ mall></div></div><div class="content-toolbar"><select v-model="selectedCourse" class="filter-sel
    ↳ ect"><option>全部课程</option><option v-for="c in courses" :key="c.id">{{c.name}}</option></sele
    ↳ ct><select v-model="mistakeDifficulty" class="filter-select"><option>全部难度</option><option>基
    ↳ 础</option><option>中等</option><option>较难</option></select><button class="filter-button" @cli
    ↳ ck="sortMistakesDescending=!sortMistakesDescending"><Filter :size="14"/>{{sortMistakesDescending
    ↳ ?'按错误次数排序':'按错误次数升序'}} <ChevronDown :size="13"/></button><span class="toolbar-spac
    ↳ er"></span><button class="btn btn-primary" @click="router.push('/quiz')"><RotateCcw :size="14"/>
    ↳ 随机练习</button></div><div class="mistake-list"> <article v-for="item in visibleMistakes" :key=
    ↳ "item.id" class="panel mistake-card"><div class="mistake-card-top"><span class="question-type">{
    ↳ {item.course}} · {{item.topic}}</span><span class="difficulty-label">{{item.difficulty}}</span>
    ↳ <button class="icon-button" :class="{saved:item.saved}" title="收藏题目" @click="toggleMistakeSa
    ↳ ved(item)"><Star :size="17" :fill="item.saved?'currentColor':'none'"/></button></div><h3>{{item.
    ↳ title}}</h3><div class="mistake-detail"><span><RotateCcw :size="13"/>答错 {{item.wrong}} 次</spa
    ↳ n><span><Clock3 :size="13"/>下次复测：{{new Date(item.nextReviewAt).toLocaleDateString('zh-CN')}
    ↳ }</span><span class="error-rate">已连续答对 {{item.correctStreak}} 次 · {{item.attempts.length}}
    ↳ } 条记录</span></div><div class="quiz-evidence mistake-evidence" v-if="item.evidence"><strong>{{
```

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// 文件：src/views/WorkspaceView.vue
    ↳ item.evidence.title}}</strong><span>{{item.evidence.section}} · 内置示例知识卡</span><p>{{item.
    ↳ evidence.excerpt}}</p></div><p class="mistake-diagnosis"><b>上次错因：</b>{{item.misconception}}
    ↳ </p><div class="mistake-actions"><button class="btn btn-soft" @click="router.push('/graph')"><Ne
    ↳ twork :size="14"/>查看知识图谱</button><button class="btn btn-primary" @click="openMistakePracti
    ↳ ce(item.questionId)"><RotateCcw :size="14"/>举一反三练习</button></div></article><div v-if="!vis
    ↳ ibleMistakes.length" class="panel empty-state"><CheckCheck :size="24"/><b>{{mistakeList.length?'
    ↳ 当前筛选下没有待复习错题':'完成测验后，答错的题会进入这里'}}</b><span>通过三道新变式题检验理解；
    ↳ 至少两次间隔复测正确后标记为掌握。</span><button class="btn btn-primary" @click="router.push('/q
    ↳ uiz')">开始一次测验 <ArrowRight :size="14"/></button></div></div></template>
    <!-- Study plan -->
    <template v-else-if="section==='plan'"><div class="plan-toolbar"><div class="plan-date"><but
    ↳ ton class="icon-button" title="前一天" @click="shiftPlanDate(-1)"><ChevronLeft :size="16"/></but
    ↳ ton><b>{{planDate}}</b><button class="icon-button" title="后一天" @click="shiftPlanDate(1)"><Che
    ↳ vronRight :size="16"/></button><button class="today-button" @click="goToday">今天</button></div>
    ↳ <div><button class="range-toggle active-tab" @click="planView='list'" :class="{active:planView==
    ↳ ='list'}">列表</button><button class="range-toggle" @click="planView='calendar'" :class="{active
    ↳ :planView==='calendar'}">日历</button><button class="btn btn-primary" @click="addPlanTask('新增
    ↳ 学习任务')"><Plus :size="14"/>新建任务</button></div></div><div class="plan-summary-strip"><span
    ↳ ><i></i>待完成 <b>{{store.taskCount}}</b> 项</span><span><CheckCheck :size="14"/>今日完成 <b>{{t
    ↳ odayTasks.filter(t=>t.done).length}}</b> 项</span><span><Target :size="14"/>每周目标 <b>{{prefer
    ↳ ences.weeklyGoalHours}}</b> 小时</span><button @click="addPlanTask('复习本周课程笔记')"><Plus :s
    ↳ ize="13"/>快速添加</button></div><div v-if="planView==='calendar'" class="panel plan-calendar"><
    ↳ label>选择日期<input v-model="selectedPlanDay" type="date" /></label><button v-for="day in calen
    ↳ darDays" :key="day.value" class="range-toggle" :class="{active:day.value===selectedPlanDay}" @cl
    ↳ ick="selectedPlanDay=day.value">{{day.label}} · {{day.count}} 项</button></div><div v-if="planV
    ↳ iew==='list'" class="plan-day-group"><div class="day-heading"><span class="day-mark">今</span><d
    ↳ iv><b>{{selectedPlanDay===new Date().toISOString().slice(0,10)?'今天':'所选日期'}}</b><small>{{p
    ↳ lanDate}}</small></div><span>{{todayTasks.filter(t=>t.done).length}} / {{todayTasks.length}} 已
    ↳ 完成</span></div><article v-for="task in todayTasks" :key="task.id" class="panel plan-task"><but
    ↳ ton class="task-check" :class="{checked:task.done}" @click="togglePlanTask(task.id)"><Check v-if
    ↳ ="task.done" :size="13"/></button><div class="plan-task-main"><b :class="{strike:task.done}">{{t
    ↳ ask.title}}</b><span><BookOpen :size="12"/>{{task.course}} <i></i><Clock3 :size="12"/>{{task.dat
    ↳ e.split('·')[1]||'09:00'}}</span></div><span class="priority-tag" :class="task.priority==='高'?
    ↳ 'priority-high':task.priority==='中'?'priority-medium':'priority-low'">{{task.priority}}优先级</
    ↳ span><button class="icon-button" @click="editTask(task)"><MoreHorizontal :size="16"/></button></
    ↳ article><button class="add-inline-task" @click="addPlanTask('完成今日学习复盘')"><Plus :size="14
    ↳ "/>添加一个任务</button></div><div v-if="planView==='list'" class="plan-day-group tomorrow"><div
    ↳  class="day-heading"><span class="day-mark">明</span><div><b>下一天</b><small>{{new Intl.DateTim
    ↳ eFormat('zh-CN',{month:'long',day:'numeric',weekday:'short'}).format(new Date(`${nextPlanDay}T12
    ↳ :00:00`))}}</small></div><span>{{nextDayTasks.length}} 项任务</span></div><article v-for="task i
    ↳ n nextDayTasks" :key="task.id" class="panel plan-task"><button class="task-check" :class="{check
    ↳ ed:task.done}" @click="togglePlanTask(task.id)"><Check v-if="task.done" :size="13"/></button><di
    ↳ v class="plan-task-main"><b :class="{strike:task.done}">{{task.title}}</b><span><BookOpen :size=
    ↳ "12"/>{{task.course}} <i></i><Clock3 :size="12"/>10:00</span></div><span class="priority-tag pri
    ↳ ority-low">{{task.priority}}优先级</span><button class="icon-button" @click="editTask(task)"><Mo
    ↳ reHorizontal :size="16"/></button></article><div v-if="nextDayTasks.length===0" class="panel emp
    ↳ ty-tomorrow">明天还没有安排，给自己留一点自主学习时间也不错。</div></div><div class="plan-local-
    ↳ note"><Info :size="13"/>计划数据已保存到账户。</div></template>
    <!-- Settings -->
    <section v-else-if="section==='settings'" class="settings-layout"><aside class="panel settin
```

// 文件：src/views/WorkspaceView.vue

```
    ↳ gs-menu"><button class="active" @click="scrollSettings('settings-profile')"><Settings2 :size="15
    ↳ "/>个人资料</button><button @click="scrollSettings('settings-goals')"><Target :size="15"/>学习目
    ↳ 标</button><button @click="scrollSettings('settings-goals')"><BellIcon/><span>通知偏好</span></b
    ↳ utton><button @click="scrollSettings('settings-privacy')"><Info :size="15"/>隐私与数据</button><
    ↳ /aside><div class="settings-main"><article id="settings-profile" class="panel settings-card"><di
    ↳ v class="settings-heading"><div><h3>个人资料</h3><p>管理你的学习身份与偏好信息。</p></div><butto
    ↳ n class="btn btn-secondary" @click="saveProfileChanges">保存更改</button></div><div class="profi
    ↳ le-edit"><div class="avatar avatar-large">{{ profileDraft.name.slice(0, 1) || '学' }}</div><div>
    ↳ <b>个人头像</b><small>当前使用姓名首字，头像上传暂未接入。</small></div></div><div class="form-g
    ↳ rid"><label>姓名<input v-model="profileDraft.name" autocomplete="name"/></label><label>学校<inpu
    ↳ t v-model="profileDraft.school" autocomplete="organization"/></label><label>专业<input v-model="
    ↳ profileDraft.major"/></label><label>年级<select v-model="profileDraft.grade"><option>大一</optio
    ↳ n><option>大二</option><option>大三</option><option>大四</option></select></label><label class="
    ↳ span-two">个人介绍<textarea v-model="profileDraft.bio" rows="3" placeholder="写一句关于你的学习
    ↳ 目标…"></textarea></label></div></article><article id="settings-goals" class="panel settings-ca
    ↳ rd"><div class="settings-heading"><div><h3>学习目标</h3><p>让智学研途依据你的目标安排节奏。</p><
    ↳ /div></div><div class="goal-setting-row"><div><b>每周学习目标</b><small>每周希望投入的学习时长</
    ↳ small></div><div class="stepper"><button @click="goal=Math.max(1,goal-1)">−</button><b>{{goal}}
    ↳ 小时</b><button @click="goal++">+</button></div></div><div class="goal-setting-row"><div><b>每日
    ↳ 专注时长</b><small>每日学习计划的默认时长</small></div><div class="stepper"><button @click="dail
    ↳ y=Math.max(1,daily-1)">−</button><b>{{daily}} 小时</b><button @click="daily++">+</button></div><
    ↳ /div><div class="goal-setting-row"><div><b>学习提醒</b><small>接收每日学习计划提醒</small></div>
    ↳ <button class="switch-control" :class="{on:notifications}" role="switch" :aria-checked="notifica
    ↳ tions" @click="notifications=!notifications"><i/></button></div></article><article id="settings-
    ↳ appearance" class="panel settings-card appearance-card"><div><h3>外观与显示</h3><p>选择适合你的
    ↳ 学习界面。</p></div><div class="theme-options"><button :class="{selected:!store.dark}" @click="s
    ↳ tore.dark&&store.toggleTheme()"><span class="theme-preview light-preview"><i/><i/><i/></span><b>
    ↳ 浅色模式</b><Check v-if="!store.dark" :size="15"/></button><button :class="{selected:store.dark}
    ↳ " @click="!store.dark&&store.toggleTheme()"><span class="theme-preview dark-preview"><i/><i/><i/
    ↳ ></span><b>深色模式</b><Check v-if="store.dark" :size="15"/></button></div></article><article id
    ↳ ="settings-privacy" class="panel settings-card"><h3>隐私与数据</h3><p>个人资料、偏好、计划、测验
    ↳ 历史和错题均保存在当前账号中。</p></article><div class="settings-local"><Info :size="14"/>课程目
    ↳ 录、图谱与题目由服务端数据库提供。</div></div></section>
    <div v-else class="panel empty-state"><CircleHelp :size="24"/><b>页面准备中</b><span>请从左
    ↳ 侧导航进入学习空间模块。</span></div>
    <Transition name="fade"><div v-if="sourceSelected" class="overlay" @click.self="sourceSelect
    ↳ ed=null"><aside class="source-drawer"><div class="drawer-top"><div><small>资料来源 · 本地模拟内
    ↳ 容</small><h3>{{sourceSelected.document}}</h3></div><button class="icon-button" @click="sourceSe
    ↳ lected=null"><X :size="17"/></button></div><div class="source-meta"><span><BookOpen :size="13"/>
    ↳ {{sourceSelected.section}}</span><span><Sparkles :size="13"/>{{sourceSelected.score}}</span></di
    ↳ v><div class="source-preview"><div class="preview-page-label">课程资料片段</div><p>…{{sourceSel
    ↳ ected.excerpt}}…</p><div class="highlighted-excerpt">{{sourceSelected.excerpt}}</div><p>以上片
    ↳ 段来自当前版本的本地演示内容，用于展示来源定位交互，不代表实际文档检索结果。</p></div><button cl
    ↳ ass="btn btn-primary full-width" @click="sourceSelected=null;router.push('/library')">在资料库中
    ↳ 查看 <ArrowUpRight :size="14"/></button></aside></div></Transition>
    <Transition name="toast"><div v-if="toast" class="toast-message"><Check :size="15"/>{{toast}
    ↳ }</div></Transition>
  </div>
</template>
```

// 文件：src/vite-env.d.ts

```
/// <reference types="vite/client" />
```